

Econ 110 Free Market Economics

Numbered Transcript

ECON 110 • Dr. David Henderson

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Lecture 1: Pillars of Economic Wisdom: Part 1

At a glance — Henderson's pillars of economic wisdom, part 1 (pillars 1-5).

- 1.1 Good morning. I'm here to give 10 talks, 10 one-hour talks, about various schools of thought that have to do with free markets. There are a lot of different schools of thought that come in their own way towards the idea that free markets are pretty good things. And we're going to see not just the schools of thought, but we're going to apply them. We're going to see what the insights are that they bring and how that matters. I'm not one of these people who says, "Oh, this one differs from that one just a little." You're going to see that each of them has something they bring to the table.
- 1.2 So, a little about myself. I was a senior economist for health policy from '82 to '84 with President Reagan's Council of Economic Advisors. And then I took on the energy portfolio in my second year there. That's me on the right. This is probably what I'm most likely to be remembered for: The Concise Encyclopedia of Economics. You can find all the entries online for free—not the book, of course, but the entries, and I'm always updating them. There are new bios to write of people and so on.
- 1.3 Let me give you the overview of the course, The Economics of Free Markets. Lectures 1 and 2 are what I call The 10 Pillars of Economic Wisdom. When I taught, for the last 20 years I taught, I started every lecture with these because they're so basic to economics, whatever school of thought you're looking at. Lectures 3 and 4 are Insights from the Austrian School. Lecture 5: Mainstream and Chicago Free-Market Economics. Lecture 6: Mainstream Economics. Lecture 7: The UCLA School. That's where I got my PhD, and it has its own little flavor it brings.
- 1.4 Lecture 8: The Hockey Stick. Now, I'm from Canada, eh, and you might think that's about that. No, it's not. It's about economic growth. I just want to have a little suspense here, so I'm not going to say more. Lecture 9 is about a school called Public Choice, and Lecture 10 is The West German Economic Miracle. It's a case after World War II, where a particular school of free market economic thinking had a huge impact and essentially created what we call the West German economic miracle.
- 1.5 Winston Churchill had this rule for people who worked under him when he was Prime Minister of Britain. "If you're going to communicate with me, I want you to put it on half a page, because you should be able to say anything important on half a page." When I read that years ago, I thought, "Can I do economics on half a page?" So this is what I came up with, and I call these The 10 Pillars of Economic Wisdom.
- 1.6 Pillar number one: TANSTAAFL—so grammatically bad—"there ain't no such thing as a free lunch." Pillar number two: Incentives matter; incentives affect behavior. Pillar number three: Economic thinking is thinking on the margin. That might sound a little funny, but it won't once we cover it. Pillar number four: The only way to create wealth is to move resources from a lower-valued to a higher-valued use. There's a corollary

there, which is that both sides gain from exchange. Pillar number five: Information is valuable and costly, and most information that's valuable is inherently decentralized.

- 1.7 Pillar number six: Every action has unintended consequences. Another way of saying that is, you can never do only one thing. Pillar number seven: The value of a good or service is subjective. Pillar number eight: Creating jobs is not the same as creating wealth. Number nine: The only way to increase a nation's real income, assuming no foreign aid, is to increase its real output. Finally, pillar number ten: Competition is a hardy weed, not a delicate flower. So in this hour, I'm going to do pillars one through five, assuming I've planned it well.
- 1.8 Pillar number one: TANSTAAFL—"There ain't no such thing as a free lunch." There are two senses of that term. In one important sense, it's always true, and in another important sense, it's usually true. Let's look at the sense in which it's always true. There are trade-offs. If you want more of this, you have to give up some of that. There's no way around it. You can't have everything. That's the sense in which it's always true.
- 1.9 Now, there was a famous president who said this: "The cost of one modern heavy bomber is this: a modern brick school in more than 30 cities. It is two electric power plants, each serving a town of 60,000 population. It is some 50 miles of concrete highway. We pay for a single fighter plane with a half million bushels of wheat. We pay for a single destroyer with new homes that could have housed more than 8,000 people." Which president said this? Someone who's very well known for his role in the military. He helped win the war in Europe: President Eisenhower. So he understood TANSTAAFL. He understood that, when you want more of something, you have to give up some of something else.
- 1.10 Now, I said there's another sense in which it's usually true. Another way of saying it is, there's a catch. Someone comes up to you on the street and says, "I'm going to buy you a free lunch." None of you were born yesterday. What do you want to know? "What's the catch? What do you want in return?" Sometimes there's no catch. I'm in my 70s, and so far, no one has ever come up to me and offered me a free lunch without there being a catch.
- 1.11 My mother taught me a little saying. "Don't look a gift horse in the mouth." I don't know, you've probably heard that. And the idea is—I found out later what it meant; I just took it at face value—it turns out, a way to judge a horse quickly, his health, is to look at his teeth. I didn't know that until some years ago. And so the idea is, someone's giving you a horse, don't question it, don't check the health. That's what my mother taught me. Well, what would we say with our understanding of there being a catch? We would say, "Always look a gift horse in the mouth." You might decide you still want to take it, but you want to do your due diligence.
- 1.12 Pillar number two: incentives matter. This is one of my favorites. It just applies all over the place. These are all true stories that I'm going to tell you. The Soviet economy was a

communist economy with completely centralized control. They didn't have a profit motive. They didn't have private property. All the factories were government-owned. So they needed a way to come up with artificial incentives.

- 1.13** This first story is about a Soviet glass factory producing glasses, the kind you drink from. And they, of course, want to have an incentive to produce output. The central planners in the Gosplan in Moscow said, "We will give you an incentive if, in a quarter, you produce more than X tons of drinking glasses." What do you think the glasses came out as? Very heavy, because that's a very quick way to produce those glasses. And then they said, "That didn't work. We will now give you an incentive. We will give you a reward if you overproduce your three-month quota by the number of glasses you produce." And what happened? The glasses came out very thin and very breakable.
- 1.14** There was another case—by the way, I'd remind you, these are all true stories from the Soviet economy. Nail, nail production. First of all, they say, "We're going to reward you for the number of nails you produce." So you get very thin nails. Then, "Oh, let's change it. Let's reward you for the weight, the tonnage of nails you produce." And so, you got very thick, heavy nails. There was a Soviet publication called Krokodil that made fun of this in a cartoon. It showed this big derrick holding up this huge two-ton nail. And someone says, "Comrades, we have produced our monthly quota." That was a way to—they didn't do that, but it is a true story that Krokodil had this cartoon because they understood it.
- 1.15** How people came from Britain to Australia, they were convicts. The people shipping them were paid by the number of convicts they took on board. That sounds reasonable, but this really bad thing happened. They got to Australia, and a lot of them had died en route. Between 1790 and 1792, 12% of them died. Why? Well, they got sick. "We're not paid to deliver live convicts," so they would just be left to die. Sometimes, if they were really sick, they'd be thrown overboard. Sometimes, they wouldn't be fed much.
- 1.16** And even though these were convicts, people cared, were somewhat humane about it. So the British government said, "Wait a minute. Let's change our way of compensating people who ship convicts. Let's have a bonus for each convict who arrives alive." And it dramatically changed it. The next year, 1793, 0.3%, three-tenths of a percent, of the people died. I mean, that's an order of magnitude drop because of that simple change in incentives.
- 1.17** This you see all over the place. I remember—I'll get to Admiral Rickover in a minute—but I remember, when I was reading economics, I was a math major. I discovered economics when I was in college. And I started reading it on my own, and no one ever said this in any of the readings, explicitly, but after about a year of reading economics on my own, I said, "You know what? Everyone's talking about incentives." That's what economists do. It was just maybe so obvious to them. Well, I'm a teacher. I want to make it obvious to you. Incentives matter.

- 1.18** There's a great example. I taught military officers at the Naval Postgraduate School from 1984 to 2017. So I liked to come up with examples that would apply to them. At the Naval Postgraduate School—as you can imagine, many of the officers were in the Navy, so some of them knew this; most of them didn't. They were pretty young. Admiral Rickover, who's known as the father of the nuclear navy, faced a real challenge back in the 1950s when he was getting submarines built. We have these submarines that we want to go very deep. Well, if you've watched *Das Boot* or any of those things, and you know that they go deep, what can happen? They can implode; the welds might not hold. So what he told the submarine welders is, "You're going to go on the maiden voyage, and we're going to go deep." That concentrated their minds.
- 1.19** A lot of you probably, maybe you've taken some economics, or you've never taken economics, but one of the big things in economics is a demand curve. Another is a supply curve. That's all about incentives. The demand curve is downward-sloping. Here's price on this axis, here's quantity on this axis; and so, at a lower price, people want to buy more. There's an incentive to buy more. The supply curve is upward-sloping. Here's quantity on this axis, here's price on this axis. At a higher price, people supplying it are willing to provide more.
- 1.20** By the way, that just simple, simple understanding leads—we'll talk about it in a later lecture—leads to understanding why price controls are so destructive, because they keep the price below what the market price would be, so people want more, demanders, and suppliers want to produce less.
- 1.21** Number three: Economic thinking is marginal thinking. It's not all or none, but more or less. So you hear people say, "Oh, when the price of gasoline goes up, people won't quit driving." And that's true. They won't quit driving, but they'll drive a little less. They might carpool, they might go to a store a little less often and do more shopping when they go—all those little marginal changes that matter.
- 1.22** I've got an example of this. Years ago, I was buying beer for my birthday party, and I was looking at the choices, and I like Heineken beer. And I saw that there were 24 for \$22, which was what I was used to. But then they had a smaller size, 18 for \$20. I'm standing there pondering, and just coincidentally, the Heineken guy who had just stocked it, looks at me, laughs because he thinks he knows what I'm thinking, and he came close. He says, "You're trying to figure out the average." And I go, "No, I'm trying to figure out the marginal." In other words, if I buy 18 Heinekens for \$20, compared to 24 Heinekens for \$22, if I go to 24, I'm buying six more Heinekens for only an additional \$2. So of course, I bought the 24.
- 1.23** Thinking on the margin—it also matters when you think about things like marginal tax rates. We economists use the term "marginal tax rate." What you probably have seen is tax brackets. So, on your last few dollars of income, what are you paying on that? So a lot of Americans will look at their federal taxes and say, "Oh, well, I'm paying roughly 12% to 14% of my income in federal income taxes, not including Social Security." But if

they look at the marginal dollars, the last dollars, they may be paying 22%, 22 cents on the dollar, or 24%, 24 cents on the dollar. That matters.

- 1.24** So let's say you're living in a nice area and you're doing all right, economically, and you get an offer to go to a less-nice area, but because of that, there's a bonus: there's higher pay. If you want to decide, figure out what you're getting to make that move, you don't look at your average tax rate; you look at your marginal tax rate. So if you're in a 22% bracket, and they're paying you an extra \$5,000, well, you're going to pay 22% of \$5,000, which is \$1,100. And of course, I haven't even included Social Security, which takes another bite. It doesn't say you should move, doesn't say you shouldn't move. It does say that that's a good way to think about making those moves, if you're looking at the tax aspect.
- 1.25** One of the most powerful tools in economics—a senior colleague of mine once called this the tool—is to look at marginal benefit versus marginal cost. So you do some activity, and you're trying to figure out how much of it to do. Maybe it's exercise. Well, you've figured out a certain amount, and then you say, "OK, what's the marginal benefit?" What's the additional benefit from, say, another 10 minutes of exercise? And what's the marginal cost from, say, another 10 minutes of exercise? It's a very powerful tool for looking at that issue, and that issue is everywhere.
- 1.26** A student is trying to figure out, "What's my marginal benefit of studying more?" I get a higher grade. What's that worth? That's the marginal benefit. What's my marginal cost? Will I give up other things? I give up leisure, I give up whatever. When I taught this to my students, I explained to them, "You might decide, once you understand the power of this tool, that you're not going to go for an A. And so, in a way, you're showing a lot of economic understanding by not going for an A because it's too costly."
- 1.27** Thinking on the margin should lead us to reconsider a famous line that probably most of our mothers taught us: "Anything worth doing is worth doing well." Let's think about that. There was an economist named James Buchanan, and we're going to run into him with the school of thought he was in. And he said once, "A real test of whether you understand economics is, do you understand what's wrong with that statement?" There are some things that are worth doing, but they're not worth doing that well.
- 1.28** Now, if you're doing welds on submarines, that's not one of them. But how about, we have three cats, and we have these kitty litters. And some mornings I'm in a hurry. And after I've cleaned out the kitty litters, there's a little that they put over the edge, and I'm in a hurry, and I leave a little kitty litter on the ground, on the floor. That's probably a good decision if I'm in a hurry. It's back to marginal benefit, marginal cost. How bad is it to leave it on the ground and then get it the next time I change the litter or clean the litter?
- 1.29** The only way to create wealth is to move it from lower-valued to higher-valued uses. So think about physical production. One of the standard examples is agriculture. What do

they do in agriculture? They take land, seeds, fertilizer, water, labor, machinery, do all those things, and everything turns out right, the output, the crop that comes out, is worth more than the inputs. They created value.

- 1.30** I live in Pacific Grove, California, and years ago, we had a Blockbuster. Remember those? And then a new way of getting videos came along, and Blockbuster changed into Trader Joe's. That resource, that outlet, that physical space, was moved from a lower-valued use—it had become lower value—to a higher-valued use.
- 1.31** Now, there's a corollary to that that really is as important as the pillar itself. And the corollary is that both sides gain from an exchange. Think about something you bought. You valued it at more than the money you gave up, or you wouldn't have bought it. The person who sold it to you valued your money as or more than what it cost him or her to make. So that person gained. Both sides gained. That is one of the most powerful principles in economics. Both sides gain from an exchange.
- 1.32** Now, one way to test a principle is to stretch it, to take an extreme example and see if the principle bends or breaks. I'm going to give you, to start with, a made-up example. You're out in the desert. You are dehydrating. If you don't get water, you die. You're crawling along on hands and knees, and in the distance, you see this sign: "Rita's Friendly Oasis." It's not a mirage. You get there. Rita's there. You say, "Rita, do you have water?" She goes, "Yes." "Oh, thank goodness. I need two quarts. And then I can rehydrate. I can make it to Addis Ababa and fly home."
- 1.33** She goes, "Fine. That'll be \$50,000." I should add, you have an American Express card, no limit. What do you say? By the way, when I asked some of my military students, the Marines said, "I'd shoot her." And I said, "Well, you don't have a gun. You don't have a gun. And she does, so that's out." What you say is, "I'll take it." Why? Because you pay \$50,000, your life is worth way more than \$50,000. So Rita gains in the exchange, \$50,000, probably minus a penny. You gain from exchange, the value of your life minus \$50,000. That's a big gain.
- 1.34** In fact, there's nothing in the principle that both sides gain from exchange that says both sides gain equally. And in fact, it might be a little surprising, but let's think it through. Who gained more? You did. Because if you value your life at more than \$100,000, you gained more than \$50,000, and she gained a penny less than \$50,000.
- 1.35** So why does this matter? It's a made-up point, but why does it matter? Because I've taken probably the most extreme example you can imagine of a monopoly. She's got a monopoly. Yet, even in that case, both sides gained. That's a pretty powerful principle. As I said, it's a made-up principle. I mean, it's a made-up example. Let's take one that isn't.
- 1.36** There are companies in the United States, Nike and others, that hire subcontractors in poor countries, that hire people there for as little as a dollar an hour, and sometimes they're 12-, 13-, 14-year-olds. Let's look at that case. Does the company that hires them

gain? Yes. Here's the tougher one, but think about it: Do the people getting hired gain? Yes, they do. How do you know? Because they're there voluntarily; they took those jobs. So that's better than the job they would have had.

- 1.37** It was interesting because someone on NPR, National Public Radio, some years ago, went down to one of those countries. And you could tell from his report on the radio, he got a surprise. He was going to find all these horror stories. And what he found was people who were saying, "You know what I used to work in? Agriculture. And you know what that paid? Less." And this was in the hot sun. Talk about sweat. So they were better off. He talked to this one person who said, "When I go back to where I came from, where I worked in agriculture, people see that I've put on 30 pounds," and unlike with me, that's a good thing. And they say, "How do I get those jobs?"
- 1.38** One other thing about that, there was a bill some years ago pushed by a US senator to fine US companies that set up and paid these kinds of wages. And this producer, I think it was rugs in Bangladesh, decided to lay off 2,000 children, just because of his fear of this bill passing. And what did they do? Started hunting through garbage, started going into prostitution.
- 1.39** See, the problem we Americans often have is, we say to ourselves, "I wouldn't want those jobs. I wouldn't want my kids to have those jobs," and that's absolutely true. But those people are in very different circumstance. No one's offering them a green card. They're in those countries. I wrote a piece on this some years ago, back in the '90s, when this had become an issue. I looked at various groups in the United States that were pushing to limit what they called sweatshops. And I pointed out, "You're looking at people who have a bunch of bad options, and they've chosen the best one of a bunch of bad options. If you take away the one they chose, you're making them worse off." So the way I put it is, "Someone who tries to get you fired is not your friend."
- 1.40** And by the way, the whole idea of gains from exchange explains why it's so hard to fight the drug war. I mean, think about it. Drugs are illegal, fine. Someone wants to buy them, someone wants to sell them. That's a voluntary exchange. So if you are, let's say, the buyer, you aren't going to go to the government and say, "He cheated me on that transaction." And so, they're not going to find—and how's that different from other crimes? Well, let's say you get robbed, you are going to go to the cops and tell them. Let's say one of your family members gets murdered. You're going to go to the cops and try to get that person caught.
- 1.41** But this is a voluntary exchange. And that's why it's so hard to fight it. And that's why they get very intrusive in the drug war, checking people's bank accounts, having, when you bring more than \$10,000 into a bank, having to fill out a special form, all that kind of stuff. So, as you can see, the whole idea of gains from exchange explains gains from exchange, why people gain, and it also explains a whole bunch of other things.

- 1.42 Pillar number five, which is one of the more subtle pillars: Information is valuable and costly, and most valuable information is inherently decentralized. Let's first of all look at the easy part of that: valuable and costly. If information weren't valuable, I wouldn't be paid to give these talks. People value it. If information weren't costly, I wouldn't have to be paid. So there's evidence right here in me, in this room, that information is valuable and costly.
- 1.43 The person who really drove this home, and we're going to get to see him when we talk about the second talk on the Austrian School, is an economist named Friedrich Hayek, who co-won the Nobel Prize in economics in 1974. This is in 1975. That's me getting his autograph. Now, he was involved in what was called the socialist calculation debate. He wrote a famous article in the 1945 American Economic Review, called "The Use of Knowledge in Society." We're going to talk about it in detail when I get to the Austrians, when I get to him.
- 1.44 His insight helps explain why the Soviet Union failed, why it was a colossal failure. Now, granted, he focused on information. Incentives mattered a lot, too, as we've already seen, but I'm going to focus on the information part. There was a book from a reporter named Scott Shane, who covered the Soviet Union. He was the bureau chief in Moscow for the Baltimore Sun. And from 1988 to 1991—so remember, in 1988, you got straight-out communism, '91, you're starting the transition. So he got to see it at a very interesting time.
- 1.45 Based on what he saw, he wrote a book called "Dismantling Utopia: How Information Ended the Soviet Union." I think it was one of the best books of the '90s. And I'm saying that, having even published my own encyclopedia. And here's what he said. He took the example of shoes. He said, "My informal survey suggested that some of the longest lines in Moscow were for shoes. At first, I assumed that the inefficient Soviet economy did not produce enough shoes. For that reason, even in the capital, Moscow, people were forced to line up for hours to buy them. Then I looked at the statistics. I was wrong. The Soviet Union was the largest producer of shoes in the world. It was turning out 800 million pairs of shoes a year, twice as many as Italy, three times as many as the United States, four times as many as China." Remember, China has a multiple of their population.
- 1.46 "Production amounted to more than three pairs of shoes per year for every Soviet man, woman, and child," so they got shoes. Here's the problem. "The problem with shoes, it turned out, was not an absolute shortage. It was a far more subtle malfunction. The comfort, the fit, the design, and the size mix of Soviet shoes were so out of sync with what people needed and wanted that they were willing to stand in line for hours to buy the occasional pair, usually imported, that they liked."
- 1.47 Here was his conclusion. "At the root of the dysfunction was the state's control of information. Prices are information." That is so key. "Prices are information—the information producers need in order to know what and how much to produce. In a

market for a product as varied in material and design as footwear, shifting prices are like sensors taped to the skin of a patient in a medical experiment. They provide a constant flow of information about consumer needs and preferences. When the state controlled prices, it deprived producers of information about demand."

- 1.48** I mean, I had this little thing I'd do in my class, when one of my students nailed it, and I'd say, "I could hug you." Well, I could hug him. By the way, he moved from The Baltimore Sun to The New York Times, and he'd read about this tough question I'd asked one of George W. Bush's officials when he came out to give a talk. It got written up, and he wanted to know if it was recorded, and I said no. Anyway, I said, "Hey, I love your book. That was great, the way you laid out Hayek's insights so well." And he said, "Who's Hayek?" So this guy didn't come to it from reading academic literature; he came to it from seeing it in practice in the Soviet Union.
- 1.49** Another example of the power of decentralized information. September 11th, 2001, was a horrible, horrible day in this country. There were actually three bright spots on that horrible day. The first one: people evacuating the Twin Towers. The Port Authority, which owned those buildings, told people, "Don't get out of the building." And people kind of used their own information and their own gut feel, and a lot of them got out. I had a friend who had a friend, has a friend, I think he's still alive, who had his own little business, and he had 20 people. And when the first plane hit, he didn't say, "Oh, grab your computers, let's leave." He said, "Let's get out." They acted on their own decentralized information. They got out, and they all lived.
- 1.50** The second thing: grounding the planes. There were approximately 2,000 planes in the air over US airspace when that hit. And the plan, the idea that the—I can't remember his name, but the head of the FAA didn't wait on the White House. He just said, "Let's get those things down." And it meant you weren't going to be landing where you thought you're going to be landing. You're going to be landing somewhere else. And the air traffic controllers moved them from one space to another. They'd never dealt with it in their lives. All those planes got down safely.
- 1.51** Then, something interesting happened. A few weeks later, someone in the FAA said, "This could happen again. Let's have a task force to think about how we would deal with it in the future, because we did seat-of-the-pants this time." So they met for a couple of times, and then they said, "You know what? No. It'll probably be different next time if there is a next time. Look how we did, dealing in real time with decentralized information. You go to a manual, you're going to lose time, and it might be the wrong advice, anyway."
- 1.52** And finally, even though it was a tragedy, United Flight 93. That was the one that crashed in Pennsylvania. What was different? Well, first of all, the advice we were always given—I used to fly a lot in those days—the advice we were always given is, if you get hijacked, go along with the hijackers. You're going to get a free trip to Cuba. You're going to have this great story to tell your grandchildren.

- 1.53** But what was different on Flight 93 was that these people had cell phones, and they found out on their cell phones—which means, by the way, they broke a federal law by getting on their cell phones—and they found out what had happened with some of the other planes. So now, no, no, no, no, it's not a free trip to Cuba. If I were to use the language of game theory, I'd say "the payoff matrix has changed." You're going to die if you don't do anything. You might die if you do something. So do something, because it might work. Of course, it didn't work for them, which was tragic, but it did work for whoever would have been attacked. People think it would have been either the White House or the Capitol building.
- 1.54** So there you have five pillars of economic wisdom, and those pillars are behind a whole lot of economics. And often it's a few pillars together. So, for example, we were talking about communism and socialism and why they fail. Well, that's the information pillar number five. It's also the incentive pillar number two.
- 1.55** Also, why does price-setting in a communist society not work? Well, it's the incentive thing again. If you set the price too low, less is supplied and more is demanded. TANSTAAFL. You've got limited resources. Even in our wealthy country, we always want more, and therefore we have to make choices and we have to make trade-offs, and there's no way around that.

Lecture 2: Pillars of Economic Wisdom: Part 2

At a glance — Pillars of economic wisdom, part 2.

- 2.1 In the first hour, we covered pillars one to five. This next hour, I want to cover pillars six to 10. Let's start with six: Every action has unintended consequences. Another way of saying that is, you can never do just one thing. You want to achieve goal X, so you take an action to do that, but typically, everything you do has other consequences. Sometimes they're good, sometimes they're bad, and they're often unintended.
- 2.2 There's this famous story, it might be apocryphal, about rats, a little town in England being overrun by rats. The town fathers or town mothers and fathers, the city council said, "We should have a way of getting people to get rid of rats." And they started thinking about what we call pillar number two: incentives matter. "Let's pay people to get rid of rats. Well, how do we pay them? They have to bring in a pair of rats' ears." So people did that. The number of rats started falling, and then the number of rats started rising. What was happening? People were breeding rats in order to get those payments.
- 2.3 That's kind of a funny story. This one is not— the Exxon Valdez, which spilled a lot of oil off the Alaska coast. When something like that happens, Congress will often say, "Let's change some law to give an incentive for that not to happen." So again, they're thinking pillar number two: incentives matter. So they said, "We're going to raise the fines for something like that, and we're going to increase the amount that the firms are liable for, civilly." It sounded like a good incentive to do.
- 2.4 Well, here's what happened. I'm going to jump ahead to this Wall Street Journal story. "Already, at least one international oil company has been parking the title to some of its oil in transit with a willing outside party, a company controlled by fugitive oil trader Marc Rich in Switzerland. Refiners are also shunning legal ownership until the oil they buy arrives at their plants, instead of accepting it in the seller's port as before."
- 2.5 Get it? In other words, if they don't own the oil in transit, they're not liable. And then at the same time, some shippers are getting ready to subdivide their fleets into single-ship companies, each with minimal assets, so there's less to take over in a lawsuit, to protect the rest of their operations from any one claim. What that meant was, you didn't get more safety, you got less safety, because you got fly-by-night operators like that guy, Marc Rich, who are using potentially less safe ships and very few assets to attach, and therefore, not as much assurance as you had.
- 2.6 Another one, it's called patient stacking in Britain. Some years ago, I don't know if you've ever seen any British TV where they talk about the NHS, the National Health Service. It's government-run. It's essentially socialized medicine. They charge zero for health services. So again, think about incentives, an incentive to go to the emergency room, even if you have some minor thing happening, because you're going to get

charged zero. So no really good way of distinguishing people with serious problems and people without. And you can line up for hours and hours.

- 2.7** Well, the government of Britain, I think it was under the Labour Party, said, "Oh, we can't have that. Let's have a rule that you can't be in the emergency room for more than four hours. There can't be a queue of more than four hours." And then somehow the hospital will be fined or hurt in some way legally if they let it go over four hours. So they did that. The hospitals did that. But the Labour Party, I think it was, didn't put any more resources into hospitals.
- 2.8** So what happened? An ambulance would show up with a patient, find out whether it would be more than a four-hour wait. If it were, keep the patient in the ambulance just outside the hospital. Then they're not in the queue, so there's no penalty. By the way, do you have an ambulance you can't use for someone else? President Nixon, on August 15th, 1971—I still remember the day, I was 20 years old—imposed a price freeze across the whole US economy. You could not raise a price or a wage by a penny without breaking the law. It was a 90-day price freeze.
- 2.9** We'll see a lot later about what's wrong with price controls in general, but I want to talk about oil and gasoline. So, we got shortages in some areas. The price of oil at the time was about \$3 a barrel. Oh, and that was a 90-day freeze. And then they went to phase two, and they allowed a little bit of an increase in prices in phase three and so on.
- 2.10** Well, while we're in phase three, OPEC, the Organization of Petroleum Exporting Countries, gets powerful and forms a cartel and uses their cartel power to raise the price of oil from \$3 a barrel to \$11 a barrel. But we have price controls on oil in the United States. So the federal government let the producers of oil raise the price to \$4.25, which isn't close to \$11, and they had price controls on gasoline. And so, what happened? We got lineups. I used to tell this story to my students. They'd say, "Oh, my parents remember that." I guess if I were teaching there still, they'd say, "My grandparents remember that." And it was because of these price controls.
- 2.11** So, Nixon is the one with the price controls, and then Ford comes in after Nixon resigns. And Ford and Congress, it was bipartisan, come up with this idea. They say, "Look, people are driving these big gas-guzzling cars." Of course they are. They were buying them, most of them, when the price of oil was \$3 and the price of gasoline reflected that. "So they're not taking account of the increased scarcity of oil." Of course they're not. You guys, the federal government, are telling them not to, by having this artificially low price. "So we're going to make them buy cars that have higher fuel economy."
- 2.12** That was CAFE. CAFE, not the stuff, not the place you go for coffee. Corporate Average Fuel Economy. So, every automobile producer was told, starting in a particular year, I think it was 1978, that your fleet of cars you sell that year are going to have to have a certain average fuel economy. If you miss by a tenth of a gallon on average, you pay a \$5 fine, which doesn't sound like much, but that's on your whole fleet. If you miss by a

mile per gallon, you pay a \$50 fine, and that could, by three million cars, get to be a big number.

- 2.13** So, what do the auto companies do? What's the quickest way to increase fuel economy without major changes in technology? It's to cut the weight of the car. And so they cut the weight of the cars by a few hundred pounds, and then what happened? We got more fatalities. And I'm going to talk later, when I talk about one particular school of thought, about these two economists who came up with an estimate of the serious number of increase in fatalities in one particular model year due to CAFE.
- 2.14** So what the government could have done is said, "You know what? We blew it. Let's get rid of the price controls on gasoline." But they didn't. They just went deeper. And by the way, they did that with a lot of other things— standards for buildings to save fuel and so on—rather than letting the price go up and letting people adjust.
- 2.15** There was a famous French economist and economic journalist named Frédéric Bastiat. He wrote a famous article titled "What Is Seen and What Is Not Seen." In every class I taught for the last 20 years I was teaching, I used this article. Here's what he said, and this completely relates to the unintended consequences part. He said, in the economic sphere, "an act, a habit, an institution, a law, produces not only one effect, but a series of effects. Of these effects, the first alone is immediate; it appears simultaneously with its cause—it is seen. The other effects emerge only subsequently. They are not seen. We are fortunate if we foresee them."
- 2.16** So again, he's getting at unintended consequences. And then he says, "There is only one difference between a bad economist and a good one. The bad economist confines himself to the visible effect; the good economist takes into account both the effect that can be seen and those effects that must be foreseen." As I said, I used this in every class I taught, this Bastiat article, and I know it had an effect on the students because I was out in the quadrangle and I saw a few of my former students huddled together, and they looked up and they saw me—remember this guy's name is Bastiat—they'd say, "There goes that Bastiat." Or at least it sounded something like that.
- 2.17** Number seven: The value of a good or service is subjective. There was a song by Sly and the Family Stone, "Different Strokes for Different Folks." That's how I summarize that principle. In other words, it's not the case that everything has the same value to everyone. In fact, if it were, think back to gains from exchange; there wouldn't be any. If you value something exactly as I value it, there's no point in exchanging.
- 2.18** And by the way, just back to that, about gains from exchange. We often hear people say, "You should give equal value for equal value." No, you shouldn't. Because if you do, you wouldn't bother. You always give up something you value less than what you're getting, and the other person gives up something he values less than what he's getting, so for both of you it's unequal. And we're going to see a lot more about that when we talk about the Austrian School, because Menger, Carl Menger, who I'm going to talk

about in the third hour, was one of three people who really came up with it and nailed this one.

- 2.19** But let me just say, this explains a difficulty that I bet a lot of you have had, which is the difficulty of buying gifts even for a loved one, even if you've lived with your wife for, oh, about 43 years, because she knows what she wants. And sometimes there's also this thing called tacit knowledge. This famous philosopher wrote a book on tacit knowledge, Michael Polanyi. She knows what she wants. It's tacit. She can't even tell you. And you're going to try to buy it for her? Good luck. And that, again, the value of a good or service is subjective.
- 2.20** Pillar number eight: Creating jobs is not the same as creating wealth. There's a famous story about a US engineer visiting China in the 1950s, and he's seeing them dig to build a big dam with shovels. By the way, when people tell this story, they say it was Milton Friedman. I looked into it. It wasn't Milton Friedman. We're going to talk about him later. It was an engineer. And he says, "You could do way better if you use, say, power bulldozers and stuff like that." And the person, the Chinese official in charge of the project, says, "Oh, no, but we want to create jobs." And the engineer says, "Oh, OK, I get it. If you really want to create jobs, don't have them use shovels. Have them use teaspoons. You'll get way more jobs."
- 2.21** And it makes the point that, of course, we aren't trying to maximize jobs. In fact, for a given price of labor, we're trying to minimize jobs for a given output or jobs in capital. Then that labor can go elsewhere and produce something else. And by the way, what would have been a good response, thinking about incentives, for the Chinese guy to say, "Wait a minute, capital is expensive, labor is cheap, we're doing the right thing," and they probably were.
- 2.22** But now back to this pillar: Creating jobs is not the same as creating wealth. Think about it this way. If we didn't destroy jobs, we couldn't increase wealth. Why do I say that? Well, technological change is usually the thing that drives the increase in wealth, and that necessarily means not as many people are going to be needed to produce a given thing.
- 2.23** So I came up with this table to make that point. These are all true numbers. This is long-distance calls in 1970, in the United States: 9.8 billion— I want to just work down that column. Switchboard operators: 421,000. Calls per day per operator: 64. Operators as a percent of the workforce: half a percent, 0.51%. Because when you made a long-distance call in 1970, you had to deal through a long-distance operator. It's kind of hard to imagine, but that's what you had to do. As a result, by the way, calls were very expensive. I remember when we got a call in 1965, we were in our small town in Manitoba, and my father got a long-distance call, and everyone got quiet. It's got to be important. And it was my aunt in Peoria, my great aunt in Peoria had died, and he had to go and close the estate. So that's 1970.

- 2.24** 1996, huge progress: 94.9 billion calls, 10 times as many. Switchboard operators: 164,000, 40% as many. Calls per day per operator, a huge multiple: 1,585. Operators as a percent of the labor force: 0.14%, one-seventh of a percent. And by the way, the work time to buy five minutes was two-thirds of an hour. You had to work two-thirds of an hour at the average wage to buy a five-minute call, and by 1996, it was one-eighth of an hour.
- 2.25** Now, this next call is made up, but to make a point. Imagine we'd had no progress, no improvement in technology, for long-distance calls, but we made the same number in 1996 anyway. So we made 94.9 billion. But with no improvement in technology, how many operators would we have needed? 4.1 million. Again, calls per day per operator: 64. Operators as a percent of the labor force would have been 3.21%, over six times as many. And of course, the work time to buy five minutes would still be back at two-thirds of an hour.
- 2.26** That's a made-up thing because, of course, if we hadn't had the technological change, we wouldn't have had 94.9 billion. It may be a much lower number. But it's to make the point— look how many more operators you would have needed if you hadn't had technological change. And that makes the point that, hey, this freed up a lot of labor, and they can go elsewhere.
- 2.27** So the story of economic growth is the story of doing more with less. So jobs per se are not a good goal. Now, later on, I'm going to be talking about how the minimum wage destroys jobs, and I'm going to be saying that's bad, but that's different because it's destroying jobs that people wanted to have and other people were willing to pay for without being subsidized to do so.
- 2.28** Pillar number nine: The only way to increase a nation's real income, absent foreign aid, is to increase its real output. So real income and real output are just opposite sides of a balance sheet. The way you make income is by producing output. There's no other way. Now, if two presidents in our history, Hoover and FDR, had understood this, we probably wouldn't call the Great Depression, the Great Depression. They're the ones who put "great" in "depression." Why?
- 2.29** Hoover, who was from the left wing of the Republican Party, thought the way you kept the economy going was to pay workers more and not let wages fall. Wages would have fallen because the money supply had fallen, something we'll see later in the course. So he got a whole bunch of major companies to the White House and said, "Please keep your wages high." And many of them followed, did that. Well, if they're keeping their wages high, they don't want to hire as many people. And so they laid people off, and we got higher unemployment.
- 2.30** FDR comes in, in the midst of the worst part of the Great Depression, and he puts into law what Hoover had tried to do just with persuasion. He puts in the National Industrial Recovery Act that forms the economy into about 500 cartels, each industry being a

cartel, and has them set wages high; they're required by law to have wages be high. Therefore, employers don't want to employ as many people. People remain unemployed for a long time.

- 2.31** That's why I say that if Hoover and FDR had understood that the only—when they're hiring fewer people, they're producing less, less output. If they'd understood the way to increase a nation's real income is to increase its real output, they wouldn't have made that mistake. They wouldn't have made those mistakes. Pretty amazing how a little economics goes a long way.
- 2.32** Pillar number 10: Competition is a hardy weed, not a delicate flower. When you read the literature on antitrust—antitrust laws are used to go after companies that are thought to be colluding or thought to be too big or whatever—you'll often read the people writing as if they think competition is like this orchid. It's really delicate, and we need the right amount of sunshine and the right amount of fertilizer and the right amount of everything, water and so on. Bad metaphor, bad analogy. Competition is a hardy weed. It just happens.
- 2.33** Why does it happen? Because let's say some firms are making huge profits. What happens next? Other people see this, and they want to enter the industry and make those profits, and that drives down the profits. So consumers gain, and we get more competition leading to lower profits, but still some profits, and consumers gaining. Profits attract competitors the way honey attracts ants.
- 2.34** Now, when my daughter was really young, there was this company that—she's 41 now—there was this company that came along called TCBY, The Country's Best Yogurt. The company that put this together would sell franchises. They'd say, "OK, you get this franchise for this market area." And you had to pay a pretty penny for it, maybe \$100,000. And they would guarantee that they wouldn't have any other TCBY franchise in that market area. So you knew you didn't have to compete with another TCBY. A whole bunch of those TCBY franchises went broke. Why? Because there's nothing magic about TCBY yogurt, frozen yogurt. Baskin-Robbins put in just a couple of yogurt machines, and suddenly they're competitors. Other competitors came along. So TCBY kept their word, didn't have any more franchises, but still, these companies lost their shirts.
- 2.35** Now, that's not the end of the story, because firms don't usually like competition. They like an easy life. And so they understand that competition is a hardy weed. So, what's a way to have less competition? It's to go to the government and say, "We don't want to compete, so let's get our government to help in preventing competition." And there are many, many examples.
- 2.36** I used to ask my students 30 years ago, "What monopoly upsets you the most?" And it was typically the cable company, because there was just one in any market area. Now, fortunately, that answer doesn't hold, but there was one, and it was because cable

companies went to municipal governments and said, "It doesn't make sense to have more than one. Let's just have one." But then satellite came along and knocked out their business. And of course, there are other things technologically that have come along.

- 2.37** Lawyers have the American Bar Association, and the American Bar Association has rules for accreditation, and they use those rules to slow the growth of new law schools. Why? Because they don't want competition. Unions will try to prevent non-union firms. They'll try to get certain laws in place so that the government has to, typically for government contracting, the government has to hire union workers. So we see it all over the place.
- 2.38** So where are we here? We've gone through 10 Pillars of Economic Wisdom. We're going to see in various ways that they're behind pretty much everything. Almost every question in economics you try to answer will draw on one or more of these pillars. That might not be all you need; you might need certain facts about the situation, but you do find these pillars coming up again and again, whether it's incentives, TANSTAAFL, information, competition being a hardy weed, and so on.
- 2.39** Let's talk about AI. AI will put a lot of people out of work, and then— here's what I think—they'll find other work. Can I say what it is? No, I can't. But what I can do is look at history. What percent of the labor force in 1900 was employed in agriculture? 40%. What percent of the labor force is employed in agriculture now, even though we export and sell to the world? 2%, because agriculture became so much more productive.
- 2.40** Now, where did those workers go? They weren't unemployed long-term. They went into manufacturing. Manufacturing grew through the '50s, '60s, and '70s. And by the way, there's this myth that we don't have manufacturing. Our manufacturing output is only 7% below its peak. But what is way below its peak is manufacturing employment, because manufacturing has become so productive. And where did those people go? Into services.
- 2.41** AI will certainly change lots of things. It's very hard to predict how—I don't want to say faith because it's more than faith—but my confidence is, people will find other work. I think the way you would mess it up is with a guaranteed annual income, because then a lot of people would just take that and not even try to adjust. I've given some talks recently to young people, and I say to them, "Make AI your friend. Figure out how to use artificial intelligence in your job or in some job you might have in that firm, and then you'll be ahead of the game." And people are doing that a lot, people who get that. Make it your friend, not just your competitor.
- 2.42** Is there active monopolies today? I recall a big story back maybe in the '80s. It was with Bell, and they had a big monopoly on the telephone and everything, and so they dismantled that. And I remember that being a really big story.

- 2.43** So, are there monopolies today? Yes. Is the telephone company a monopoly? No longer, because there's so much more competition there than there was. And again, a way to see that is, for example, with cell phones, and what does a long-distance call cost you now? Zero, essentially. But there are still some pretty active monopolies protected by the government. If they are private monopolies, the only way they can hold on to it is by continually improving the product or continually dropping the price or both.
- 2.44** So, people think of Amazon, and the only way they can do this is by making it so much better. So I order lots of vitamins, and I finally gave up the other day on ordering vitamins from other companies that I won't mention because they'd say, "We're going to charge you \$9 to ship it, and it's going to take 10 days." And I found out— maybe it's lower quality, I don't know—but I go to Amazon and no shipping charge because I'm a subscriber for the year, and I get it the next day. I mean, there's no way, if that's a monopoly, I want more of them. They're so good in that respect. Walmart is another one that's quite good in that respect.
- 2.45** So I think the typical monopoly will be, say, lawyers—we'll talk a little more about that later when we talk about some of the economists who've discussed those things. Let me tell you a whole success story about monopoly. Up until the late 1970s, the Interstate Commerce Commission would not allow a trucking company to move goods from point A to point B without a license. And so, the company would apply for that license. And who would complain? Who would show up at the hearing? Other companies already shipping between A and B. And they'd say, "No, no, shouldn't do that."
- 2.46** There was this thing called public convenience and necessity. When I was actually a junior economist at the Council of Economic Advisors, before the Reagan thing, but back under Nixon, I was involved with transportation regulation. And people even had an acronym for it, PC&N, public convenience and necessity. And it was used to prevent all kinds of people from shipping from A to B.
- 2.47** Also, let's say you had a license to ship goods from A to B, and you want to come back to A, well, you'd like to fill your truck with goods in A and B. You can't because you didn't have a license. So we had a lot of empty trucks running around. And then Carter, President Carter—well, Ford started a little, but President Carter got into high gear to reduce regulation, and then that kept going during the Reagan administration. And we have so much more competition now in trucking.
- 2.48** Another example is airlines. You've seen those pictures of people traveling that people put them on Facebook and so on. "Oh, wasn't travel so nice in 1965? And everyone wore suits." Yeah. And they paid what for many of us would be two months' wages for that trip. And so, two companies came along that figured out a way around it. One was Southwest Airlines. The Civil Aeronautics Board, which had limited the number of

airlines in 1938 and had not allowed a new one since then, regulated interstate travel. They couldn't regulate intrastate travel.

- 2.49** So Herb Kelleher started Southwest Airlines, and he went between three cities in Texas: San Antonio, Dallas, and Houston, at very low prices. California, where I moved to go to graduate school in 1972, had PSA, Pacific Southwest, just within California. I remember once paying to fly from LA to Oakland in the mid '70s, \$8.
- 2.50** What happened was this: Senator Kennedy, Ted Kennedy, wanted an issue because he wanted to run in 1980 for the Democratic nomination for President. He hired a guy as a staffer, a Harvard law school professor, and he gave him two options. He said, "What should I choose as my issues?" One of the options that the guy gave him was the deregulation of airlines. It looked right. Kennedy chose it and ran with it. And so they got this guy who was chairman of the Civil Aeronautics Board, an economist named Alfred Kahn, who, instead of saying, "Let's use the law in the most restrictive way we can," went the opposite way and said, "Let's allow as much competition as we can within the law." So if an airline comes and wants to fly a route, they used to have to get permission; you're in. An airline comes and wants to drop prices, you may.
- 2.51** Then we get the actual bill in 1978 that deregulates. We have much, much lower fares now. Are we crunched in our seats? Yeah, we are. But when airlines try to increase that space a little and charge for it, most of us consumers say—by our behavior, not by our words—we'd rather be crunched in those seats. So there was just a huge amount of deregulation that got rid of a lot of monopoly in the US economy back then.

Lecture 3: Austrian School: Part 1

At a glance — The Austrian School, part 1.

- 3.1** In this hour, I want to talk about the Austrian School of Economics. Now there were a lot of people in that school. For this hour, I'm going to talk about two of them, Carl Menger and—I think I'll say this correctly—Eugen von Böhm-Bawerk, both Austrians, very important in the development of economics. They were part of, especially Carl Menger, and we're going to see in a couple of minutes, was part of what's called the Marginal Revolution.
- 3.2** Prior to the Marginal Revolution, economists tended to buy into what's called the labor theory of value. Adam Smith, who wrote the famous "An Inquiry Into the Nature and Causes of the Wealth of Nations" in 1776, believed in the labor theory of value. David Ricardo, in the early 1800s, believed the labor theory of value. Karl Marx based his whole edifice, his whole system of thought, on the labor theory of value. So what is it?
- 3.3** The labor theory of value says the value of something derives from the amount of labor that goes into it. I want to let that sink in, to see if there's even an inkling of what might be wrong with that, but we'll see. So the labor theory of value says the value of something is based on the amount of labor that goes into it. Well, then what happened was the Marginal Revolution of 1871. Three people in three different countries came up with it, independent of each other, with basically the same idea.
- 3.4** I think Menger said it best. William Stanley Jevons in Britain, Léon Walras in France, Carl Menger in Austria. And here was the basic—there were two basic insights. Value is subjective. Does that sound familiar? That's pillar number seven. There's no such thing as intrinsic value. Value depends on the person. What you value will be different from what I value. Even if we value the same thing, the value we put on it will differ between us.
- 3.5** And there was another part—and here's where the marginal comes in—each unit is less valuable than the previous unit. You go to Taco Tuesday at a restaurant, and you're trying to decide how many to buy. You start with two. The third one, even though it's priced the same, you value a little less. The fourth one, you value less. So at some point, of course, the price exceeds the value, the marginal value, so you quit buying them. Hopefully soon, hopefully without way overeating.
- 3.6** Now, what was so interesting was that this insight, which, if you think about it, seems obvious, wasn't obvious even to the great Adam Smith. And here's why I say that. In "The Wealth of Nations," Adam Smith talked about what has become known as the diamond-water paradox. Here's what he said: "Water is so much more valuable than diamonds. We need it for our very survival. And yet, diamonds, per some weight, are priced at a huge multiple of water." He couldn't figure out why. What's wrong here?

- 3.7** Well, guess what? Menger and the other marginalists did figure it out. Water overall, the overall amount of water in the world, is way more valuable than the overall number of diamonds. But the marginal value, a little bit of water, is worth very little, because there's so much of it. The marginal diamond is worth a lot because there's so few of them. So actually, if you were to do a demand curve—I think I can draw a demand curve without freaking you out— here's the quantity of water and the quantity of diamonds. Here's the price. Here's the demand for diamonds. Here's the demand for water.
- 3.8** Why is the price you're willing to pay for that first unit of water so much higher than the first unit of diamonds? Because without that first unit, you die. But we got so much water, we're down here. So the value of a marginal unit of water is very low. Whereas diamonds are still expensive to produce, so we're probably here, with the supply curve there, and the price is high versus a very low price here. So overall, if you think of the overall value of water, it's very high. The overall value of diamonds is very low, but the marginal diamond is more valuable than the marginal unit of water.
- 3.9** So, they resolved that, and Menger was huge in that. So Menger is also thought of as the founder of the Austrian School of Economics. He refuted the labor theory of value because—well, I'll say more in a minute. And he used subjective value to show that both sides gain from exchange.
- 3.10** He also showed the value of middlemen. I live in Monterey. I live 20 minutes from Castroville, where they grow artichokes. When my daughter was about four, we were driving by and she asked my wife, "Is this where the artichokes live?" We buy artichokes. Do we drive even that 20 miles to buy artichokes? No, we don't. We go to Safeway. They are the middleman. And you use middlemen, middlewomen, whatever you want to call them, all the time. And he pointed out the value of that.
- 3.11** He also, as I said, refuted the labor theory of value. Capital and labor are used to produce goods. What, he said, if capital and labor are expensive but don't produce much value—right there, they're expensive, but they're not producing value, so there's something wrong with either a labor or a capital theory of value. One of his examples was using capital to store up drinking water in regions abounding in freshwater springs. You build up big reservoirs of drinking water where you've already got springs providing it essentially for free.
- 3.12** And I want to get slightly political here and give my modern example, which is the high-speed rail in California, which I think, by the way, if it ever gets built, will be called the medium-speed rail. It's used up lots of capital and lots of labor, like tens of billions. What's its value so far? Zero. There are no rides yet.
- 3.13** Even very smart people can get hung up on this, and that's why I've titled this next slide, "Are you smarter than a Harvard philosopher?" Because if you aren't, you soon will be, at least on this issue. This Harvard philosopher, I saw him speak up at Stanford a

couple of years ago, a very smart guy, but here's what he said in one of his books. Well, let me give a background. He said this. In 1965—he grew up in Minnesota—he could go to a Minnesota Twins game and pay \$3 for one of the best seats. The Twins' major star at the time was a guy named Harmon Killebrew, who earned \$120,000 per year.

- 3.14** Here's what he wrote in a book in 2012: "The business of baseball has changed a lot since then. The current star player for the Minnesota Twins, Joe Mauer, recently [in 2010] signed an eight-year contract for \$184 million. At \$23 million a year, Mauer makes more per game than Killebrew made in an entire season. Not surprisingly, ticket prices have soared. A box seat at a Twins game is now \$72."
- 3.15** So are you smarter—I'm not saying on philosophy, but on economics—than Michael Sandel? He doesn't quite come out and say it, but it seems, I read the page a few times, it seems that he's thinking that higher player salaries led to higher ticket prices. Is that what happened? A hint here: don't get sidetracked by inflation. Yes, there was a lot of inflation between the 1960s and 2015. But ticket prices and player salaries adjusted for inflation are much higher now. So that's not it. What is it?
- 3.16** Let's say those players were getting that huge money that Joe Mauer got, but people didn't value going to the games much. That wouldn't last long. What's driving the salaries of Joe Mauer and the other players is demand, demand to see them play. It's not that the labor cost is driving value; it's that value is driving players' salaries. So if you remember that, and you end up in a room with Michael Sandel, you've got something, you've got something on him.
- 3.17** Now, Böhm-Bawerk also was very big on refuting the labor theory of value. His example was wine. Wine grows more valuable with age, even with no added labor. It's valuable as it ages. Now, I think there are even easier examples, and when I went through their work, I couldn't find them using this, even though I had a vague recall they'd used this, but I couldn't find it.
- 3.18** But here's mine: two people are out trying to find gold. One pans for it all day in a river in California. At the end of the day, has enough little bits that it adds up to one ounce of gold, which is \$5,000, but he spent 8 hours doing it. Another guy, on his way to do that, trips over a stone, picks it up, it's a one-ounce piece of gold. What's the amount of labor that went into that stone? Nothing, essentially. So labor doesn't drive value. Value is not based on labor.
- 3.19** And by the way, what that showed was that the whole Marx edifice came crumbling down. And I have a friend and economist colleague who, with another friend and colleague, did a paper in the *Journal of Political Economy*, one of the top journals in economics, recently, showing that the economics profession just—Marx was almost nothing in the economics profession until the Russian Revolution. That's what drove Marxian economics: a government funding a huge amount of study of Marx. So this was well understood after 1871.

- 3.20** Now, Böhm-Bawerk also talked about what he called the "roundaboutness and productivity of capital." He argued that, when you introduce a lot of capital, you'll make the process of production more roundabout. It takes more time, but capital also makes it more productive. And that was part of his theory about why interest rates exist.
- 3.21** So Böhm-Bawerk had a theory of interest, why interest rates are positive. There are two factors. One is what's called time preference. I want it now. You want it now. And because I want it now, I'm willing to pay a premium to get it now rather than later. And because you want it now, you charge me a premium for me to get it now rather than later. So that's why interest rates are positive. Even in a world with zero inflation, interest rates would be positive. And then another driver is that production is roundabout. It takes time.
- 3.22** Menger was also very important in understanding the origin of money. Think about how a barter economy would work. I have a cow. You have furniture. I want to sell my cow. You want to sell your furniture. Unfortunately, I don't want your furniture. That's not a deal. The fancy term economists use for that is that we don't have the double coincidence of wants. I want a, I have a cow that you don't want. You have furniture that I don't want. What can we use so that we don't have that problem? Money.
- 3.23** I sell my cow for money. I take that money, and I buy furniture, or I buy something else. You sell your furniture for money. You take that money. Maybe you buy a cow. Maybe you buy something else. Money is a way around that, and very primitive societies have had money, even before we had paper money, even before we had gold, although gold emerged because it was such a great way of storing value, but it didn't have to be gold.
- 3.24** Milton Friedman, whom we'll talk about later, talked about, I think it was the island of Yap, where they used stones as money, and the bigger the stone, the more value. So you would sell something valuable, and they'd move this great big stone from where it was to your place, and it worked.
- 3.25** In prisoner-of-war camps in the Second World War, there was this economist who wrote an article in a major journal. He was a prisoner of war in a German prison camp in World War II, and he talked about what emerged as the money was cigarettes. They'd get the Red Cross packages every month, and some of the people would hold onto cigarettes and use them to buy other things from other people in the prison camp.
- 3.26** Now Menger also has a section in his "Principles of Economics" about how prices get formed. I'm going to go through some numerical examples on this. If you see this big thick book, this was all these readings from Menger. I was the discussion leader some years ago at this conference on Menger's work, and I remember saying to the group, and I got a lot of agreement, that if I'd read this before teaching supply and demand, I would have taught this first before teaching supply and demand. It's a real basic input.
- 3.27** So Menger has this example: one seller selling one item and one potential buyer. So, A has 100 bushels of grain, and he would be willing to trade that. He'd be indifferent

between that 100 bushels of grain and 40 bottles of wine. B has 40 bottles of wine, and he would be willing—he'd be indifferent between 40 bottles of wine and 80 bushels of grain. Is there a potential trade there, and at what price?

- 3.28** The answer is yes, and here's why: B is willing to take anything more than 80 to give up his wine. He's indifferent between 40 and 80. So if you offer him 81 and he thinks that's the only game in town, he'll do it. A is willing to give up something less than 100 bushels for 41, let's say 99. So, there will be trade if these are the only games in town, and the price will be somewhere between 81 and 99. We can't tell where; we can't tell where that price will settle. What we can say is that it will be between these two boundaries. Where it settles will depend in part on negotiating skills. Who's better, say, at threatening to walk away? OK. So that's the first one.
- 3.29** For some reason, I don't know why, he shifts from wine to horses. So A has one horse, and he is indifferent; he'd be indifferent between having that horse and having 10 bushels of grain. B_1 doesn't have a horse. He has 80 bushels of grain, and he'd be indifferent between having those 80 bushels and having a horse. Is there a trade there? Yeah. He'd be willing to give up as much as 79 bushels of grain. He'd be willing to take as little as 11 bushels for one horse. So there could be a trade where the price is, say, between 11 and 79.
- 3.30** Then Menger complicates it with competition on the buyer's side. So we've still got one guy with a horse, but now there's person B_2 , and he has 30 bushels of grain, and he'd be willing to give up to 30 bushels of grain for one horse. Well, let's see. A looks at this situation and says, "Hmm, you'd be willing to give up as much as 79, B_1 . You'd be willing to give up up to 29, B_2 ." Who's he going to say no to? B_2 , because I can get more from B_1 . But notice what that does. That narrows the price range. And this is why this is important for a discussion of price formation, because now there's no way he would accept 11 because he could get 29. So, now the price will be somewhere between 30 and 79. And B_1 gets it.
- 3.31** And then, Menger says, let's consider B_3 . B_3 , his trade-off between a horse and wheat is one to 50. So he'd be willing to give up, say, 49. He has grain, he doesn't have a horse. He'd be willing to give up 49 for one horse. So, A looks at that and says, "Well, you'd be willing to give up 49, no way I'm taking less than 49. And in fact, no way I'm taking less than 50 because I got this guy willing to pay 80, so the price settles between 50 and 79."
- 3.32** So, notice that, with more buyers, more potential buyers of one good, the price range narrows. And that was his key insight there. And then he way complicates it, that I'm going to get really wrapped around the axle if I try to do it, where you have him with multiple units. But what it shows then is that with—I'll just say it in words—with multiple units, more horses, he will satisfy the person willing to pay the most, and then the person willing to pay the second most, and so on. Again, we're working our way down a demand curve. Another way of thinking about it, because we're talking about

subjective value and thinking on the margin, the Marginal Revolution, is that the marginal value to each person differs.

- 3.33** Here's what he said—and then he introduced multiple sellers, and I'd get really involved if I did that—"But under competition, where no single competitor has the power to regulate," that is, set the price, "by himself either the price or quantity of a good traded, each individual competitor desires even the smallest profit, and the exploitation of existing possibilities of making such profits is no longer neglected. Competition leads, therefore, to large-scale production with its tendency to make many small profits and with its high degree of economy, since the smaller the profit on each unit, the more dangerous becomes every uneconomic waste, and the brisker the competition, the less possible becomes an unthinking continuation of business according to old established methods."
- 3.34** There's a lot in there to unpack. Let me just do a little of it. When you had the one person and no competitor on the seller's side, the person just went for the highest price he could get. But a case I didn't show, because it was going to get complicated with numbers, but you can imagine it, when you have more sellers, the price is going to be lower because of competition, and you get more people satisfied. So you get even those marginal buyers somewhat satisfied.
- 3.35** The way I think of this, when I was reading this, I thought, you know what? This is Walmart. They have managed to squeeze out little bits of gain from selling a huge amount of product to a lot of people. And therefore, it actually means there's less waste than otherwise because of that. And that's why—and this gets back to the thing about competition being a hardy weed—people worry that, say, Amazon has a monopoly or Walmart has a monopoly. But what was our concern about a monopoly? Typically it's been, they charge high prices.
- 3.36** What's Walmart's strategy? Charge low prices. What's Amazon's strategy? Charge somewhat low prices and very quick delivery. And that's why I don't worry about them. I don't worry about, say, competition disappearing. They have to keep getting good at being good and getting better in order to keep their place. In fact, I would hazard a guess that, say, even five years from now, Walmart won't have as high a market share as it has now, and neither will Amazon. Why? Because there'll be other things that come along that are competitive with that.
- 3.37** There's one other thing I want to talk about in Menger, that he talks about. Before I do, I want to just make this distinction that Adam Smith made. Adam Smith distinguished between what he called use value and exchange value. The use value is the value to you in using it. The exchange value is what you can get from it.
- 3.38** Menger had this example of shoemakers and tailors, and I can find it very quickly and read it to you. It's kind of clever. He says, "On the other hand, goods become altered in many instances in such a way that their exchange value, which previously was the

economic form of value to the economizing individuals possessing them, recedes as compared with their use value." And here's this example. "Thus, innkeepers and grocers usually employ foods having some external defect for their own consumption, since the defect in these goods causes them to lose their exchange value almost completely, while their use value often remains the same, or is at any rate not diminished to the same extent as their exchange value. The same phenomenon can be observed in other trades. Shoemakers, especially in smaller villages, often wear badly fitting shoes. Tailors often wear imperfectly cut clothes, and hatters often wear hats in whose production some slight accident has occurred."

- 3.39 So, you kind of use the bad stuff for yourself. There's that famous line that "the shoemaker's kids go barefoot," right, which is the extreme. But I think that's a great insight about that, once you start thinking about marginal value, you get that. That's it for the first set of Austrians. The next time we're going to talk about Ludwig von Mises and Friedrich Hayek. Thank you.
- 3.40 So, going back to before the Marginal Revolution in 1871, and there is the labor theory of value. If we applied those today, how much would a mother make?
- 3.41 How about, let's not apply the labor theory of value because it's wrong. Let's just ask: how much does a mother make, and then maybe we can go back to it. A mother gets two main things if she's a full-time mom. She gets the pleasure of the kids, and she gets to get lots of food and a place to live and so on, without paying for it. So there's an implicit exchange being made.
- 3.42 And by the way, one of the things that was huge in women entering the labor force, which really got going after World War II, was time-saving devices in the home. My mother, who didn't go full-time in the workforce, had this old washing machine with this little wringer and she had to run the things through it. And she kept telling me when I was young, "Don't put your arm there," and whatever. That took a lot of time to do that wash. And then—actually, we never got there; she died in 1969. But if she'd lived another few years, she would have had a washing machine and a dryer and so on. Dishwashing machines saved huge time, ovens being better and so on. So that was a major factor in women going out into the workforce and making serious money. I don't know if that exactly answers you.
- 3.43 There was one other thing I was going to say about that. There's this historian named Robert Caro who wrote a series of books on LBJ. He has essentially spent, let me do the math, 45 years now, writing books on LBJ. He is the most thorough historian. He's accounting for, in one part, like, why was LBJ so popular with his constituents when he was a member of Congress from the Hill Country in rural Texas?
- 3.44 Here's why that mattered. He has an eight-page segment in there. The subtitle in this book on LBJ—one of his books on LBJ—is titled "Sad Irons," I-R-O-N-S. And it talked about women, say, living on a hill. They had no electricity, and it's 90-plus degrees out

in the summer, and it's Monday, and Monday's wash day, and they go down to the well and pull up the water, multiple buckets of water. They have to heat it over a wood stove inside, and then they do the wash by hand, and then they have to heat the irons on the stove and use that to iron the shirts and so on. That's why it was called Sad Irons. People felt sad, women especially, when Monday came, and they knew they had to do this all day long in 90-degree weather. So when rural electrification came along, man, you're our hero.

- 3.45 And so anyway, the mother's work then, for a given amount of time—again, here's what's wrong with the labor theory of value— for way less time, she could do as good a job or better job of doing the laundry than she could when she had to haul all this water and do all this heating and take eight hours. So again, further evidence against the labor theory of value.

Lecture 4: Austrian School: Part 2

At a glance — The Austrian School, part 2.

- 4.1 Good morning. I had an interesting driver this morning. We got talking about talks. He's taking people who give talks, and I told him this joke I'd heard years ago that I still like, and this is before the era of the web. This guy drives around, he gives the same talk everywhere, and he's got the same driver, and they've been doing this for years.
- 4.2 So they're on the way to a gig. Remember, there's no web. They can't tell what he looks like in advance. And the driver says to him, "Sir, I've got a great idea. Let's change uniforms. You wear my driver's uniform. I'll wear your suit, and I'll go up and give your talk. I know it." The speaker goes, "Hey, that's a great idea." So, they do that, and the driver delivers the talk pitch-perfect. First question comes up, and the driver goes, to the questioner, "Sir, that question is so basic, I'm going to let my chauffeur answer it."
- 4.3 Before we get into the Austrians, the later Austrians, I want to talk about something that I thought about afterwards that I want to just clarify—the whole idea of thinking on the margin, and what is the margin. Think of it as the increment, the next unit, the additional thing, in other words, as opposed to total. That's the way to think of it. And remember, I mentioned, you can think through your taxes and think of, well, should I make that next dollar? You don't keep the average amount, the average rate of your taxes; you keep net of marginal. So if your tax bracket is 24%, you keep 76 cents of that dollar.
- 4.4 I have a Substack, and I posted one last night because I found this in a Wall Street Journal article last week. It's not about taxes, but it's about something very related. For people on Medicare, if they get Part B, which pays for doctors, their premium for Part B depends on their adjusted gross income. There was this great article in The Wall Street Journal last week that laid it out, where, what she had, the reporter had this table, and I think my sub-line after my title was, "Don't earn that extra dollar." Here's why.
- 4.5 If you were an adjusted gross income and you're filing jointly of \$209,000, and you make an extra dollar, your Medicare Part B premium goes up by \$81 per month, which is over \$970. So, on that next dollar, you pay your whatever it is, 24 cents, but you lose \$970. So your implicit tax rate is—let me do this—90,000% on that dollar. It's crazy. So the margin can really matter in some situations.
- 4.6 Now, on to Lecture 4 on the Austrians, Ludwig von Mises, Friedrich Hayek, and a later Austrian who's still alive named Israel Kirzner. First of all, Ludwig von Mises. He wrote a book in 1912—if you do the math, he was born in 1881, you see he was 31; that's pretty young—called "The Theory of Money and Credit." When I was at UCLA in the early '70s, we weren't getting a lot of the Austrians covered. So, some of the students formed this group to go through his book, line by line. And it was just amazing what this

guy knew back in 1912. And it was used as a major textbook in Europe, a money and banking textbook, for the next two decades.

- 4.7 In it, Mises expanded Austrian marginal utility theory to money, pointing out that money is not demanded for its own sake; it's demanded for its usefulness in purchasing other goods. In that same book, Mises also argued that business cycles are caused by the uncontrolled expansion of bank credit. One of the big debates he was in, and he was the guy who was really early on this— if he had been paid attention to, we might not have had socialism— and it was on economic calculation under socialism.
- 4.8 He wrote a piece called "Economic Calculation in the Socialist Commonwealth" in 1920. He argued that under socialism, that is, central planning, economic calculation cannot be rational. And he gave an example of a railroad. Assume the central planners have decided that a railroad should be built. But because you have socialism, because you have government ownership and no markets, you don't have prices. So they don't have prices to guide them about where to build the railroad. They don't know. Moreover, they don't have prices to guide them about how to build a railroad, what materials should they use, what should be the amounts of capital and labor, all those kinds of things. That was the problem he pointed out.
- 4.9 Here was his bottom line: "Thus in the socialist commonwealth, every economic change becomes an undertaking whose success can neither be appraised in advance or even later retrospectively determined. There is only groping in the dark. Socialism is the abolition of rational economy."
- 4.10 In the late '40s, he published what's thought of as his magnum opus: "Human Action." I read it through, cover to cover—it's close to a thousand pages—when I was 19, and I can't say I understood the whole thing, but I understood 90% of it. It brought together all of his contributions from his earlier work. One of his major claims was that economic truths are derived from self-evident axioms and cannot be empirically tested.
- 4.11 Now, he failed to persuade many economists outside the Austrian School. Most economists, mainstream economists, and also just the people in most of the free market schools of thought I'm talking about, think you can test them, think you can say, "OK, here's your theory. What would it predict about the world? Let's see if the world matches the predictions." But he was very firm on the idea that they're self-evident.
- 4.12 He wrote a short book in 1947 called "Planned Chaos," and it was basically how economic planning leads to chaos. And one of the big parts of it is how one intervention causes a problem that then leads the government to intervene to solve that problem, and then that intervention causes another problem. He took the example of fixing the price of milk below the market level. You get a shortage. What can you do? Well, you could get rid of the price control, and then you'd get rid of the shortage. But what if the government says, "No, we want to keep the prices of the inputs lower, so

we will set the price of cows lower, and maybe the price of feed for the cows lower." Well, then that causes a shortage of cows. That causes a shortage of feed and so on.

4.13 Nixon and Ford and Carter had price controls on gasoline. That led us to consume more gasoline than we would have had. It caused us not to downscale to smaller cars than we would have at a higher price. So what did the federal government do? They dictated to car producers that they had to hit some average number of miles per gallon on their cars for a year. And of course, that caused, as I mentioned, there to be lower-mass, lighter cars, which led to more fatalities. I'm going to say more about that later.

4.14 I'll take another example. I'm from California, which is one of the most heavily regulated states in the country. Lots of regulations. A few years ago, the California government banned the use of single-use plastic bags. They said, "These are filling the garbage dumps, and they're flying around in the air," and so on. How would you adjust to that? Produce plastic bags that aren't single-use. Produce heavier plastic bags that can be reused.

4.15 Some of us, like me, who's Mr. Cheapskate, do that. Some of us do that, but lots of people throw them away. So, behold, about a year ago, the state government figures out that there's actually more plastic being wasted—fewer bags, but each one weighs a multiple of what the earlier ones weighed. And now—and it's so like what Mises talked about—they don't say, "Oh, let's get rid of these regulations altogether." Now they have banned any kind of plastic bag at, say, a restaurant or a supermarket and so on.

4.16 One of the big things about Mises, and this is in "Planned Chaos," he had some great lines that just lay out. He said, "In a market economy, the consumers are supreme." In other words, producers, even producers as big as Walmart and Amazon, have to pay attention to what consumers want. If they don't, they're dead. And then he said, "The market is a democracy in which every penny gives a right to vote." You're voting with your money.

4.17 "The richer man casts more ballots than the poorer fellow, but to be rich and to earn a higher income is, in the market economy, already the outcome of a previous election." After all, how did that person get rich? Well, either by inheritance or by producing something people wanted. I'm excluding theft. He did not defend theft or fraud.

4.18 And then he said—and he already, by the way, I'll explain why he uses this professor line in this quote—already, it's clear that in academia, there are a lot of people, not mainly in economics departments, but in other departments, knocking the free market. And here's what he has to say to them. "The consumers have made their choices and determined the income of the shoe manufacturer, the movie star, and the welder."

4.19 In other words, the reason the movie star is making so much money is that people want to see the movie. And the shoe manufacturer, if successful, is making a reasonable amount of money because people want the shoes. The welder, if the welder is a good welder, is making money because people want welds. And then he says, "Who is

Professor X to arrogate to himself the privilege of overthrowing their decision?" In other words, the market is the result of decisions, not just of producers, but also consumers.

- 4.20 Mises moved from—he was Jewish, and he got the heck out of Austria and moved to Switzerland, and then moved over here. And so, he became fairly familiar with our laws in the 1940s. And one of the things he talked about was our laws regarding labor unions. He said, "In most non-socialist countries, the labor unions are granted special rights. They are permitted to prevent non-members from working. They are allowed to call a strike and, when on strike, are virtually free to employ violence against all those who are prepared to continue working, that is, the strike-breakers." You'll often have police say, "No, we don't want to interfere when the strikers are going after the strike-breakers, because that would be taking sides." No, it isn't; it's just enforcing people's rights, law and order.
- 4.21 One last note on Mises: Mises was also a strong proponent of laissez-faire. Laissez-faire means keep the government out; let people do what they want to do peacefully; the government's there maybe to enforce property rights, enforce contracts, a few other things, and that's about it. He advocated that the government not intervene anywhere in the economy. Interestingly, though, even Mises made some striking exceptions to this view.
- 4.22 For example, he believed that military conscription could be justified in wartime. I went through his section of "Human Action" where he talked about that before, when preparing this talk. I've got to tell you, I could get into it at length, but I've got better things to do. I just found his argument fairly unpersuasive. We're going to see more about the argument on conscription when I get to the Chicago School.
- 4.23 The next person I want to talk about is Friedrich Hayek. I showed you that picture of me when I had different-colored hair and different-length hair, getting his autograph back in 1975. Hayek was a student of von Mises, and the way I like to put it, with this series of articles, but especially one article, he drove the final intellectual nail into the coffin of socialism. So, Mises had done a pretty good job. Hayek amped it up by talking explicitly about the role of information. And he did that in his famous article, "The Use of Knowledge in Society," which I'm going to get into at length. He also wrote a book in 1944 called "The Road to Serfdom," and I'm going to talk about that. And he wrote a book in around 1960, called "The Constitution of Liberty."
- 4.24 Another guy, he's very famous, Daniel Yergin, who wrote a book called "Commanding Heights," called Hayek the "preeminent economist of the last half of the 20th century." That's high praise indeed. Some people say it's Milton Friedman. That's a contest I can handle. I think they're one and two, and I'm not sure who's one and who's two.
- 4.25 One of the points that Hayek talked about at length, because it was so important, is that the market is a spontaneous order. When I was in my teens and early 20s, and I

read this, I kept not getting it. Spontaneous, to me, meant fast, and that's not what he meant. He means it comes about organically. It doesn't mean it comes about fast, but it comes about organically, without any government planning it. So he means unplanned. The market was not designed by anyone, but evolved slowly as a result of human actions.

- 4.26** Hayek also did early work on business cycles. The market, of course, doesn't work perfectly. Nothing does. What causes the market, asked Hayek, to fail to coordinate people's plans, so that at times large numbers of people are unemployed? One cause, he said, was increases in the money supply by the central bank. Such increases, he argued, in a book called "Prices and Production," would drive down interest rates, making credit artificially cheap. Businessmen would then make capital investments that they would not have made, had they understood that they were getting a distorted price signal from the credit market.
- 4.27** An interest rate is a price. But capital investments are not homogeneous. Long-term investments are more sensitive to interest rates than short-term ones, just as long-term bonds are more interest-sensitive than short-term treasury bills. Therefore, he concluded, artificially low interest rates not only cause investment to be artificially high, but also cause "malinvestment," too much investment in long-term projects relative to short-term ones, and the boom then turns into a bust. Hayek saw the bust as a healthy and necessary readjustment, but it's ideal not to have them. And so, the way to avoid busts is to avoid the booms that cause them.
- 4.28** Now, this is controversial within the economics profession. The reasoning makes sense to me, but as I mentioned a few minutes ago, one way to test a theory is to look at its implications. So I was at a conference about 10 years ago where a relatively well-known modern Austrian named Roger Garrison was speaking. And I asked Roger, "Can you point to the kinds of industries that will expand artificially, due to artificially low interest rates?" That way, we could have a test of his business cycle theory. And his answer was no. So I don't know what to do with it. I would love for modern Austrians to really grapple with that, and we may get somewhere.
- 4.29** Now I want to turn to Hayek's article, "The Use of Knowledge in Society." He wrote it in 1945 in the American Economic Review, which at the time was the leading US economics journal—and still, arguably, the leading one. To put that in perspective, this article would never be accepted by that journal today. There are no equations. There are no data sets. There's no econometrics. It's just tight reasoning with colorful examples.
- 4.30** So, he made this famous statement—and I'm going to tell you why I'm using the term local knowledge—he said, "The peculiar character of the problem of a rational economic order is determined precisely by the fact that the knowledge of the circumstances of which we must make use never exists in concentrated or integrated

form, but solely as the dispersed bits of incomplete and frequently contradictory knowledge which all the separate individuals possess."

- 4.31 In other words, each of you has knowledge in your head about what you want as a consumer, about what you can do as a producer. Other people don't have it; they don't have, in their heads or their computers, your knowledge. And even the most powerful computers in the world, even the best AI, can change that a little on the margin, but can't really do much about it.
- 4.32 I talked earlier about tacit knowledge. A lot of the knowledge we have is tacit. We couldn't even express it to you, but we have it. One way I think this shows up is that sometimes I'll go to make a call. Someone will ask me a phone number of someone, and I'll think about it, and I can't think of it. And I'll pull out my phone, and I start pushing the buttons, and I have it. It's amazing. It was this tacit knowledge that my fingers knew or something like that. By the way, I do want to challenge him a little. Knowledge can't be contradictory. In other words, if you think A, and I think not A, one of us is wrong.
- 4.33 And let me just explain local knowledge. The term that economists use now in the Hayekian tradition is local knowledge. We have knowledge about the things around us that no central planning agency could have. I'm not a golfer, but I sometimes watch golf. I remember lefty Phil Mickelson was on this course. He played a lot, and he shot this beautiful shot, and the microphone was near him, and I heard him yell out, "Local knowledge." And so, I go online right away. Was Phil a major in economics? And the answer is no. But he got the term right. It was local knowledge of that golf course.
- 4.34 He says, "The economic problem of society is thus not merely a problem of how to allocate given resources— if 'given' is taken to mean given to a single mind which deliberately solves the problem set by these 'data.' It is rather a problem of how to secure the best use of resources known to any of the members of society, for ends whose relative importance only these individuals know. Or, to put it briefly, it is a problem of the utilization of knowledge which is not given to anyone in its totality."
- 4.35 So think about your local knowledge. Think about your job, the things you've picked up all along the way in your job. Think about, as I mentioned earlier, your knowledge about what you want to buy, where sometimes you don't even know until you get to the store. Now, think about how that would work if you had to get a planner's permission to do anything.
- 4.36 By the way, I said I live in California. I live in Pacific Grove, which is right next door to Carmel, and those two cities have very similar regulations. We see this with building a house in California. We have a weekly called the Carmel Pine Cone, and every week there's a story about someone wanting this color roof or this type of roof or this garage. And there's this huge uproar. Sometimes they get permission, and often they don't. They're not allowed to act on their local knowledge.

- 4.37 I had one—well, I'll get to it when I talk about something else. But I'll just mention, I had one a few years ago, where I wanted to cut a branch off a tree that was threatening our house, and I had to figure out whether the branch was under 4 inches or over 4 inches. If it was over 4, I had to get a permit, and I might not get it. You know what I did? I went to a hardware store in a different town to buy one of those saws, so that the hardware store in our town wouldn't remember me. And on Labor Day morning at 6 AM, I went out and sawed the thing off, and this cop comes by, and he waves. And I measured, and it was three and three-quarters. But this is crazy. This is crazy, having to live this way.
- 4.38 So again, Hayek is talking about how the world works, how local knowledge works, but he's also saying, implicitly and sometimes explicitly, we should be allowed to act on our local knowledge. And then he says, "We need to remember only how much we have to learn in any occupation after we have completed our theoretical training, how big a part of our working life we spend learning particular jobs, and how valuable an asset in all walks of life is knowledge of people, of local conditions, and of special circumstances."
- 4.39 I've taught this article to all my students for my last roughly 20 years of teaching. And by the way, they had trouble with his Germanic writing style, these long sentences. So I wrote out notes on each paragraph to give them, to walk them through, and then it started working very well. And here's a question I asked my military officer students. You graduate from the naval academy or you graduate from officer candidate school or ROTC, and you get into your first job. On a scale of 1 to 10, where 10 is you are it, 1 is you know nothing, what did you know in your first job? The typical answer I'd get back is 2 or 3. You learn it on the job. Local knowledge.
- 4.40 He gives some great examples. "To know of and to put to use a machine not fully employed, or somebody's skill which could be better utilized, or to be aware of a surplus stock which can be drawn upon during an interruption of supplies, is socially quite as useful as the knowledge of better alternative techniques." We tend to hold scientists up there, but it's also the very practical people who really matter a lot.
- 4.41 "And the shipper who earns his living from using otherwise empty or half-filled journeys of tramp-steamers, or the estate agent," he's writing now in Britain, so he means real estate agent, "whose whole knowledge is almost exclusively one of temporary opportunities, or the arbitrageur," someone we're going to meet later in this hour, "who gains from local differences of commodity prices, all are performing eminently useful functions based on special knowledge of circumstances of the fleeting moment not known to others."
- 4.42 Now, he was criticizing economic planning in the sense of central planning, and he wanted to make sure people understood that he wasn't arguing against planning. So he said, "This is not a dispute about whether planning is to be done or not. It is a dispute as to whether planning is to be done centrally by one authority for the whole economic

system, or is to be divided among many individuals." So, more on what he called circumstances of time and place, and what we economists in the Hayekian tradition call local knowledge.

4.43 "It follows from this that central planning, based on statistical information, cannot by its nature take direct account of these circumstances of time and place, and that the central planner will have to find some way or other in which the decisions depending on them can be left to the 'man on the spot.'" So now we have an apparent dilemma. "We need decentralization because only thus can we ensure that the knowledge of the particular circumstances of time and place will be promptly used." By the way, this isn't me talking, this is Hayek talking.

4.44 "But the 'man on the spot' cannot decide solely on the basis of his limited but intimate knowledge of the facts of his immediate surroundings. There still remains the problem of communicating to him such further information as he needs to fit his decisions into the whole pattern of changes of the larger economic system." So, there's a dilemma here. It looks like it might be chaos. Each of us gets to act on the basis of his or her own knowledge. But how does that mesh? Isn't there just chaos?

4.45 So, I call this the price system to the rescue. Free markets to the rescue. He gives this example of tin. The price of tin rises. Is it due to a decrease in supply or an increase in demand? Let me show you a demand and supply curve. Quantity of tin per year. Price of tin. Let's start out with the demand for tin and the supply of tin. This price, P_0 , is the price at which the quantity of demand equals the quantity of supply. At that price, Q_0 will be the amount both sold and bought.

4.46 Now, the price of tin rises, so something must have changed. Either the demand went up because they found new uses, or the supply fell. Maybe there was a strike in Bolivia where they produced tin, and we don't know. So for example, if it's the demand rising, we get this price, P_1 , and more tin produced. If there's a strike that's causing the supply of tin to fall, we get S_1 , and with the original demand curve, this price. Both are above P_0 . So, the price of tin rises.

4.47 If there were a central planner who wanted to do a good job, what would that central planner want users of tin to do? Use less. But how would the central planner know: should we use less tin in tin pots, or should we use less tin in tin cans? The central planner wouldn't know that. But each user has his local knowledge. Each user says, "I can tell, if the price of tin rises, how to cut back on my use of tin." It does not matter for our purpose, and it is very significant that it does not matter which of these two causes has made tin more scarce. Because either way, whatever it is, I, as the tin-user, don't need to know that. I can adjust optimally in a way that no central planner could do, because he uses the word marvel.

4.48 He says, "The marvel is that in a case like that, of a scarcity of one raw material, without an order being issued, without more than perhaps a handful of people knowing

the cause, tens of thousands of people whose identity could not be ascertained by months of investigation are made to use the material or its products more sparingly; that is, they move in the right direction. This is enough of a marvel, even if in a constantly changing world, not all will hit it off so perfectly that their profit rates will always be maintained at the same constant or normal level."

- 4.49 And then he explains his use of the word marvel. "I have deliberately used the word 'marvel' to shock the reader out of the complacency with which we often take the working of this mechanism for granted. I am convinced that if it were the result," the price system that does this, "if it were the result of deliberate human design, and if the people guided by the price changes understood that their decisions have significance far beyond their immediate aim, this net mechanism would have been acclaimed as one of the greatest triumphs of the human mind." Or as I once put it when I was teaching this article, maybe the free market should have won the Nobel Prize in economics every year it was offered. So, that's Hayek's "Use of Knowledge in Society." I think of it as one of the 10 most important articles in the 20th century in economics.
- 4.50 Now, as I said, there's this Germanic kind of writing style. Hayek had a buddy named Leonard Read, who was the head of something called the Foundation for Economic Education. He always liked to boil things down in a way that made it more understandable. So he wrote a famous article in 1958 called "I, Pencil." And it's a little hokey, but I taught this to 30-year-old military officers who didn't think it was hokey. They loved it.
- 4.51 The idea is that the pencil is talking, and the pencil says, "No one knows how to make me." And that's a literally true statement. No one knows how to make me. It's the result of a complicated division of labor where the inputs into this little simple No. 2 Ticonderoga pencil come from about five or six different countries, and he names the countries and lays it out. And this whole price system just meshes these people's decisions. So it doesn't have all of what Hayek has, in his "Use of Knowledge in Society," but it's quite understandable.
- 4.52 There was a famous economic analyst named Howard Stern who illustrated this. I call this "Howard Stern on local knowledge." This was way back. He hosted Julie Cialini, '95 Playmate of the Year, and Stacy Sanches, '96 Playmate of the Year. And he asked them 10 questions. The first six were about things in current events and politics and so on and science, stuff like that. First three questions. "Who is the President of Russia?" Julie: "Gorbachev." Stacy: "Gretzky." Correct answer: "Boris Yeltsin." I mean, everyone knows Wayne Gretzky is the Prime Minister of Canada. Who doesn't know that?
- 4.53 "Define the meaning of NAACP." "Something, something for certified pianists. It's some police organization." Correct answer: "National Association for the Advancement of Colored People." Question: "Who is the inventor of the light bulb?" Julie: "I know Edison invented the telephone, but I can't remember the light bulb guy." Stacy: "I don't know." Correct answer: "Thomas Edison." Alexander Graham Bell was the phone guy.

- 4.54 Next three: "Who's the Speaker of the House?" Julie: "Gore, something or other."
Stacy: "Bill Clinton." Correct answer: "Newt Gingrich." Question: "Define the meaning of the letters CIA." Julie: "I don't know." Stacy: "Certified Investigation Association."
Correct answer: "Central Intelligence Agency." Question: "What is the center of our solar system?" Maybe a little hint in there, in the word solar. Julie: "The equator."
Stacy: "The moon." Correct answer: "The sun."
- 4.55 Then, Howard turns for his last four questions to questions that relate broadly to the industry they're in. "What did the initials D. K. stand for?" Both knew it was the fashion designer, Donna Karan. "What is Cristal?" Both knew it was an elite champagne. By the way, I didn't know that. I buy the cheap stuff. "What car company has a model known as a 911?" Both knew it was a Porsche. "Whose face is on the \$100 bill?" Both knew it was Ben Franklin.
- 4.56 What is this telling you? Here's a perfect example of two people having their local knowledge, knowing the kind of knowledge that matters to them. I've got to say that I've often wondered, when I look at our school system, I think our government school system is a massive failure. And yet, it hasn't resulted in as bad a result as I would have expected. And I think it has to do with local knowledge.
- 4.57 I don't know if you ever watched Jay Leno. He called it Jay Walking, where he'd go out and ask people questions about the world, and they wouldn't know much. "When was the Civil War?" "1700," that kind of thing. Well, it doesn't matter much to them. What matters is the knowledge they get in their particular job. I mean, I think it's sad that they don't know these basic things. But what's really neat is that we still have a very productive economy without their knowing that.
- 4.58 One of Hayek's other books that's very important in his contribution—and it's not just totally about economics, it's also about political systems—is a 1944 book called "The Road to Serfdom." A friend, a mentor, gave me my copy for Christmas when I was 18 years old, and I started reading through it, and I couldn't stand it because, in my view, he was too good-willed toward socialists. He dedicated it to "socialists of all parties," and he didn't do it ironically. He was trying to persuade them. And I guess I matured a lot because I then picked it up again at age 19, and I loved it.
- 4.59 His big point was that the roots of fascism and Nazism were starting to take hold in Great Britain. Because fascism and Nazism were all about—I mean, put the Holocaust aside, which is a huge thing to put aside—but they were all about the government making itself the decider on everything and people having virtually no power in their own lives. And he was seeing that happen with Great Britain, and this was his warning.
- 4.60 It was written the way Hayek writes, which was sometimes hard for people to access because of that Germanic writing style. The Reader's Digest digested it down to a very short pamphlet with pictures to make the point. That sold hundreds of thousands of copies, and people got it. And there were some reversals in a good way in Great Britain

after the war. There was this central planning of property usage, and they backed off a little. There was a proposal to have the federal government in Britain tell people what jobs they could have, literally say, "You can't do this, you've got to do that." They backed off that, big time, and so it had a big effect.

- 4.61 And here's what he said about government power. The government's power is "a power which is newly created and which, in a competitive society, nobody possesses." Hayek points out that the power a multimillionaire—there weren't billionaires at this time—"who may be my neighbor and perhaps my employer" has over him, "is very much less than that which the smallest fonctionnaire," smallest government bureaucrat, "possesses, who wields the coercive power of the state and whose discretion it depends whether and how I'm allowed to live or to work."
- 4.62 I gave my example of cutting tree branches. The city official in Pacific Grove had this power over me that this major company, Home Depot, didn't. That's where I went and bought that little saw. They have zero power over me. Amazon has zero power over me. Walmart has zero power over me. But a bureaucrat whose name I don't even know in a town of 15,000 people has more power than they have over me. And that's, fortunately, just about cutting tree branches.
- 4.63 But how about if they decide whether you can set up a company? As I mentioned in a previous lecture, governments often decide, "You can't be in this business. You can't have this job. You can't start this company." One example, in California, we have forest fires. You know who they often use to help with forest fires in an emergency? Prisoners. They let them out of prison for a while, and they fight fires. They seem to think they do it well. They go back into prison. They serve their term. They come out. Guess who's not allowed to be a firefighter? Someone with a prison record. That makes no sense.
- 4.64 So, more on government power over the economy. He says, "Whoever controls all economic activity controls the means for all our ends, and must therefore decide which are to be satisfied and which not." And I think this is so often forgotten about socialism. If the government has total power over the economy, it decides everything.
- 4.65 He also had a chapter, and it was titled, "Why the Worst Get on Top." Why do we have some of the worst people in our country be in top positions in politics? He lays out why. He says, basically, they're willing to do fairly extreme things to keep their power, and they're willing to be demagogues and so on to get people spun up in favor of their ideas.
- 4.66 And he quotes the late University of Chicago economist Frank Knight's memorable statement, and I'm going to talk about Knight soon—his memorable statement that "the probability of the people in power being individuals who would dislike the possession and exercise of power is on a level with the probability that an extremely tender-hearted person would get the job of whipping-master in a slave plantation."

There's what's called self-selection. Who wants to be on top? Who gets to be on top? People who are pretty ruthless.

- 4.67 Now, there was, I should say an economic journalist named Henry Hazlitt, but he was almost an economist in his own right. By the way, if I had to recommend one book for people who don't know economics, it's a book written by Henry Hazlitt in 1946. It's still relevant. You read the chapters and you feel like he wrote about something I read last week, and it's called "Economics in One Lesson." So Hazlitt was friends with both Mises and Hayek. Hayek moved to the United States. Mises had moved to the United States.
- 4.68 And I remember hearing this story. I met Hazlitt once; I didn't hear it from him, but I heard it from someone who knew him, that he was asked the difference in temperament between Mises and Hayek. And here's how Hazlitt answered. "You ask both of them: 'What color is that cab?'" This is in New York. "'What color is that cab?' And Hayek would say, 'Well, it's yellow, of course.' And Mises would say, 'Any fool who cannot see that that cab is yellow!'" So it's just like a difference in temperament that the two had. And I think that's part of why Hayek got more recognition, even though Ludwig von Mises was fully as good.
- 4.69 And finally, I want to talk about a modern Austrian. He's probably 90 by now, but he's still alive, named Israel Kirzner. And he had a few—his major thing he brought to the table was on entrepreneurship. I remember going through, just randomly, seven textbooks, seven economics, the introductory to economics textbooks that various publishers had sent me because I, as someone asked to use a textbook, if I choose that for 40 people, they make a lot of money.
- 4.70 I went through, and I looked in the index for the term "entrepreneurship," and in many cases, I didn't even find it. So I paged through, and sometimes they'd talk about it, sometimes they wouldn't. But it just was, at best, shoehorned in there in a paragraph. Why? Well, because in the standard textbook, the ideal kind of competition is what economists call perfect competition. What is perfect competition? Everyone produces the same thing. Everyone sells it for the same price. There's no advertising, there's no innovation, and I'm sounding like Ben Stein. How's that great?
- 4.71 Well, Kirzner said, "They've left out the role of the entrepreneur." And the key in entrepreneurship is what he called alertness, alertness to opportunities. And one of the big ones he talks about is arbitrage. An entrepreneur sees that the good is selling for \$10 in this market, and over in this market it's selling for \$15, and he calculates the cost of shipping it, and it's \$2. So he buys it for \$10, adds \$2 to ship, sells it for \$15, and makes \$3. It's that kind of thing that actually makes competition work, because other entrepreneurs see the same thing, and they go and buy it. And now maybe they bid up the price to \$11 here, and they bid up the price to \$14 here, and so on. So entrepreneurship, especially arbitrage, is a very big part of the competitive story.

- 4.72 He also emphasized the idea of competition as a process, not the kind of end state you get with this idea of perfect competition. And he talked about two main aspects of that — the role of advertising and the alleged waste from competition. The alert entrepreneur must also alert potential buyers to the presence and, ideally, the attractiveness of the items he's selling. In other words, it's not a given that people know what's out there. Advertising tells them what's out there.
- 4.73 Doing so uses resources, but those resources are not wasted. Kirzner writes that, "Selling effort, including advertising, that alters the opportunities perceived by consumers, constitutes an entirely normal avenue of competitive-entrepreneurial activity." There's also this claim out there that competition is wasteful because people make mistakes.
- 4.74 Here's what he says about that: "The truth is that until the newly competing entrepreneur has tested his hunch about the lowest cost at which he can produce, we simply do not know what organization of industry is best. To describe the competitive process as wasteful because it corrects mistakes only after they occur, seems similar to ascribing the ailment to the medicine which heals it, or even to blaming the diagnostic procedure for the disease it identifies."
- 4.75 In other words, mistakes are part of the deal. Mistakes are part of the process in finding out what works and what doesn't. So that, in a nutshell, is the modern Austrians, modern in the sense they did everything in the 20th century, starting with Mises, then Hayek, and Israel Kirzner into the 21st century.
- 4.76 Is there people that have more of a tacit knowledge versus a local knowledge, because that's the way their brains thinks? Like I said, in patterns or rhythmic understandings of things?
- 4.77 Here's how I'll reframe it and answer that, I think, this way. Are there degrees of tacit knowledge? I think there are. By the way, though, tacit knowledge is local knowledge. It's just part of—so I know that I want a ride back to my hotel at the end of today. That's not tacit knowledge, but it's local knowledge. I don't know what I'll have for lunch, but when I see the menu, I will know. There's kind of tacit knowledge. I might have been able to tell you, but I didn't and couldn't maybe, until I saw it. That's what I would do.
- 4.78 I don't have a question, but I want to make a comment about what you were speaking on, Howard Stern. It's like, people of today, everybody's a genius when it comes to politics. But then when you ask them, "Recite the preamble to the Constitution," and the majority of them would be looking at you like, "Huh? What is that?" Or the rest of them don't know what it is. Or you ask them, "Since you know so much, then what's the three branches of the government?" They don't have a clue. They might name one. "Oh, the legislature."

4.79 They're struggling, standing there, two, three minutes trying to think of the other two. And it just amazed me when you said that the people know, but they don't know. There's certain things they can pick up and say, "I know this," like the basic stuff that you were explaining the ladies got all right. But they speak on stuff that they really don't even study on, but they act like they're experts at it.

4.80 Yes. So I take it that your claim that everyone's an expert on politics was ironic. Yes, that's right. They think they are, but they're not. And by the way, when we get into the school of thought on what's called public choice, we'll talk more about why people are very ignorant of what's going on politically and how it's rational for them to be ignorant.

Lecture 5: Chicago School

At a glance — The Chicago School.

- 5.1 The next free market school of thought I want to talk about is the Chicago School, named after, you might expect, the University of Chicago. There were a lot of major players. I could have had double this list, but I want to focus on what I regard as the most important and interesting ones. The first one was an economist named Frank Knight, who had a long career from the early 1920s to the early 1970s.
- 5.2 By the way, I went down to the University of Chicago when I was 19 and knocked on Milton Friedman's door, and he talked to me for 10 minutes. And then I went to Frank Knight's door, and I heard him kind of yelling at someone on the phone through his door. I'm not the kind of person who backs down. And I didn't knock. And I still regret that because, a year later, he died. I shouldn't say, "He died, laugh, laugh," but anyway. So I never got to talk to him.
- 5.3 Milton Friedman—I'm going to talk a lot about him—George Stigler, Gary Becker, Ronald Coase, and Sam Peltzman. Let me start with Frank Knight. He wrote a major book in 1921 called "Risk, Uncertainty, and Profit." One of the big things he laid out is that, when you see high profits in a firm, sometimes it has nothing to do with a monopoly. It has to do with risk. It's a particularly risky enterprise. It could fail easily. There could be a decline in demand. There could be foreign competition, whatever. Meanwhile, the people in that business have to make high profits to compensate them for taking that risk.
- 5.4 He was very clear on what risk meant. He was very specific. Risk, in his view, meant that you could assign an actual probability to things. So it was a 0.5 probability this would happen, a 0.4 probability that would happen. As opposed to what? As opposed to uncertainty. Uncertainty means you know it's not a sure thing, but you don't really have a clue about the probabilities.
- 5.5 I remember this study came out in a British journal, whose name escapes me, claiming that various vaccines led to autism. The Lancet was the name of the journal. And apparently, some people refuted it, said, "This is bad data," this and that. And so there was this discussion on Facebook I got into, where people were saying that those data have been refuted, and I wasn't challenging that. But they were saying, "Therefore, vaccines don't cause autism." And I said, "Maybe they don't, but I don't know." And this philosopher said to me, "Well, you're an economist. You put probabilities on things. What's the probability it causes?" I said, "I don't have a clue." "Wait a minute, you're an economist." "I don't have a clue." I said, "It's uncertainty." He didn't get it.
- 5.6 Frank Knight educated a generation of economists. So his writing was important, but probably more important was who his students were. In his class, one particular year in the 1930s, his PhD class, were Milton Friedman—who I'm going to talk about—George

Stigler, and another economist who was a friend of mine but didn't make as big a splash. But that was in one class. And by the way, Milton Friedman's wife, Rose Director, and that was her maiden name, was Director, and because Frank Knight organized them alphabetically, Milton sat beside Rose and ultimately married her.

- 5.7** Frank Knight had this kind of—he was very controversial in things he wrote, and he liked to make extreme statements. And so he wrote this article, which I had a copy of in my library—it was just a typed-out copy, it wasn't in a journal or anything like that—titled "The Case for Communism: From the Standpoint of an Ex-Liberal." Now, that title requires a lot of unpacking. Liberal to him meant classical liberal, which is what we today generally call libertarian. He wasn't making a case for communism at all. He was expressing his despair about the way the world would go.
- 5.8** Well, here's an interesting story. A student in my class—this was about 20 years ago—came to me at the end of the quarter and said, "Professor, my great-uncle was an economist at the University of Chicago." I said, "Really? What was his name?" He goes, "Frank Knight. But I didn't want to tell you because he's a bit of a black sheep in our family." I said, "Let me guess why." I rotated my chair, pulled his article out. "Was it this, 'The Case for Communism from the Standpoint of an Ex-Liberal'?" He goes, "Yeah, we thought he was a communist."
- 5.9** I go, "No, that was just his style. In fact, he educated Milton Friedman. He educated George Stigler. He educated a lot of people who had a substantial role in advising the Reagan administration. You might argue that, without Frank Knight, we don't get Reagan's 1981 tax cut." And this guy went away feeling so good about his great-uncle.
- 5.10** Frank Knight also was known—and I should have brought it as a prop—Frank Knight was known for, when he was working on an article, wearing this green poker hat, poker shade thing. He was very famous for it. In the mail one day, years later, shows up the green poker hat. He had retrieved it from his great-uncle's estate, so I still have it.
- 5.11** So, on to Milton Friedman. Here's a picture of me with Milton. Milton was shorter than me and bald. This was at a conference, actually, on Austrian economics in South Royalton, Vermont, in June 1975. I'm the one with the wonderful color combination of outfits, with my arms folded and my hair curly and long. That's Milton. I want to say a lot about Milton because, as far as the modern Austrian School, meaning the school of thought after, say, 1945, I regard him as the most important person for many reasons, both his contributions to economics and his contribution to public policy.
- 5.12** Milton Friedman revived what was called the quantity theory of money. I'll get into that in a minute. Let me tell you the background. John Maynard Keynes had revolutionized economics, the area of macroeconomics, the theory of the economy as a whole, things like inflation, unemployment, economic growth, from about the mid '30s to when his "General Theory" book was published, to the early 1970s. Milton Friedman harkened back to the monetarism that had dominated before Keynes and had lost some respect.

- 5.13** The quantity theory of money says, $MV = PY$, where M is the supply of money, V is the velocity of money, how quickly it turns over, P is the price level, usually measured by the consumer price index, and Y is real output. We call it real GDP, real gross domestic product. And that's not a controversial equation. It's actually definitional. This has to be true.
- 5.14** What do I mean then, when I say he revived the quantity theory of money? Well, in the Keynesian view, velocity moved passively, and you couldn't predict where it was going. In Friedman's view, velocity was more stable, and velocity is the inverse of the demand for money, basically. It's more stable. So if it's more stable, then you can say things about what happens when M goes up, when the money supply goes up, and if V is roughly constant, well, then this must go up. And then it's separated into price going up, price level going up, which is inflation, or real GDP going up, which is economic growth. That was what Milton spent a lot of his career on in the '50s and '60s.
- 5.15** So he wrote a book with a fellow economist named Anna J. Schwartz called "A Monetary History of the United States, 1867 to 1960." It's about a 600-page book, and they just track the money supply over that rough century and the various things that were happening, and correlating it with what was happening to inflation, what was happening to economic growth, and so on.
- 5.16** And the best chapter, the most interesting, exciting chapter, was the one that I reread a couple of times when I was studying for my area exam in monetary theory, was the chapter on the Great Depression. Because if you know some of the history of that, a lot of people attribute the Great Depression to the stock market crash in 1929. And they show that, no, that was not a major factor at all. The major factor was that the money supply fell by 30%. Well, if money supply falls by 30% and prices don't immediately fall by much, real output has to fall. So, they don't have the last word on the Great Depression. There's been more work done since then by an economist named George Selgin, whose book I reviewed that's really excellent. But nevertheless, they have a large part of the story of the Great Depression.
- 5.17** And then you have to bring in things they didn't talk about, but I talked about in one of my first two lectures, which is that both Hoover and Roosevelt, Hoover with persuasion, Roosevelt with legislation, tried to keep P high and wages high. Well, if you do that, you prevent them from falling, and you're going to get more of a reduction in real output. So that was a huge contribution.
- 5.18** The other thing Friedman is really well known for in this area is inflation. He made this famous statement, "Inflation is always and everywhere a monetary phenomenon." Why? Well, let's first of all explain, look at the way you can make that tautological. Because inflation necessarily means a drop in the value of money, by definition, that makes it a monetary phenomenon.

- 5.19 But he meant more than that. He meant, you will not see inflation anywhere that's sustained, that goes on for a few years, without going back and seeing that the money supply went up. So, you see that this goes up, say, 10%, so we have 10% inflation in a year. The odds are almost 100% that it's because M went up. Again, that's kind of common sense now, but he made it common sense. The Keynesians really hadn't talked much about inflation at all.
- 5.20 By the way, one of the things he did, and others in the Friedman tradition have done, is look across countries and look at what happens when you have very high inflation, and there it's really obvious. So there was the famous German hyperinflation of 1921 and '23, when prices were doubling every week, and sometimes over a few days. What was happening to the money supply? It was going up roughly comparably. So it wasn't a surprise.
- 5.21 There was a famous story—well, here's a true story, and then there's maybe the apocryphal story. The true story is that there were so many Reichsmarks out there, the German money supply, that people carried them around in a barrel to buy a few things—sorry, a wheelbarrow. Now, the apocryphal story is, a guy carries his money to a store, goes in to buy something, leaves the barrel out there with the money, comes out, the money is on the ground, the barrel has been stolen. That's how low-value the money was. That's apocryphal. But if you look at the data, it's quite believable.
- 5.22 And then there have been inflations since then that are extreme, like Zimbabwe, again. They even have, someone gave one to me, a trillion-dollar bill, and I'm not wealthy because of it. It's worth a penny or so. And so he nailed that one, and it shows up most easily in the hyperinflations.
- 5.23 The other thing that came along—and this was years after he predicted it, which is what made him so credible—was stagflation. That was the term people came up with in the early '70s: stagnation combined with inflation. Stagnation meaning either negative growth, GDP falls, or low growth, the rate of growth of GDP falls, combined with inflation. And he argued, "We can have high inflation and slow growth, even recession." The Keynesians had, in their model, no way for this to be true.
- 5.24 So he made his presidential address to the American Economic Association in 1967, saying that, and in the early '70s, we got that. We got the major recession of 1973 to '75, when we had high inflation and actually negative growth. The seat-of-the-pants definition of recession, by the way, it's not exact, but it's very close, is two quarters in a row of negative economic growth, so six months of negative economic growth. So Friedman explains something before the fact, which is always a way to be credible. And Friedman himself will say, "It wasn't that I was so persuasive; it was the data were so persuasive."
- 5.25 So back to this thing about inflation—he put together a series on PBS in 1980 called "Free to Choose," and they made a book based on the series, he and his wife, Rose

Friedman. She was an economist in her own right, by the way, but she had kids and never finished her PhD, but she rewrote much of what he wrote. I think she was a better writer.

- 5.26** There was this dramatic moment in the "Free to Choose" series. He's at the Bureau of—it's episode 9 out of 10 on inflation—and he's at the Bureau of Engraving and Printing, where the things are churning out money. And he has this dramatic moment. He says, "So all we have to do to stop inflation is stop printing money." And he hammers this little red button and then presses stop.
- 5.27** And so, when I was preparing this lecture last week, I thought, I'd heard that someone asked him, "How did you get permission to do that?" With a twinkle in his eye, he said, "I didn't." But then I thought, "Well, I'm going to check ChatGPT," and I'm one of those people who pays \$20 a month for the better version. And it says, "Well, the story is apocryphal. We can't back it up." So I don't know. So that's Milton Friedman on inflation, recession, stagflation. He really did revolutionize macroeconomics, monetary economics.
- 5.28** There's something else he did, though, that in a way—certainly for a certain group of people, namely healthy young men—had a bigger effect. And that was his role in helping end the draft. There was a University of Chicago conference in 1966, where all the major players on both sides of the issue of the draft were there. There was Senator Edward Kennedy, who was a major supporter of the draft; Margaret Mead, the anthropologist who supported the draft; Milton Friedman, who was an opponent of the draft; a young congressman from Illinois who later became Secretary of Defense in the Bush administration—oh, my goodness, his name's escaping me. Anyway, he was there, and he was a congressman who was against the draft. And there was a Democratic congressman from Wisconsin named Robert Kastenmeier, who was against the draft.
- 5.29** Here's what Friedman said about that conference. "I have attended many conferences. I have never attended any other that had so dramatic effect on the participants. A straw poll taken at the outset of the conference recorded two-thirds of the participants in favor of the draft; a similar poll at the end, two-thirds opposed." In other words, there was a shift of one-third over a four-day conference. "I believe that this conference was the key event that started the ball rolling decisively toward ending the draft."
- 5.30** Now, what economists had—so, Milton Friedman was basically in favor of freedom. He said people should be able to be free to choose their own occupation and not be drafted into a business, into a place where they might actually be killed. But economists also had an economic argument. And by the way, I was a math major, and I was up in Canada, where I wasn't subject to the draft, but I hated the idea of the draft, and I was reading the American literature on it. And that's where I came across this economic argument.

- 5.31 And it was one of the things that made me, on the margin, think maybe I might want to be an economist, because they came up with this clever and correct argument that I'd never heard outside of economics, and the economic argument was: for a given-size military, the draft is more expensive. Why is that? How come? When you have a draft, the military can pay less because they don't have to persuade people with money. So the budgetary cost of a military in which a substantial number of people are drafted is lower than the budgetary cost of one where they have to pay people to volunteer.
- 5.32 So what are economists getting at? They're getting at something called opportunity cost. The cost is not just the budget, not just the money spent to hire people. It's the loss in output and loss in well-being by people who are in the military involuntarily, compared to where they would have been.
- 5.33 Of course, the famous example was Elvis Presley. I was seven when I remember reading that Elvis Presley had been drafted. His income at the time, in those years' dollars, 1958, was about a million a year, although Colonel Parker took a lot of it, but that's another story. It was about a million dollars. Think about what that'd be worth today, and think of all the output he was producing that people valued as consumers. And he goes into the military and that's lost. That's the extreme example.
- 5.34 There are more prosaic examples, someone who would have, say, gone on to the NFL, and you lose those couple of years, and you've lost a lot. The average amount of time as an NFL player is about three and a half years. Or someone who might have gone on to become a doctor and has to put that off. Or even just someone who had a business he wanted to start that he now doesn't get to start. And so when you add up all those opportunities lost, economists showed that the opportunity cost of a drafted military is higher than the opportunity cost of a volunteer military, because who shows up in a volunteer military? People who want to be there, and therefore that means who are attracted by those pays.
- 5.35 There was a push, by the way, to reintroduce the draft in 1979, and I testified against it before Senator Sam Nunn. And in researching for my testimony, I looked at every bill to reintroduce the draft. There were about 10 of them, and every single bill had in it a substantial drop in first-term pay. That's what they do. Why pay them when we don't have to?
- 5.36 Friedman, as always, said it very well at the conference. I took this out of the transcript: "The real cost of conscripting a soldier who would not voluntarily serve on present terms is not his pay and the cost of his keep; it is the amount for which he would be willing to serve. He is paying the difference. This is the extra cost to him that must be added to the cost borne by the rest of us. When he is forced to serve, we are in effect imposing on him a tax in kind equal in value to the difference between what it would take to attract him and the military pay he actually receives. This implicit tax in kind should be added to the explicit taxes imposed on the rest of us to get the real cost of our armed forces."

- 5.37** I just remembered the name of the congressman, Donald Rumsfeld, who was against the draft. So that's Friedman. Friedman had a buddy who was with him in Frank Knight's class named George Stigler. George Stigler got his PhD from the University of Chicago in the 1930s. He spent from 1947 to 1958 at Columbia University, and back to Chicago in 1958.
- 5.38** By the way, one of the great things about being in this business, from really starting to pay real attention when I was 19 and now being 75 and talking to people, means you learn lots of neat stories. So this older colleague of mine was in Stigler's class at Columbia in the early 1950s. And he knocked on Stigler's door, and Stigler told him to come in. And Stigler is on the phone. And he says—I'll try to do my Stigler voice—"Yes. Yes. Oh, thank you very much." Hangs up the phone and says to my friend, Pat, "I just found out I got tenure. I never have to work again." Anyway, of course, he did work.
- 5.39** So, some of Stigler's contributions: Stigler was one of the first people to look at the effects of the minimum wage. I want to show you, with the diagram, what he was saying. Here's the quantity of labor—this is unskilled labor—and the wage rate. Here's the demand for labor. Let's say here's the supply of labor, and here's the wage that equates the quantity demanded to quantity supplied. Let's say, in some low-wage state like Mississippi, and this is unskilled labor, let's say it's \$6 an hour.
- 5.40** So what Stigler pointed out is that if the government raises the minimum—I'm taking today's dollars—to \$7.25, it can't guarantee a job. All the government can do is guarantee that, if you have that job, that's what you'll get paid. But in fact, it's that guarantee that causes some of these people not to have that job, to lose their job or not be hired in the first place. And Stigler, it was his article in the American Economic Review that laid that out nicely.
- 5.41** He also wrote an article, a popular article, on rent control, with his buddy Friedman, cleverly titled "Roofs or Ceilings?" and it was, as I said, on rent control. The idea is, if you get ceilings on rents, you might not get roofs. The idea here, here's the quantity of rental housing, here's the price of rental housing, rent. Here's the demand. Here's the supply. Here's what the rent would be. Here's the amount that's rented.
- 5.42** If the government imposes a rent ceiling that's below what the rent would be, you get this much demanded, but this much supplied, and you get a shortage. That shortage will show up in a very low vacancy rate, but also, with landlords being able to pick and choose, they don't have to maintain as much. There'll be more people wanting it at that price than get it. And so they don't have to maintain as much, and they can be pickier. I'll talk about that a little later, when I talk about the UCLA School.
- 5.43** Stigler was also one of the first people, one of the first economists, to really study regulation and not just advocate it or talk about it theoretically. He started that whole thing. He pushed this idea. He became President of the American Economic Association in 1964, and his speech to the American Economic Association was titled "The

Economist and the State." And here's the gist. Let's actually study the effects of economic regulation, rather than engaging solely in a priori speculation.

- 5.44 Stigler had a way with words. I think he was one of the best writers in economics ever. Here's what he said. By the way, he tracked the way economists talked about regulation through the previous century and a half. He said, "The economic role of the state has managed to hold the attention of scholars for over two centuries without arousing their curiosity. Economists have refused either to leave the problem alone or to work on it." They just keep talking about it. And Stigler said, "Let's study it. We've got data. Let's study it." And he did that. He did some of the early work in the economics of regulation in the 1960s and '70s.
- 5.45 Now, I talked a few minutes ago, when I talked about the draft, about opportunity cost. And I'm just laying this out because, again, Stigler had such a way with words that I just thought this was beautiful. "An economist is a person who, reading of the confinement of Edmond Dantès in a small cell, laments his lost alternative product." By the way, he said that in a context where he's saying, "We economists are kind of narrow."
- 5.46 And I'm saying, "No, George," and I knew him, "George, you guys are wide. You're pointing out something that should be pointed out. You lose a lot." I mean, Dantès is, of course, a fictional character in "The Count of Monte Cristo." When I was trying to remember it, I thought it was about Oscar Wilde. I was wrong because that's a real guy, but anyway, the point is that could be a huge lost alternative product.
- 5.47 And then imagine, in the extreme—I hope this never happens—but imagine Elon Musk going to prison. We won't get—a friend of mine locally in Phoenix took me out on his \$100,000 Tesla on Sunday night. Oh, my goodness. He says, "Let's go to this place," and he keys it in or says it in words to this person, not a real person, of course, and it takes us there. He's just got his hands off his wheel. I mean, it is incredible. We're going to be, he says, a few years from, not decades, a few years from being able to set it and go to sleep. I'm looking forward to that. I might start driving long distances. Anyway, opportunity cost is a big deal.
- 5.48 Now, as I said, Stigler had a way with words. I talked about it on the first day, my "Concise Encyclopedia of Economics." I wrote almost all the bios in that encyclopedia. I wrote one on Stigler. The stuff he said was so memorable and so good, I had a little segment at the end that I didn't have for any of the other 100 or so economists called "Stigler at Large," him talking about issues. I want to quote two of the five that I used.
- 5.49 "Sears, Roebuck and Company, and Montgomery Ward made a good deal of money in the process of improving our rural marketing structure, but I am convinced that they did more for the poor farmers of America than the sum total of the federal agricultural support programs of the last five decades." I mean, think about it. You're out on a farm in the middle of nowhere, and this catalog comes along, and you can order from it, and the US Postal Service delivers it in a week or two. That's incredible.

- 5.50 And then another one. "Am I to admire a man who injures me in an awkward and mistaken attempt to protect me?" He's talking about a government official. "And to despise a man who, to earn a good income, performs for me some great and lasting service?" What Stigler is pointing to is, there's this overall prejudice against the free market and in favor of government.
- 5.51 You'll hear people talk about someone who "served" in government. When I get introduced, and I give a speech, and someone says I "served" in the Reagan administration, I usually take a minute to correct him. I think I did. I worked in the Reagan administration. You be the judge of whether I served. There's no market test of my service, the way there is for the plumber.
- 5.52 An economist who came along in the '50s, who studied under Friedman, was Gary Becker. His PhD dissertation was huge and was made into a book. I think he published it as a dissertation in '55, and the book was in 1957. It was called "The Economics of Discrimination." Up until that point, people had kind of a Marxist view of discrimination. The person being discriminated against obviously loses; everyone understood that. But the Marxist talked as if the person discriminating gains. Becker thought that through and said, "No." Think about why.
- 5.53 Take an employer who's white, who's looking at a white man and a black man, and they are both productive. It's sometimes hard to find productive people. He takes the white man over the black man, even though the black man might be a little more productive. He loses out, so discrimination hurts the discriminator. Of course, that leaves the question: why do they discriminate? Well, because they've got those prejudiced tastes, and it's worth it to them.
- 5.54 But Becker's point is that free markets make discrimination costly, more so than regulated markets do. So if you continue to discriminate and lose out those profitable hiring opportunities, you don't do as well as an employer who doesn't discriminate. And so, you tend to lose market share to the employer who doesn't discriminate. And again, it doesn't mean you won't. It means that the free market makes you bear a cost for discriminating on grounds that don't have to do with productivity.
- 5.55 We talked earlier, Becker has, we talked earlier about the role of women in the workforce. He was one of the major innovators in the economic literature in that, talking in the '60s about how all these things in the home made it more easy for women to leave the home and work part-time or full-time.
- 5.56 Another major player who lived past the age of 100. Now, he was from Britain. He spent a lot of his productive years at the University of Virginia, but he came to the University of Chicago in the early '60s. He wrote a famous article in the Journal of Law and Economics about the Federal Communications Commission. The bottom line was, "We don't need it." And here was the argument that people had made for it. They said

—this was when you had radio and TV, not other stuff since then—"That spectrum is limited. It's scarce, and so we have to have the government allocate it."

- 5.57** And his argument was, "Everything's scarce." TANSTAAFL, we don't have everything we want. Steel is scarce. Should the government allocate steel? No, it goes to the highest bidder. So maybe you need an FCC to say, "We're going to enforce these property rights," but you don't need an FCC to allocate them. He saw that as just obvious, and a lot of people didn't. Now, essentially, the whole economics profession sees that as being obvious. And that was him. He wrote another article called "The Problem of Social Cost" in the Journal of Law and Economics the next year.
- 5.58** There was this famous British economist named Arthur Pigou. And there's this concept in economics called externalities that I haven't talked about yet, so I want to just take a minute on that. What's called a negative externality is a cost imposed on person B, or persons B, C, and D, by person A, that is not taken account of by person A in his action. A standard example is pollution, where I'm producing a product, and pollution goes in the air and floats down and harms people, and I don't take account of that. Arthur Pigou said, "Well, a solution for that is, tax the polluter at a level that somehow is similar to the cost imposed by the polluter."
- 5.59** By the way, we rarely see that. Governments tend to come in and just regulate directly. You can understand why Pigou would say that when there's a polluter and, say, a thousand people being polluted. But a lot of the issues in law are ones where there's someone here imposing a cost on someone here, and that's just one and one.
- 5.60** Coase's argument was, well, if it's one person imposing a cost on someone else, they can get together and negotiate. "How much is it worth to you for me to impose that cost? If I'm doing something that's worth \$10 to me and the cost is \$12, I'm not going to be able to pay you, and I'll stop doing it. On the other hand, if it's worth \$20 to me and just \$12 to you, I'll pay you \$12 and I'll be better off." And he went through a whole lot of legal cases in that article—that's why it was in the Journal of Law and Economics—to look at how things like that have been handled over the years.
- 5.61** But the last thing I want to mention is his article, "The Lighthouse in Economics." In that article, he goes through—a little background: economists tended to see the lighthouse as the quintessential example of something government should handle, because how would you make money with a lighthouse? Would you turn it off when a ship goes by that didn't pay? That'd be pretty ugly. And so, they speculated a lot about why you needed government to provide it. So in a way, they were like the people that Stigler criticized. They're speculating. They aren't looking at the facts.
- 5.62** Coase comes along and says, "Gee, did we have private lighthouses?" And the answer was yes. Britain had a lot of private lighthouses, and they didn't turn them off when ships went by that didn't pay. How did they get them to pay? When they came in to port, and maybe the customs officials were there, the customs people were told,

"Charge them this," and they took it and turned it over to the people who owned the lighthouse. Now, granted, that's a government role, but still we had profitable, privately-provided lighthouses. It was Coase who uncovered that, looking at the facts that other people speculated about.

- 5.63** One of the areas that deals with this in economics is called public finance. And there was this textbook I used, which was very good in many ways, but the real irony was, the picture on the cover was about when you need government and what they should do and what they should provide. The picture on the cover was of a lighthouse. And the author, a guy named Joe Stiglitz, who later won the Nobel Prize in economics, I thought, "Did he ever read Coase?" Did it ever occur to him the irony of showing on the cover something that he thinks the government needs to provide that the government didn't provide?
- 5.64** One of Stigler's really productive students was an economist named Sam Peltzman. He got his PhD under Stigler in the mid-1960s, and he went on his first job to UCLA. I had read, a couple of years before showing up at UCLA in 1972, an article Peltzman had written on airline deregulation when he was a graduate student, and it was one of the better articles. He was one of the movers and shakers in pushing for deregulating airlines, which we already talked about.
- 5.65** So, when I got to the campus, I like to knock on the doors of the people whose work I had read. And so, I go to Peltzman's office, and he has a bumper sticker on the door—remember, this is UCLA—and the bumper sticker says, "University of Chicago at Los Angeles." So Los Angeles was thought of—the way people put it down in a way was, it's a farm school, for the University of Chicago, which we'll see there's slight evidence for, but not mainly. We'll see when we talk about the Chicago, the UCLA School tomorrow; they had their own stuff.
- 5.66** So Peltzman did some path-breaking work starting at UCLA, and then he went back to the mother church. He went back to University of Chicago. So I got lucky. I took his class my first year at UCLA in his last year there, and it was one of the best classes. And what he did that made such a big, big splash was an article called "An Evaluation of Consumer Protection: The 1962 Drug Amendments," in the Journal of Political Economy, one of the top five journals, 1973.
- 5.67** Here's what he did. He noted that there were these amendments to the Food and Drug Act in 1962; it was called the Kefauver-Harris Amendment or Amendments, I've forgotten which, that said—up till then, the Food and Drug Administration could approve a drug if the evidence showed that it was safe, but they never judged whether it was efficacious. We had a system where the drug company submits their paperwork demonstrating safety, and the FDA has to decide within, I've forgotten whether it's three months or six months, whether to allow it. If they don't say anything, it's allowed.

- 5.68** Well, something happened in the early '60s with a drug called thalidomide. Thalidomide was taken by women in pregnancy for morning sickness, and it had this really bad side effect. Sometimes the babies were born without limbs. Mostly that happened in Europe and Australia. It happened slightly here. Maybe they brought the drugs over. But a woman at the FDA named Frances Kelsey, who was a fellow Canadian, was awarded a medal of freedom for stopping it. I actually have a co-author who's in his late '80s who knew her. He said she stopped it because she lost the paperwork. She said if her head wasn't attached, she wouldn't have known where it was. And so it was just pure luck that it was stopped.
- 5.69** But it was in response to the thalidomide tragedy. Do you see any irony here? They introduced an efficacy requirement because of the thalidomide tragedy. Thalidomide was efficacious; it was just unsafe. So if you'd had a requirement for efficacy, it still wouldn't have stopped it. It was just the requirement of safety didn't stop it. So Sam says, "OK, that raises the cost of introducing a new drug because now we have to do further tests with large numbers of people to establish efficacy."
- 5.70** And remember, when you're doing double-blind tests, and it is efficacious, half the people are missing out on this thing that can help with their disease, so that's a big deal. He showed how it added a few years to the development process before the drugs were allowed. The result was a 60% drop in new drugs annually, relative to the trend. And by the way, it delayed new drugs by two years or more. That number is now closer to 10 years. So that was a big, big deal. An economist who did more careful work after that found evidence supporting basically what he said.
- 5.71** He started this work at UCLA, finished it at Chicago, his work on the regulation of car safety. As a result, a fellow economist friend and I started calling him Seatbelt Sam. There was a law passed in 1966 requiring seatbelts, but requiring a whole lot of other things. Padded steering wheels—no big thing like in the 1957 Cadillac, sticking out your head in the center of the steering wheel, things like that. And what he found was that fatalities of occupants of cars didn't fall relative to trend, but fatalities of motorcyclists and pedestrians rose relative to trend.
- 5.72** So, what's going on here? Here was his argument, and later work has backed it up. You're a driver. Because you have a seat belt, because you have a padded dashboard and so on, you're safer in a given accident. So what do you do? Drive more intensely. I'll give you an example from my life. We have one daughter. She was quite young. I'm driving along, going 70 in a 65, and we have this station wagon, and she wants to take her seat belt off and go back into the back thing and get something. I said, "OK, Karen, but do it fast." I'm going 70; what do you think I did? I cut my speed. Right there, there's evidence that if an occupant of a car is less safe suddenly in an accident, I adjust my driving behavior.
- 5.73** Because of that work that Peltzman did, there's a new term in the literature. It's called the Peltzman effect. You make things safer, and people adjust their behavior—I should

have added this to my thing; I just read it a couple of weeks ago. They're making these football helmets better, and what they're already finding is people take more chances because the football helmets will save them in a particular situation. So they will do it more intensely. It could be that, on net, they're safer, but they're not as much safer as you'd expect if you didn't adjust for the difference in behavior. So that, in a nutshell, is the Chicago School.

- 5.74** I've heard over hundreds and thousands of times about the Great Depression, so basically, are you saying it wasn't as bad as they put it out to be?
- 5.75** No. Was the Great Depression as bad as they put it out to be? It was. The issue is, what caused it? What made it great, so to speak? What made it last long? There are two things. Friedman and Schwartz's contribution was to say that it was the decrease in the money supply because of the failure of the banks. And then the other thing that I talked about early in this time was the government keeping wages and prices high, rather than letting them fall, so unemployment stayed high.
- 5.76** One more thing to say about that. This is strange to say to an American audience, but back then, we had a very primitive bank system. Here's what I mean by that. We had restrictions on branch banking. So in many states, you have one bank, and that's it. You can't have more. And so what that means is, the banks could not diversify across states. So when you had crop failures in the Midwest in the late '20s, early '30s, and the banks that were lending to farmers went bankrupt, if they'd been part of a national system where, say, the ones in New York are fine, and they just transfer some money to the bank in Iowa, it wouldn't have been as bad.
- 5.77** I'll give you an example. I come from Canada, and Canada had a very sophisticated banking system, eight major banks, zero bank failures. The United States had something like 35,000 banks, mainly small; 10,000 of them failed. Those are rough numbers. So, yeah, that's what I'm saying.
- 5.78** So we've learned all this stuff about socialism. I think it's political, but why did we still elect the mayor of New York? Haven't we learned anything?
- 5.79** Why do we elect a socialist as mayor of New York? Haven't we learned anything? "What you mean 'we', Kemosabe?" People here have learned something, I think. Here's the thing. Socialism was shown to be a failure to the public in general in the late '80s, early '90s. Have you looked at the demographics, the age of the people who voted for Mamdani? They were, at best, two years old when that happened. They weren't paying a lot of attention. So they don't know about a lot of the failures of socialism. That's the first thing.
- 5.80** Second thing is, Mamdani calls himself a democratic socialist, and he is. That doesn't mean we're going to get socialism in New York. There are checks and balances on him. And so, one of the things a lot of people say is, "Oh, people are going to see now how bad socialism is." No, they aren't, because he's not going to get a lot of what he wants.

He's facing a \$12-billion annual budget deficit with a \$119-billion budget. He's going to have to figure out that before he manages to expand childcare and give it free to everyone, before he makes buses free and so on. We aren't going to see this incredible experiment. And thank goodness we're not. Thank goodness we're not.

- 5.81** The other thing is rent control, which we talked about. He likes it; he wants more of it. There's a rent control board in New York; it has nine members. Two of them came up for renewal last month. Eric Adams, the outgoing mayor, renewed them because they wanted to at least allow a 3% increase, which is in the legislation. It allows them to do that. He appointed two more to this board of nine, who want to allow a 3% increase. That makes four. And there was already one on the board who wanted to allow a 3% increase. So that's five out of nine. That's a majority. They're going to vote to allow that increase, and good.
- 5.82** Now, when the couple of terms are up in two years, that's when Mamdani can act, and he has a four-year term, so we'll see. But the point is, it's not going to be this great experiment where we get to see how bad that goes, how badly that goes, because he has constraints on him. Also, I've never understood this. I've written a couple of things about Mamdani. He wants Kathy Hochul, the governor of New York, to raise the corporate income tax on companies in New York and give the money to him. Wait a minute. I'm in Syracuse, and you're going to get the money? I can't imagine she'll go along with it, but maybe she will. I don't know.
- 5.83** Do you ever foresee another—I don't want to say Great Depression happening to that extent, but do you ever foresee—because I know there was the housing market crash, I believe in '09, '08. But do you ever foresee it happening to some extent like that again?
- 5.84** Do I foresee another depression, anything close to the Great Depression? No, I don't. Here's why: the unemployment rate in the Great Depression hit its peak at 25%. The closest we came was in the '73, '75 recession, the '81, '82 recession, and the 2007, 2009 recession. I don't have the numbers exact, but they were all around 10%. That's a pretty big difference. So no, I think they understand enough about not letting the money supply fall.
- 5.85** Having said that, let me tell you a true story. At Milton Friedman's 70th or 80th—he lasted till he was 92, I'm going to say 80th birthday—Ben Bernanke, who was chairman of the Fed, made this little tribute to him. He said, "Milton, you're right. We caused the Great Depression. We're sorry. We won't do it again." But then he's chairman of the Fed, and what's he doing? He persuades Congress in October 2008 to let them start paying interest on reserves to banks holding reserves. What does that do? Well, now a bank says, "Gee, I could lend it out and take a risk, or I could get a sure thing by holding the reserves and having the government pay me." So what should have happened is that the money supply should have expanded a little to prevent that, and it didn't.

5.86 So there's a guy who said, "We understand," but he didn't totally understand. But again, do I think we'll get something like 25% unemployment? No, I don't. Could we get 12? I think so. One last thing to add, it goes back to incentives. One thing not to do is to increase the length of time in which you give people unemployment insurance, because it pays them to stay unemployed. They get pickier about the kind of job they'll take.

Lecture 6: Mainstream Economics

At a glance — Mainstream economics.

- 6.1** Where have we gone so far? Well, we've looked at the Austrian School and the Chicago School, but there are more free market economists out there, even in what's called the mainstream. There's a big world out there. So what I want to do in this lecture is do a sampling of the mainstream. I could spend 10 hours on this, but of course, that wouldn't make sense. I just want to hit some of the highlights of people who are not thought of as part of the Chicago School, the Austrian School, or what we're later going to see as the UCLA School, just doing good economics that ends up becoming very critical of various government regulations and various government programs, and also implicitly or explicitly, very positive about free markets.
- 6.2** So, I'm going to talk about the Brookings Institution. It's kind of thought of as a left-of-center organization in DC, where there are two economists, both friends of mine who do work in this area. Harvard, where you have Edward Glaeser and the late Martin Feldstein. Dartmouth, where you have Douglas Irwin. Columbia University, Jagdish Bhagwati. UC Irvine, David Neumark, and Michael Clemens. And he's moved around, so I don't have his latest affiliation.
- 6.3** So I want to start with Robert Crandall. I've already talked about him. He's the one who did a study in the Journal of Law and Economics in 1989 with John Graham of Harvard. What they looked at, I've talked about, was that CAFE, the Corporate Average Fuel Economy law, reduced the weight of cars in the late 1980s and would continue to do so in the '90s by about 500 pounds on average. To put that in perspective, that's about a 10% to 15% drop in weight. The lighter the car, all other things equal, the more in danger the inhabitants are in a crash. That, they estimated, made lifetime fatalities for each model year 2,200 to 3,900 higher than otherwise over the assumed 10-year life of the cars. That's a substantial number.
- 6.4** Now, Crandall also looked at—we talked earlier about monopoly in the economy and how government is one of the major, the major, creator of monopoly. I happened to mention monopoly in the legal profession, with governments keeping people out from being lawyers. Crandall, in an article he wrote, said, "We need more lawyers," not something you hear every day, but he pointed out the role of the American Bar Association in accreditation and how slowly they move in allowing and accrediting new law schools. And of course, if they move slowly on that, the law schools aren't turning out people who are accredited, and we don't get as many lawyers. So there's a huge premium to existing lawyers.
- 6.5** I talked earlier about externalities and how this British economist named Arthur Pigou, about a century ago, advocated taxes to use against externalities, negative externalities. But governments tend to use what's called command and control. They

like to get in there and tell companies how to do things. So Crandall wrote a piece on pollution controls in "The Concise Encyclopedia of Economics," edited by that noted economist, David R. Henderson.

- 6.6 He gave this actual, real-world example. In St. Louis, there was a paper products firm and a brewery. They both put out pollutants that were somewhat comparable. The paper products firm, you could reduce a ton of pollutant for \$4. The brewery, it would cost \$600 to reduce a ton of pollutants. Under the regulation by the Environmental Protection Agency, the EPA required each of these to reduce pollution by some amount.
- 6.7 Here was the point that Crandall made: what if they'd been able to trade? What if you gave them some amount they have to reduce pollution, but if the pollutants are comparable in their harm, let them trade to either buy the right to pollute or sell the right to pollute. So let's take a simple example where each has to reduce pollution by a ton, and they can't trade. So, this one reduces pollution by a ton, and it costs \$4. This one reduces pollution by a ton, and it costs \$600. Total cost: \$604.
- 6.8 But what if you allowed trade? Then, think back to Menger, the brewery would go to the paper products guy and say, "I'll pay something less than \$600," and the paper products guy would say, "I'll charge something more than \$4." There's a lot of room in there. Let's say it works out to \$300. But I want to focus on the cost of reducing pollution, not on who shares these benefits. Clearly, they're going to be able to buy the right to pollute by a ton by having this one reduced by another ton. So this now will reduce by two tons for \$8. This one will reduce by zero because it's too expensive. So the overall cost is \$8. That's a \$596 drop in the cost. And he argued in that article in my encyclopedia that we could do that across a wide range of things—let polluting entities trade their way out of it, and we can get the least-cost avoider doing the avoiding.
- 6.9 Crandall has a colleague at Brookings named Clifford Winston. While I was preparing these, I came across an article he wrote just last month in The Wall Street Journal, in which he argued that private airports could beat snow delays. Here was his argument: we don't have private airports; we have government-owned airports. But what if we had private airports as they do in Britain? Maggie Thatcher made that change. Are there any airports for which you could heat the runways so they don't have ice on them, and charge appropriately so that the airlines would pay to be able to take off a little sooner?
- 6.10 They did this cost-benefit analysis and found that almost all the airports in the United States, even if you had private for-profit ownership, would not pay. But the one airport in which it would was Boston. And he pointed out, if you got that, you would save X million dollars a year. So again, thinking about how you introduce markets and market incentives.

- 6.11** He wrote an article in the summer of 1998 about economic adjustment to deregulation—Winston did—and here's what he said: "This paper suggests, however, that industries are likely to behave quite similarly when it comes to adjusting to deregulation, and that their adjustment, while time-consuming, will raise consumer welfare significantly, even at first, and increasingly over time. Markets will become more competitive." Think about deregulation of airlines. Think about deregulation of trucking. "Firms will develop innovations to become more efficient and more responsive to consumers. The benefits to society will grow as the adjustment continues."
- 6.12** And then he went on. "It is time to refocus the debate directly on that evidence and broaden our vision of where markets can replace governments in determining the most efficient way to create and sell products and services. This process is taking place in telecommunications, electricity, and cable TV." Remember, he's writing this in 1998. "Looking ahead, government control over prices, entry, and exit in ocean transportation clearly warrants the deregulatory reforms proposed by the Clinton administration. Similar proposals were also warranted at the state and local level for taxis, limousines, and occupational licensure."
- 6.13** Well, taxis, what have we got now? Uber and Lyft. So that's happened, and that covers limousines also.
- 6.14** Occupational licensure, what that means is licensing people to practice in certain occupations. How many occupations do you think there are in the United States in which, at least in one state, you need government permission to practice? The answer is 800. What percent of the labor force is employed in those occupations? Approximately 25%, whereas in 1950, it was 5%, so it's growing.
- 6.15** I'm not just talking about occupational licensure for doctors and lawyers. I'm talking about it for tree surgeons, haircutters. There's this thing that's done instead of haircutting, which is cornrow braiding. In some states, the government says, "You've got to take a degree or a course in cosmetology in order to be a cornrow braider." Now, what's one of the main things they talk about in cosmetology? How to use chemicals safely. What is not used in cornrow braiding? Chemicals. It's essentially a way of keeping people out from competing with the standard haircutters.
- 6.16** Also, notice his statement about ocean transportation. There was this law passed in the early 1920s called the Jones Act. If you ship anything from point A to point B in the United States, it has to be in a ship built in America and crewed by Americans. Well, that's had some unintended but completely predictable effects.
- 6.17** In Maryland, they sometimes want salt—no, sorry, in Massachusetts, they sometimes want salt for their roads during the winter. They could get it from Louisiana. But if it were shipped by ship, it would be very expensive. So they get it from another country much further away, because it has to be shipped between two points in the United States to be subject to the Jones Act. If it comes into the United States from another

country, it's not subject to the Jones Act. So it creates all these weird things that, again, are predictable.

6.18 Another one: a ship comes into the Baltimore harbor, and it would, say, deliver the stuff intended for the places around Baltimore, but then go on to, say, Boston or Charleston, and break-bulk there to deliver to those areas. Well, then, if it's going from Baltimore to Boston or Charleston, that's A to B, that's within the United States, that's the Jones Act. So what they do instead is come into one of the harbors, break-bulk, break it all down, put on trucks, and have trucks go hundreds and hundreds of miles, creating congestion on the roads instead of being able to ship it by ship.

6.19 There's also something comparable for people on pleasure cruises, like Carnival cruises and so on. I don't know if you've ever noticed, if you get on one of those, you never go from a city in the United States to a city in the United States. For example, if you get on one on the coast I live on, California, you'll go from San Francisco to either Victoria or Vancouver. And then you can go from Vancouver to Juneau, because, again, it's going from one country to another country, one country to another country.

6.20 You'll never go from San Francisco to Seattle, let's say, because then they'd have to use US-built ships, crewed by American crews. Instead, they can have these ships crewed by people from third-world countries who work their tails off and are very happy. They are making money that's very good compared to what they could make at home. Those are the kinds of things economists all over the place write about, whether or not they're part of one of these schools.

6.21 Another one is Ed Glaeser of Harvard. His big thing is deregulating housing. I have a friend and former co-blogger, Brian Caplan, who wrote a graphic novel, which is really—it sounds weird—it's not really a novel, but it's very full of information titled "Build, Baby, Build." And what he does is showing that the way governments have restricted housing in this country, especially on the coasts, causes housing to be roughly twice as expensive as it would be. So the average household spends about 20% of its budget on housing. If we had complete deregulation of housing, it would spend, say, 10% of its budget. And there, right there, is a 10% increase, actually, if you do the math, an 11% increase, in their standard of living.

6.22 One of the guys who really got this going was Ed Glaeser. Here's what people used to argue. I used to hear this from real estate agents in California: "The reason housing is so expensive in California is that they aren't making any more land." The land is the constraint.

6.23 Glaeser goes, "I wonder if that's true." What he does is, he looks at two comparable houses in coastal California, comparable in terms of the value of the location and the size of the house, all those things. But one is on this plot of land of a certain size, and the other is on a plot of land that's double the size. So if the scarcity of land is what's driving that big differential in price, you should see the one on the big piece of land

selling for way more than the one on the small piece of land because it's got way more land, and that's the scarce resource.

- 6.24** Wrong. What you see is—at least in the data he looked at in the '90s— that the one with double the land sells for maybe \$10,000 more. I gave a talk on this to a bunch of developers who understand this, and usually, I know when I'll get the applause lines, I plan for it, and I said, "California suffers not from a scarcity of land, but from a scarcity of building permits." They broke into applause, and that's exactly right. And so he was the guy who estimated that.
- 6.25** Another economist, again, not one of these schools, but a prominent economist, was Martin Feldstein, who died a couple of years ago. He was my boss. He was the chairman of the Council of Economic Advisors in the early '80s when I was his employee. He did a lot of work in the '70s, '80s, and '90s on the effect of marginal tax rates, which we've already talked about, on people's actions.
- 6.26** Before he wrote about this, people focused on the labor supply, and that's what I focused on when I talked about it. You have a higher marginal tax rate than your average. You respond by not working quite as hard, as if your tax rate were lower. But his key insight was that people respond on many margins, not just on the labor supply margin. There are ways to reduce taxable income that don't necessarily involve working less.
- 6.27** One is to shift investment to tax-favored forms. For example, under our Constitution, the federal government is not allowed to tax people's interest on municipal bonds. They're tax-exempt. So what you see is people in very high tax brackets buying disproportionately municipal bonds. And of course, the municipalities can get a break on the interest rate because they don't have to pay as much to be competitive with taxable bonds. So maybe where the typical interest rate on a federal bond is 3.5%, the typical investment on a fairly high-rated municipal bond is 2%, because people are willing to invest there to save on taxes. So that's a shift of investment to a tax-favored form that has nothing to do with the money supply. It does mean that the federal government will collect less money than it would.
- 6.28** Another way to save on taxes is to reduce money income and replace it with non-taxable perks, for example, more generous health insurance. The federal government and the state governments do not tax employers' contributions to their employees' health insurance. If you want to give employees a raise, one way to do it is to give them a little more generous health insurance. Maybe cut the co-insurance rate, maybe cut the co-payment, maybe cut the deductible.
- 6.29** Another way, of course, is to defer capital gains. You are not taxed on capital gains until you realize them, until you sell the asset. So people often will hold off on selling their assets. And what they found, by the way, is when the tax rate on capital gains fell,

capital gains realizations, capital gains sales went up so much that, for a couple of years, the government collected more at a lower tax rate.

- 6.30** Now, one thing that Feldstein emphasized was what's called the deadweight loss from high marginal tax rates. What is deadweight loss? It's a measure of the distortion that the tax causes. So, for example, if the government imposes a sales tax, say, of 10%, then—and it had 0% sales tax before—then someone who would have bought the good won't buy the good because it's not worth it with that tax. And so, there's this gain from exchange that is lost.
- 6.31** And there might even be a deadweight loss, even though you don't pay anything. I'll give you the example I used to use. I've gotten higher income and less patience, so I don't use it anymore. But when I flew into Winnipeg every year and rented a car to go to my cottage, I had to pay a tax per day if I got the rental car at the airport. But if I paid \$20 and got a taxi cab to downtown Winnipeg, I could avoid a \$10-a-day tax over 17 days. So I could pay \$20 and spend 20 minutes to save \$170, and that was worth it. So there's a case where there was a deadweight loss of \$20 plus the value of my time, and I paid nothing. So a thing can cause a deadweight loss without people actually paying any of it.
- 6.32** The other thing that's a well-known theorem in economics—and Marty Feldstein talked about this—is the deadweight loss is not proportional to the tax rate; it's proportional to the square of the tax rate. And there's a little proof that I am not going to run through with you, but it is. So, what that means is, if the government, say, doubled the marginal tax rate, it would quadruple the deadweight loss.
- 6.33** Another economist—again, not one of these schools, but has been very important in the area of international trade—is Douglas Irwin from Dartmouth. He's done a lot of work on trade and tariffs. He did a book on the Smoot-Hawley tariff. I don't know if you've heard of the Smoot-Hawley tariff. If you see that movie with Ben Stein as the teacher, he asked them about Smoot-Hawley, and the scene is a scream. It's someone's day off. Oh, my goodness. This is happening. "Ferris Bueller's Day Off." Thank you.
- 6.34** Anyway, he puts a lot of weight, Ben Stein does, he's the son of Herb Stein, by the way, who was my first boss at the Council of Economic Advisors when I was a summer intern. Anyway, he puts a lot of weight on Smoot-Hawley causing the Great Depression. Doug Irwin wrote a book that I reviewed when it came out, saying no, it didn't cause it, but it did make it worse. And it made it worse in two ways. We had a little less trade, where we are getting fewer imports, so we're worse off. And it led to retaliation around the world. And that was what made it worse than otherwise.
- 6.35** Doug Irwin also emphasized that protection—that means, we use the word protection in a funny way in economics; it means protecting industries from competition—protection is rarely temporary because of the constituencies it creates. You put a tariff

on some good coming into here, and suddenly you have a lobby group of the producers here who want to keep the tariff and make their life easier and be able to raise their prices.

- 6.36** My favorite example of this is the so-called Chicken Tax of 1964. Lyndon B. Johnson was the President at the time. Two countries in Europe—France and Germany—imposed tariffs on chickens that our producers sent there. So LBJ imposed a temporary tariff on light trucks, because they were producing a lot of light trucks like pickups and sending them here, a temporary tariff of 25%, whereas the tariff before that had been 2.5%, so he ten-tupled it. That was supposed to be a temporary tax. The tariff on light trucks today is 25%, so that temporary tariff has lasted 62 years.
- 6.37** Now, one of the arguments, economists argue, by the way, almost all economists are pro-free trade. Economists argue that even if other countries impose tariffs on us, we're better off not retaliating. And in fact, we would be better off having complete free trade no matter what they do, cut our tariffs to zero the way Britain did in the last half of the 19th century. But politically, that's hard to achieve. Why? Because you're going to get all these firms in America who are protected from competition by tariffs, who are going to lobby to oppose that. Who's going to lobby to favor it? Just us consumers who don't pay a lot of attention.
- 6.38** So the idea is to have what are called reciprocal trade agreements. Get people together, get other countries together with us, and say, "We'll cut the tariffs on these goods from you, if you cut the tariffs on our goods to you." And then we can get producers of those goods, exporters of those goods in America, to be lobbyists for that. One of the main groups of lobbyists, historically, meaning in the last number of decades, has been agriculture, has been farmers. And so, that's the argument for having reciprocal trade agreements.
- 6.39** One of the other things Doug Irwin emphasizes, as do many economists, is that trade deficits, in other words, where we buy more of their stuff than they buy of our stuff, are not evidence of unfair trade. They just aren't. In fact, trade deficits are the opposite of capital account surpluses. If people in those countries end up with more of our money because we buy more from them than they buy from us, what do they do with it? They don't just sit on it. They buy our bonds. They buy our stocks. They invest directly.
- 6.40** And even they might hold on to the dollars. You might think that's bad. No, that's good, because it costs our government about 10 to 15 cents to produce a Benjamin, to produce a \$100 bill. And if they're holding on to it, we got \$100-worth of stuff for 15 cents. And if you're worried about that reduction in the Benjamins, reduction in the amount of money in our economy, go back to $MV = PQ$, and you think you'll have a mini recession because M is falling, it's like that old Jay Leno ad for Doritos in 1989. "Go ahead and crunch them. We'll make more." We can make up for that amount of money lost. By the way, that's me talking, not Doug Irwin, but he would agree with that.

- 6.41** Now, another economist—again, not part of one of those schools, but definitely in the mainstream—is an economist who grew up in India named Jagdish Bhagwati. He's pro-free trade, as, as I said, are almost all economists. He did the early work of spelling it out very rigorously, clearly. One of the things he came up with that I studied at UCLA in my graduate program was what's called the targeting principle. People were justifying tariffs on the grounds that it would solve, or it would help solve, some domestic distortion, either due, say, to private monopoly, or due to some other government program.
- 6.42** And what he and his co-authors showed was that the first best policy, the best policy, was to target—to use domestic policy on a domestic issue, not mess it up by having tariffs on trade. He also did a path-breaking book with his wife, Padma Desai, on India's industrialization. India had, before 1991, a highly regulated economy where you needed permission to do almost anything. You're producing goods in a certain factory, and you want to shift to producing another kind of good, you've got to get permission. Permission takes months, at least. You want foreign exchange, and they've got foreign exchange controls to be able to buy something abroad, you need to get the government to let you buy foreign exchange, US dollars, so you can buy those things. All these ways they hampered the economy.
- 6.43** And they used this also to get domestic production of various items that, in a free trade world, India wouldn't be producing. One of them probably was cars. They produced these fairly crappy cars that cost a fair amount and didn't work that well. There was this old line— you drive along and it would rattle. And there was this old line that the only thing that doesn't make noise in the car is the horn. Permission was required for everything, and that was a path-breaking book at the time.
- 6.44** Another economist—again, not part of those schools—was David Neumark. He pointed out, and we've already talked about this, that increases in minimum wages reduce employment of unskilled workers, especially teenagers, young adults, and the least educated. And with another economist named William Wascher, he critiqued a famous study that had come along in the early '90s, the Card-Krueger study. That study was path-breaking, and I'll tell you what was wrong with it, but at first it was path-breaking.
- 6.45** These two economists, David Card, a fellow Canadian, and Alan Krueger looked at where the minimum wage had gone up in New Jersey and hadn't gone up in Pennsylvania. They said, "Let's look at fast food restaurants because they employ a lot of low-skill workers." They found what you'd expect to find with the standard story is that employment in New Jersey would fall relative to employment in Pennsylvania. If they're both going up, the trend will be lower in New Jersey than in Pennsylvania, for example.
- 6.46** Well, what Neumark and Wascher did was really examine the data that Card and Krueger had used. They found that—Card and Krueger had their research assistants call up a fast food restaurant and say, "How many people are employed?" And whoever

answers tells you that answer. That's pretty crappy data compared to what you could do, which is to get establishment data that are reported to the Bureau of Labor Statistics. And that's what they did, and they found that relationship went away.

- 6.47** Neumark has also pointed out longer-run effects of the minimum wage, such as reduced hiring and reduced training. Part of the package of compensation is not just the wage, but the training. You're getting training for a better job. But if you have to pay a higher wage, you might cut back on the training, and therefore people, it looks good to them, but it's not, because they're not climbing the ladder as quickly.
- 6.48** He also pointed out that minimum wages are not well targeted at reducing poverty. Why? A lot of the minimum wage workers are in relatively high income households. They're teenagers whose parents are making good money. And so it's just not at all targeted at reducing poverty. In fact, Neumark told a story about his daughter coming home and being excited that the minimum wage had gone up because she was working in a fast food restaurant. He goes, "Honey, what do you think I'm studying? That's not good. Maybe it's good for you, but it's not good for that poor worker from a poor area."
- 6.49** Another economist—again, not part of these schools—is Michael A. Clemens, and his big issue is immigration. We've talked about exchange and why both sides gain from exchange. Well, that's true across borders. It's also true with immigrants. If someone immigrates here and works, and a higher percentage of people who immigrate here work than the percent of Americans, they're doing something that other people value, so both sides gain. And so economists have tried to estimate those gains.
- 6.50** What Clemens did is, he wrote this famous article titled "Trillion-Dollar Bills on the Sidewalk?" And let me go back to the old joke. Two economists are walking along. One of them sees a \$20 bill on the sidewalk, or as might be the case, a \$5 bill, and says, "Hey, there's a \$20 bill." And the other economist says, "Well, there couldn't be because someone would have picked it up." Well, that's where he gets this title, "Trillion-Dollar Bills on the Sidewalk?"
- 6.51** And what he does is starts out by discussing the efficiency gain from eliminating international barriers other than to movement of labor. So all policy barriers to merchandise trade, in other words, all tariffs and quotas and so on, all these numbers are a percent of GDP. So this one study estimates that GDP, world GDP, would go up by 1.8% if you had total free trade. Another one says 4.1%. Another one says just 0.3%. But they're all in the ballpark of single digits, and typically low single digits, per cent. And capital flow is not a huge benefit there.
- 6.52** But now let's look at getting rid of all barriers to labor mobility, and this is the percent increase in world GDP. This one study finds a 147.3% increase in world GDP. Others find lower percents, but they're all very high— 96.5%, 67%, 122%. Why? Well, imagine someone in a poor country in Africa or a poor country in Asia who has limited skills, and that person comes to Canada or Japan or Britain or United States. Especially if it's

United States, that person's pay will typically go up by five to 15 times immediately. If it's immediate, it's not because they're building skills, it's because they're so much more productive here with what's available than where they were.

- 6.53** An economist friend of mine named Brian Caplan, who wrote a different graphic novel on immigration, says, "It shouldn't be hard to imagine that. Think about if someone tried to be a farmer in the Antarctic. His productivity would be very low. So he comes over here, and it's suddenly much higher." That's the extreme, but people in those countries have very low productivity because of the system they're in, not because of them.
- 6.54** Now, it is true that the immigrant gets the biggest percent of the gain, but we gain, too. Mothers gain who want to be in the labor force and hire nannies. I gain if I get someone to do my gardening. There are all kinds of ways we gain. And in fact, my friend Pat, who's the one who was in Stigler's office when he found out he got tenure, and he knew I was arguing for more immigration on humanitarian grounds, let those poor people in, he said, "Henderson, you're arguing all wrong. Here's how you argue." And he was a pretty wealthy man. He said, "I should have five servants, and you should have two." In other words, there are all kinds of ways we could use low-skilled labor, where we're better off and they're better off.
- 6.55** Now, by the way, one caution about these numbers. They're extrapolated like crazy. They're probably wrong, but they're probably not very wrong. The one way they could be kind of wrong is, maybe a lot of those people wouldn't move even if they had the ability—even if it were allowed. But if it were allowed and they wanted to move, these numbers are plausible.
- 6.56** The last economist I want to talk about, I don't know where to place him. He's not on the mainstream—well, I'll tell you who he is: Thomas Sowell. He's a 95-year-old colleague at the Hoover Institution, where I'm affiliated, and he's still going strong. It's hard to place him. He's got his PhD from the University of Chicago. He was on the UCLA faculty through most of the '70s when I was there. He's his own man, but he's not thought of as mainstream; I just didn't know where to put him.
- 6.57** He's both an innovator and a popularizer. So a lot of the things he writes are things based on other people's research, but he does it so well and so clearly. I want to give an example. His clarity in examining data compares same with same. Here's what he writes about. "Data from the World Bank that show that trade was expanding as tariff rates were falling said that the ratio of incomes of the 20 highest-income nations to the 20 lowest-income nations increased from 23-to-one to 36-to-one." In other words, wait a minute, you're getting trade expanding, but the lowest-income nations are falling? This seems counterintuitive because various studies have shown that poor nations gain even more from freer trade than richer nations.

- 6.58** Well, it should seem counterintuitive because Sowell spots the problem with the comparison. He says, "But such statistics are grossly misleading because neither the top 20 nations nor the bottom 20 nations were the same in 2000 as in 1960. Comparing the same 20 nations in 1960 and in 2000 shows that the ratio of the income of the most prosperous nations to that of the most poverty-stricken nations declined from 23-to-one to less than 10-to-one. In other words, our intuition that the poor countries should do better, held. Expanded international trade is one of the many ways poor nations have risen out of the bottom 20." So there's mobility in and out of who's in the top 20.
- 6.59** Well, there's another way he looked at mobility. And again, these data aren't original with him. It's just, he digs them out and points out the importance of them, and that is wealth mobility. Sowell also finds—by the way, this is a quote from a paper I wrote on his work—Sowell also finds great mobility among the wealthiest people in America. In the famous Forbes list, they started listing the 400 richest Americans in 1982. In 1982, there were 14 Rockefellers, 28 DuPonts, and 11 Hunts. Twenty years later, there were only three Rockefellers, one Hunt, and no DuPonts. In other words, there's a huge amount of mobility. Where was Elon Musk on that list? Where was Jeff Bezos on that list? There's huge mobility.
- 6.60** And Sowell, by the way, who got his PhD under George Stigler, also has a way with words. He said, "It may be fashionable for journalists to refer to 'the whim of the marketplace,' as if that were something different from the desires of people, just as it was once fashionable to refer to 'production for use, rather than profit,' as if profits could be made by producing things that people cannot use or do not want to use. The real contrast is between choices made by individuals for themselves and choices made for them by others who presume to define what these individuals really need."
- 6.61** So, what you will find is that Sowell is almost always taking the position, taking the side, of the common person trying to do well, rather than the elitist who tries to say what they should have and often tries to use government coercion to make that happen. Sowell also wrote on the minimum wage, and he talked about something that economists have known for a long time called survivor bias. So again, he didn't come up with this, but he's the one who pointed out it applies in this context, and he's the one who said it so well.
- 6.62** So, he noted that surveys of employers before and after a minimum wage increase can survey only those businesses that survived in both periods. You can't survey, you can't say, "How many low-skilled workers are you hiring?", to the business that went out of business? And maybe one of the reasons it went out of business is the minimum wage went up.
- 6.63** He used this great analogy, because of what happens if you forget that fact. He said, "Using such research methods, you could survey people who have played Russian roulette and prove from their experience that it is a harmless activity, since those for whom it was not harmless are unlikely to be around to be interviewed. Thus, you would

have, quote, 'refuted' the myth that Russian roulette is dangerous." In another context, he talks about how people have this idea that it's very easy for big companies to make money because we look at big companies and they're making money. What do we not see? The ones who went out of business; they went bankrupt.

6.64 So he had this great, again, a survivor principle point. He said, "Using the same methodology, we could prove,"—and just interviewing people who were around—"we could prove that no one was killed during World War II." So that's my survey of people not in those schools of thought I've talked about, people mainly in the mainstream, plus Thomas Sowell, whom I didn't know where to put.

Lecture 7: UCLA School

At a glance — The UCLA School.

- 7.1 So far, we've talked about the schools of thought, the Austrian School of thought, the University of Chicago School of thought. And then I went to more mainstream economists, and now I'm going back to another school of thought, the UCLA School of thought.
- 7.2 I got my PhD at UCLA, and it was when I was reading work by those people when I was an undergrad, and when I took a year off to study economics on my own, that their names—I would read journal articles and I'd say, "Oh, I like this article." I'd write down the name of the person. I'd write down where the person was. Chicago and UCLA and a couple of others kept coming up. So, I applied there, and I got accepted.
- 7.3 Now, the main players at UCLA, in my view, were Armen Alchian and Harold Demsetz. I'm going to talk a lot about their work. Other important players, well, there are a number, but the two I'm going to focus on are Jack Hirshleifer and Benjamin Klein.
- 7.4 Now, the first Alchian article I read was published in 1968, and I read it when I was 18, as an undergrad. I was a math major. I wasn't an econ major. It's titled "The Economic and Social Impact of Zero Tuition." He's at UCLA, which charges zero, or very close to zero, tuition at the time for in-state students, and he's criticizing that. And of course, I'm in a college, I'm in a government-funded university, and everyone around me seems to believe in government funding college. He had a different view of it.
- 7.5 Here was his point: if you look at who goes to college and graduates, and what their incomes are compared to incomes of people who don't go to college, they're substantially higher. So, isn't it strange to tax everyone to give money to people who are going to be higher-income, even if currently they're not high-income?
- 7.6 And he had a great analogy, an analogy between the college student and a person sitting in a pool of untapped oil. So think of the person sitting on top of a piece of land that has oil underneath, but hasn't put the money together yet to drill to get the oil. But if he does, he's going to be a very wealthy person. What's the analogy? Well, the college student is sitting on a reservoir of oil in here, in a sense. If he taps it by going to college, he gets a higher income.
- 7.7 We don't generally say that we should subsidize people who are sitting on a pile of oil. Why should we subsidize people who are sitting on a pile of mental oil? He said, at most, that would justify lending the money, but lending the money not at an artificially low interest rate, as student loans are now, but at a real interest rate that reflects risk. So he tells the story. There was this famous black person who worked with the UN named Ralph Bunche. And in fact, the building I was in, the building the economics department I was in at UCLA, was called Bunche Hall, named after him.

- 7.8** Here's a story—this is a quote from the article that I just mentioned by Armen Alchian, "The Economic and Social Impact of Zero Tuition"—he says, "One poor, 'uneducated' resident of Watts, upon hearing Ralph Bunche say that he could not have had a college education unless tuition were free, opined, 'Perhaps it is time he repay us out of his higher income for that privilege granted him by taxes on us Negroes who never went to college.'" So that guy got it.
- 7.9** By the way, I read this in my second year or early third year of college. In Canada, the standard degree was a three-year degree. I was just so taken by it that, along came this commission at the time, just coincidentally, looking into whether we should increase funds for higher education. I basically took Armen Alchian's article, wrote it out in my own words, and submitted it, arguing that they should charge us more.
- 7.10** OK, now, on to Armen Alchian and who he was as an economist. The way I like to put it is, he did rigor without math. If you talk to most economists today, they think that, in order to be rigorous, you have to be mathematical. What I learned the hard way, by writing out answers to problem sets that he assigned in class, is, you can be rigorous without math. I would use language very carelessly, and he'd write in the margin, "no," and he'd explain the exact language I should use. So it's rigor without math.
- 7.11** One of his big contributions was in the area of the economics of property rights. And the short version is, you tell me—him talking— you tell me the rules, and I'll predict behavior. And he wrote, along with a man named William Meckling, this article in 1960, saying, "Differences in goals will not explain differences in the ways individuals pursue those goals." Incentives matter, property rights matter.
- 7.12** And he gave the famous example of the pricing of Rose Bowl tickets. Rose Bowl was a big deal in Southern California; it still is. And those tickets were priced very low, very below market levels. So there was a huge lineup of people to get them. A lot of people would say, "Well, that's crazy." Economists look at that and say, "Why not charge a market-clearing price?" Alchian's explanation was, "Well, we don't have clear property rights. If they charge a higher price, the people setting the price don't get the money. It's the Rose Bowl Association gets the money, and they're just employees of the Rose Bowl Association." So they have an incentive to price low. So there'll be a big lineup, and then they can give some tickets to their friends. And that was—it's kind of obvious in a way, but that was an interesting insight.
- 7.13** One of the students at UCLA, we became friends later, he came in as a student after I left, named Thomas Hazlett, who later became, for a while, chief economist at the Federal Communications Commission, used that way of thinking to understand the FCC's allocation of operating licenses. The FCC gave away operating licenses for radio and TV, rather than charging for them.
- 7.14** Here's what I wrote in one of my books about his insight. "Years later, Tom Hazlett, an economist who also studied under Alchian, used Alchian's insight about the parking

fee." Oh, sorry, the parking fee, not the Rose Bowl fee. They also underpriced parking, so they could hand out parking to their friends at UCLA, "about the parking fee to explain why Michigan congressman John Dingell, who, at the time headed the House Commerce Committee, strongly opposed auctioning off valuable electromagnetic spectrum to broadcasters and other users. Dingell wanted instead to give away the licenses for specific uses and limited periods of time. Hazlett pointed out that, if the government auctioned the spectrum to the highest bidder, the billions of dollars gained by the federal government would not benefit Dingell."

- 7.15** "Moreover, the Federal Communications Commission's power to allocate spectrum would drop to zero. When something is allocated to the highest bidder, the bidders, not the auctioneer, determine who gets it. Therefore, Dingell, who had a big oversight role over the FCC, would find his power over allocation dropping to zero also. Without that power, people wouldn't invite him to dinners and hunting parties as frequently, would contribute less to his campaign fund, and would return his calls less quickly, if at all. In short, he would be a less important man in Washington, and probably a less wealthy one."
- 7.16** Now, back to Alchian and the effect of zero tuition. One of the things he pointed out with zero tuition at government-owned California universities is students, as a result, have very little power. They don't matter much if they're not paying. He also, related to that, introduced the idea of the residual claimant. Let me explain with my dry cleaner example.
- 7.17** Well, first of all, let me say what a residual claimant is. A residual claimant is someone who gets a claim on the residual, the difference between revenues and costs. It's the owner, typically. You have a firm, the residual claimant is the owner. So if you think about a mom-and-pop dry cleaner, typically a very small operation, the residual claimant is the owner. Sometimes the owner is the one you deal with as the manager, or at least is just behind the scenes and very easy to access.
- 7.18** So here's my true story about dealing with the dry cleaner once: I had a very nice shirt. I had worn it only a few times. I took it into the dry cleaner, and when I went to pick it up, it was lost. Being an economist, I calculated what was the leftover value of that, and it wasn't what I paid for it. It was 80% of what I paid for it. So I paid 80 bucks, and I asked them for 64 bucks. And it was just a young employee. What do you think the young employee did? Gave me 80 bucks. Why? Because, and by the way, let's say that hadn't happened. Let's say that person argued with me. I'd say, "I want to see your manager." Let's say the manager argued with me. "I want to see the owner," and probably the owner is the manager.
- 7.19** Why would they do that? Because the owner is the residual claimant, has to worry about the future stream of revenue and the future stream of costs. If the word gets out, and this is way before the internet, but still, I could tell friends in a small area, if the word gets out that they didn't come through, they lose business. So that's a very

powerful incentive. Again, it gets back to property rights. Property rights give very powerful incentives. And one of the things that comes out again and again in Alchian's work is that strong property rights solve problems.

7.20 Let me ask three questions. Should public schools teach evolution or intelligent design? Should insurance companies cover contraception? Should I be able to take off my shoes in your living room? Those don't sound like similar questions, especially the last one. But in a way, they are, if you think about property rights.

7.21 The problem with the public schools is that no one owns them, and so you have conflict between the various groups of parents who want things one way versus the other way. If we had completely private schools, people would pay their money and have their kid go to the school that teaches what they want taught.

7.22 Should insurance companies cover contraception? I have no idea. But with good property rights and no regulations on whether they do, they can decide whether the benefits from covering contraception are worth the costs, and people can choose insurance companies accordingly. My own gut feel is, they wouldn't because contraception is so cheap. It's almost like if you had auto insurance covering your oil change. It really wouldn't make sense.

7.23 Should I be able to take off my shoes in your living room? That one's obvious to everyone. Who should get to decide? You, who owns the living room, and you're going to say, "Well, how much do I value David's company?" If I say I can't take off my shoes, do I leave? So, how much do you value my company? And you trade off that with your discomfort at my having my shoes off. In other words, property rights are the answer to all three of those problems, and that's what they have in common.

7.24 There was an economist, who I became friends with at the University of Rochester, my first academic job, who was later at the University of Chicago, and he hated smoking. And this is in the days when you could smoke in the hallways or in your offices. This wasn't ancient history; this was the 1980s. There was another economist there named Bob Lucas, who, by the way, later won the Nobel Prize. Conversations with Lucas were always really interesting. Bob Barro hated smoking. He didn't want smoking in his office. He had a sign on his office door, "No smoking, except for Bob Lucas." Bob Lucas was almost a chain-smoker, but Bob Barro gained so much from the conversations that he was willing to put up with that. Notice how property rights solved that problem.

7.25 Now, one of the things that Alchian worked on, and we saw this with Gary Becker, with the Chicago school, is how the profit motive makes discrimination costly. Here's what Alchian said: "Discrimination in choosing employees by reason of race, creed, sex, beauty, or age will be more pronounced in not-for-profit firms than in business firms." Because in not-for-profit firms, there's not a residual claimant. There's not someone who gets the profit from getting the most productive person. In not-for-profit firms, there's not a residual claimant. In for-profit business firms, there is a residual claimant.

- 7.26** One of the stories, an economist named Steven Globerman and I wrote a short book on the UCLA School of Economics. And we told the story of a guy named Murray Wax, whom I met back in the '90s, and it's just a beautiful illustration of this. Murray Wax had been a member of the Communist Party in the 1940s. He was a dedicated academic. He was past his communist time, but that was on the record that he'd been a member of the Communist Party. And he got a position teaching at the City Colleges of Chicago.
- 7.27** However, when he went to sign the papers, someone had looked into his history and said—remember, this is just before the McCarthy era, but there was the anti-red era at the time— and they said, "No, we won't hire you." He went to a company, a for-profit company, called the Tony Company, where he used his writing skills to write ad copy, and they had no problem with that. Why? They're trading off. Well, he's a communist. How much is that going to matter when he's writing ad copy for our products? Not much. And he's good.
- 7.28** So here's what he said. I interviewed him about this when I was on sabbatical at Washington University in St. Louis, and I ran into him. I called him up later to get the exact story, and here's what he said: "I had absorbed all this Marxist teaching, but until then I hadn't realized this paradox. The corporations didn't care about my communist background, but academia—which I had thought of as mine—was willing not to hire me for reasons totally unrelated to my teaching ability." That just beautifully illustrates what Alchian wrote about.
- 7.29** And by the way, this harkens back to something Voltaire said centuries ago. It's one of my favorite quotes from Voltaire, and that has nothing to do with the fact that his birthday is the same as mine. He says, "Go into the London Stock Exchange, and you will see representatives of all nations gathered there for the service of mankind. There, the Jew, the Mohammedan [Muslim], and the Christian deal with each other as if they were of the same religion, and give the name of infidel only to those who go bankrupt." In other words, the religion just doesn't matter in this for-profit world.
- 7.30** Alchian wrote an article delving into the data on this, with an economist at the University of Chicago named Reuben Kessel. Here's what they said. They're looking at for-profit firms, but some firms are regulated in the amount of profit they can make. They said, we expect to see a systematic difference, and here's why. "If regulated monopolists are able to earn more than the permissible pecuniary rate of return," able in the sense they could, but then the regulation prevents it, "then inefficiency is a free good, because the alternative to inefficiency is the same pecuniary rate of return and no inefficiency."
- 7.31** So they got some margin to play with because they can't make more than some level. They aren't allowed to. So they thought that, well, this might show up. There's been a reasonable amount of anti-Semitism in the United States over the decades, certainly back then. I mean, Harvard often didn't even allow Jews to be students. But this is just

after that, after Harvard started allowing Jews. They looked at 224 non-Jewish and 128 Jewish MBA graduates from Harvard Business School employed in 10 major industry categories.

- 7.32** The two most heavily regulated categories in terms of profits allowed were transportation, communication, and, of course, public utilities, and finance, insurance, and real estate. Thirty-six percent of the MBAs were Jewish. The representation in those two categories of firms was 18%, in other words, half of their overall representation. What did they compute as the probability of this happening by chance? Less than 0.0005, or less than five in 10,000. So that fit what they expected to find.
- 7.33** Alchian and Harold Demsetz, the other player I talked about, also wrote a piece on landlord discrimination, looking at rent control in Chicago. Remember what I showed with rent control? If it keeps rent below the free market level, which is the whole point, then you're going to have a shortage, you're going to have a lineup, and you're going to have landlords having now easy discretion about who to take and who not to take. They've got a lineup of people. If the person is a white landlord who doesn't like black people, it would have cost him something to discriminate against black people in a free market with no rent control, but it costs him nothing in a free market where he can pick and choose among qualified tenants.
- 7.34** So, here's what they did: they looked through the newspapers in Chicago— rent control was imposed in many major cities during World War II, and Chicago had it. As the war went on, and the rents stayed constant, but there was underlying inflation that wasn't allowed, the rents got further and further out of whack with what the free market rent would have been, so it was easier and easier to discriminate. So here's what they found.
- 7.35** "The percentage of apartment for rent advertisements specifying that the apartment was for rent only on a 'restricted' basis"—restricted in those days, it would say that in the ad, and everyone knew what that meant; it was completely legal in those to discriminate on basis of race—"or only if the renter purchased the furniture"—that was another way around the rent control, say, "Oh, yeah, we'll rent at this rate, but charge a premium for the furniture"—"rose from a pre-war low of 10% to a wartime high of 90% during the period of World War II, when rent control effectively created queues of prospective renters." Again, it fits this point about what happens when discrimination is costless.
- 7.36** Thomas Hazlett, who we've already seen a few minutes ago, used this in an article he did for my encyclopedia on apartheid in South Africa. The conventional wisdom was that apartheid was devised by affluent whites to suppress poor blacks. That was absolutely incorrect. He pointed out that white mine owners were among the strongest opponents of apartheid because it prevented them from hiring lower-wage but productive black workers. Hazlett notes that the white mine owners' "self-interest was

so powerful that it led the chamber [of mines] to finance the first lawsuits and political campaigns against segregationist legislation."

- 7.37** Now, on to more Demsetz and some of his contributions. Demsetz, one of his big things, was on the origin of property rights. "A primary function of property rights is that of guiding incentives to achieve a greater internalization of externalities." I talked about externalities earlier. An externality, and we usually talk about negative ones, is a cost imposed on someone not taken into account by the person imposing it. And internalizing it means somehow having that person taken into account.
- 7.38** "Every cost and benefit associated with social interdependence is a potential externality. One condition is necessary to make costs and benefits externalities. The cost of a transaction and the rights between the parties (internalization) must exceed the gains from internalization." In other words, if it's too costly to internalize, they don't get internalized.
- 7.39** With that thought, he then went and looked at the literature on Aboriginal Canadians and Native Americans, and he found some really interesting literature in the anthropology literature. This is interesting because, if you talk to most economists today, they'll put down sociologists, they'll put down anthropologists, but especially anthropologists often have a lot to say about history that economists otherwise are pretty ignorant of.
- 7.40** And he found this anthropologist who wrote in 1915, a guy named Frank Speck, who, Demsetz's quote, "had discovered that the Indians of the Labrador Peninsula in Canada had a long-established tradition of property and land." Another anthropologist named Eleanor Leacock studied Labrador Indians and wrote up her findings in 1954. And Demsetz wrote, "Leacock clearly established the fact that a close relationship existed, both historically and geographically, between the development of private rights in land and the development of the commercial fur trade."
- 7.41** Let me go on to say what Demsetz says about that. I said Demsetz beats Garrett Hardin to the punch. I'll explain that in a minute. Demsetz said in this 1967 article, "Because of the lack of control over hunting by others, it is in no person's interest to invest in increasing or maintaining the stock of game. Overly intensive hunting takes place. Thus a successful hunt is viewed as imposing external costs on subsequent hunters—costs that are not taken into account fully in the determination of the extent of hunting and of animal husbandry." What's the solution for that? Property rights. So if the various Indian tribes can just say, "This is our area; we're hunting this area. This other area is yours; you're hunting that area," then they're doing what's called internalizing what would have been an externality and solving this problem.
- 7.42** There was a famous scientist, biologist named Garrett Hardin, who wrote an article in Science, the magazine, in 1968, called "The Tragedy of the Commons." It's one of the most cited articles ever from Science magazine. He basically says that if you don't have

private ownership, you're going to have overuse of the commons, of what's in common. Because if I hold back in order to let, say, the grass grow so the cattle can graze, well, you're going to have your cattle graze, so I'm not going to hold back. Well, what's interesting is that this quote I just gave from Demsetz is saying the same thing. Garrett Hardin gets credit for this. I had him do a piece on this for my encyclopedia. And when my co-author and I researched this book, I realized, "Oh, my god, 1967 is before 1968." So it was Demsetz, really, who came up with this.

- 7.43** And so, here's what Demsetz said on Leacock, on the fur trade: "The geographical or distributional evidence collected by Leacock indicates an unmistakable correlation between early centers of fur trade and the oldest and most complete development of the private hunting territory." In other words, with the fur trade being so valuable, and it being relatively easy to section off, "This is your area, this is my area," they actually had property rights.
- 7.44** He then compared that—he found an anthropologist comparing that—to the American Southwest. Tribes agreed to hunt in their own well-defined areas. Since furry animals aren't migratory, the agreed-upon territorial rights had value. They don't move around a lot. Conversely, grazing animals in the Southwest wandered all over the land, so that territorial rights didn't have as much value. Put differently, in the Southwest, the costs of enclosing grazing animals in a specific geographical area were prohibitively high. And this was before barbed wire was invented. Barbed wire came along just after the Civil War, and that completely changed things. It made it much easier to establish property rights.
- 7.45** Another area in which Demsetz contributed to economic thinking was on the issue of concentration and profits. Let me give a little background. The leading school of thought from, say, the '50s to the late '60s, in the area of economics called industrial organization, is— let me tell you the empirical regularity, and then tell you what the thought is about that. The empirical regularity was that industries that were highly concentrated, meaning the top four firms had a large percent of the market, tended to have higher profit rates.
- 7.46** The leading view on that at the time was that it makes sense because they have more market power. They're colluding either implicitly or explicitly, and so they're using this market power to get higher profits. Demsetz challenged that view, and it was the result of one conversation he had at the Quadrangle Club. That's the faculty club at the University of Chicago. He was at Chicago for a large part of his career before he went to UCLA.
- 7.47** Someone in this conversation pointed out that, in the auto industry at the time, this was in the '60s—as you know, there were three firms, General Motors, Ford, and Chrysler—General Motors was the only company making serious money. At the time, General Motors was by far the largest auto company in the United States. So that got Demsetz thinking. If they're the ones making money, and of course, it's concentrated,

and they're large, they're a large part of those three, the overall profitability in that industry is going to be high, because they're such a big part of it. But if this other view is correct, that there's implicit monopolizing going on, why aren't the other two firms making good money?

7.48 So Demsetz came up with an alternate hypothesis. His alternate hypothesis was, maybe General Motors is doing a good job. Ever heard of that? Maybe they're getting economies of scale. Maybe they're figuring out how to do things better, maybe a slightly better product. I still remember this TV show, I think it was on PBS, I saw in the late '70s, where Henry Ford II, who was managing Ford Motor Company, was asked why General Motors was doing so much better than Ford. He said, "Well, it's obvious. They've got a better product." Later evidence that Demsetz found and that other people found supported that insight.

7.49 One of the big issues in economics is the role of antitrust. Antitrust came along with the Sherman Act in the late 19th century, and it basically has restrictions against monopolizing an industry. And whereas most economists favor antitrust and don't have much skepticism about it, Demsetz was not one of those. Once he came up with what he just came up with, that I talked about, he started thinking, "Well, if they go after companies for making high profits, maybe they're going after them for being successful, and it's not a result of monopolization."

7.50 So here's what he said on that issue in 1974: "I do not suggest that we abandon the search for private conspiracy, but I do think it is time to pay much less attention to the structure of industry and virtually no attention to the notion of nongovernmental barriers to entry. A commitment to the machinery of competitive organization requires that we generally accept the consequences of effective competition. For antitrust, this means that market share and profits can be expected to shift in favor of successful rivals."

7.51 In other words, profits are not necessarily the result of monopolization. Profits are a return to doing things successfully. In competition, sometimes someone wins. So, he's been one of the skeptics of antitrust. And by the way, he and a whole lot of other people who've written in this area had an effect. Antitrust was really at its peak in terms of enforcement in the 1960s. This work comes along in the '60s and '70s.

7.52 And an academic entrepreneur, who I'm not going to talk about in another context, he's just one-of-a-kind, named Henry Manne, who read the work of Demsetz, knew Demsetz, knew all these people, knew Alchian, started teaching economics to federal judges. And this is among the things they taught them. You have an antitrust case come along. You're a judge. You were a lawyer before. You were never an economist. What does the economics say?

7.53 They were successful in the sense that federal judges now often will look at more the economics to figure out, is this really a monopoly, or is this the result of successful

competition? There was a recent case just in the last month where Judge Boasberg, B-O-A-S-B-E-R-G, turned down a suit—and I'm forgetting the company. It might have been Meta; it was a major, major company—saying, "Look, yeah, they're making good money now, but they're doing it by doing a whole bunch of things right. That is not evidence of a monopoly." So this has had a major, major effect.

- 7.54** One of Demsetz's other contributions was his discussion of what he called the nirvana approach. Let me give the background on that. Economists often find what they think is a problem with the free market, and then conclude that the government should step in. And what they often do is compare the actual market with the ideal government, rather than with the actual government. Demsetz called that the nirvana approach; assume that government can create nirvana.
- 7.55** By the way, economists—this has become a very widely used idea now. People call it the nirvana fallacy. That's mistaken. It's the nirvana approach. But the nirvana approach consists of three fallacies: the "grass is greener" fallacy, the "free lunch" fallacy, and the "people could be different" fallacy.
- 7.56** The "grass is greener" fallacy says, "Oh, the grass here, the free market, we just assume that the grass over there, with the government regulation or whatever they're going to do, is greener, but we just assume it. We don't show it." By the way, we're going to talk about this when we talk about public choice: if you look at why we get this result in the free market that some people don't like, it has to do with incentives. Maybe there are externalities not being internalized or whatever. Well, when we look at government officials, don't they have incentives? Don't they respond to incentives? And are those incentives things that drive them to the right conclusion? So, for example, if they have some government regulation that causes some problem, do they lose money? No, they might not even lose their job.
- 7.57** The second fallacy is the "free lunch" fallacy. So one of the problems that economists often point out is a problem with markets, is the absence of certain kinds of information. Well, information is costly. So they're assuming the opposite of TANSTAAFL. They're assuming there is a free lunch, and there's not. And when the government does something with regulation, they're going to have to figure out the information, and they also have a cost to getting the information. And by the way, going back to incentives, they have less incentive to get good information.
- 7.58** And finally, the "people could be different" fallacy. In other words, I see this problem, and I think there's a government solution, because people should be different than they are. And so, a standard one is, "I think"—and this is before Social Security came along—"I think people aren't saving enough for their old age. They should be different. They aren't. So I'm going to make them save for their old age," and now we get the problem we have with Social Security.

- 7.59 So in a way, the nirvana approach would be like the judge in a skating contest between two performers, performer A and performer B. The judge sees performer A alone, and not B, and on that basis, awards the medal to performer B. That would make no sense, but that is really what people do. And it's not hard to find examples. In our chapter on this, in our book on the UCLA School, which, by the way, the PDF is free online at the Fraser Institute, we give examples of people who literally, and these are smart economists who literally write about how we should do this, but they don't have any guideline about why government would do the right thing, relative to what the market is doing.
- 7.60 The next economist I want to talk about is Jack Hirshleifer and his work on recovery from adversity. He was writing a lot of work in the late '50s and early '60s for the RAND Corporation. The RAND Corporation was a few miles away from UCLA. Here's why this mattered at the time: people were really worried, rightly so, about the effects of nuclear war. And a lot of people said, "Well, if we had nuclear war, that's the end. It's all over, if, say, a city gets hit, and half the city is wiped out or whatever." And Jack Hirshleifer said, "Hm, I wonder if that's true."
- 7.61 Here's what he found. The historical—let me just say, I'll get to his examples. He said, "The historical review here suggests an extraordinary resiliency of human populations and social structures. It is, of course, impossible to prove that social breakdown will never occur in the aftermath of disaster, especially when we contemplate the unprecedented catastrophe of nuclear war. But the lurid picture of post-disaster regression to savagery, that staple of fiction and of popular thought, can draw no support from the historical record."
- 7.62 In other words, if you look at novels and films and so on, it's all dog-eat-dog after the thing hits. He goes, "Well, we had some pretty adverse things in Germany after World War II, with Dresden and with Hamburg just being wiped out. How did that work?" And he went into detail and looked at it. Actually, there was an amazing amount of cooperation within days, getting things going again, getting the streetcars going, people trading and people working with each other.
- 7.63 And then, he also looked at the Soviet Union after war communism. So Lenin takes over in the second revolution, the October Revolution. The first one that overthrew the Tsar was, of course, not by the Bolsheviks, but the Bolsheviks then overthrew the Mensheviks, and took over and had what was called war communism, complete government control of the economy. It was a manmade disaster. In 1922, industrial output was only 20% of its level in 1913.
- 7.64 So Lenin says, "We're not going to last long if we do this." So they put in what's called the New Economic Policy. It ended compulsory requisitions of food, legalized private exchange, abandoned central planning, ended conscription of labor. And by 1923, industrial output was 75% above its 1920 level, and agricultural output was up by 17%. So if you have private property, relatively free markets, no conscription of labor, not a

lot of central planning of the economy, the economy can recover quickly. In my last talk, I'm going to talk about Germany after World War II, which is a great example of all this.

- 7.65** By the way, none of this is to say that Jack Hirshleifer was saying, "Oh, nuclear war, no problem." He was just saying, "Let's avoid it. But if we get this bad thing, let's not just say, 'Oh, it's all over.' Let's figure out how to maybe don't have price controls, enforce private property and so on, and we'll recover more quickly."
- 7.66** The final economist I want to talk about from the UCLA School is Benjamin Klein. I took his course, and this was the first time, I think, that I saw an economist who took a serious look at the idea and the role of advertising, because if you see the literature at the time, in the '60s and '70s, when I was studying in the '70s, people dumped on advertising: "What's the point of advertising sex when you're selling cars or all these things that don't seem to have much to do with the car?"
- 7.67** Klein had an answer to that. Here was his answer: "Advertising can be seen as a stream of sunk-cost investments that will pay off for a company only if that company stays in business long enough and can charge a sufficiently high price to recapture those investments. To stay in business for a long time, the company needs to deliver products whose quality is commensurate with the prices charged. A company seeking to cut corners and cheat on quality runs the risk of severely damaging its brand name."
- 7.68** In other words, what they're doing with advertising, whatever is the content of the advertising, the advertising is expensive. So if we're trying to build a brand name with expensive advertising and we screw up, we've lost a lot. And so, the fact that we are sinking these costs, these non-recoverable costs, into advertising, and we're communicating that to the market, people in the market say, "Oh, I guess they're expected to be around for a long time." And so, they will buy from them, and the company will try to stay around for a long time. So that, in a nutshell, is the UCLA School.
- 7.69** So, going back to discrimination in your story with Murray Wax.
- 7.70** Yes.
- 7.71** And just how it did not pan out with the history, not hiring him because of his Marxist background in history and beliefs. But now he should have been hired due to his—
- 7.72** Ability.
- 7.73** Ability. Right. So currently, we have DEI, which is pretty much the same thing. I've always thought it wasn't good, just because we're not hiring for race, sex, abilities, and such. Instead, maybe create jobs with positions that align with the different abilities, if that's the case, or whatever their talent is. I'm agreeing with the economists who overall think that we get better results and profitability from companies that hire the

right people that align with the right jobs and everything. My question is: how do current economists look upon DEI?

7.74 How do current economists look upon DEI? I haven't seen polling data on that. I can tell you how economists who are against discrimination on legal or gender grounds look at it, which is that it's a bad idea, because it says you're supposed to hire to some extent based on gender. You're supposed to hire to some extent based on race. And that's a step backward from where we—it's backward in a different way. It's a different version of Jim Crow, in a sense, where now the discrimination is against white people and white men in particular.

7.75 So I think that's how you would look at it, if you don't want discrimination based on race or gender. But I don't know. I haven't seen polling data. And part of why I think I haven't is a lot of people are afraid to speak out. If you look at academia, it's still stuck in this thing where it can be damaging to your career if you speak out about it. Anyway, that's my answer in a nutshell.

Lecture 8: The Hockey Stick

At a glance — 'The hockey stick' — Joel Mokyr and long-run growth.

- 8.1 In this lecture, I want to talk about what I call Joel Mokyr, the Industrial Revolution, and the hockey stick. This is not my Canadian heritage showing up. This is something very different. Economists often talk about the hockey stick.
- 8.2 Back in 1996, I made a deal with The Wall Street Journal to try to save them a 24-hour cycle. I said I could wake up in the early morning on the day the Nobel Prize in economics was announced and know within a couple of hours whether I knew enough about the winner to write up a piece and send it to them that morning. I've done it successfully, I think, 24 out of the last 30 years. I did it for Joel Mokyr. I had heard about him. I knew about his work. Almost everything I know about him, I learned that day or in the three months since. Because he won the Nobel Prize along with two other people, and I wrote up what he won it for. And it's very exciting to me, and I'll tell you why.
- 8.3 Economics has become very technical, very mathematical, a lot of theorems and proofs and stuff like that. Joel Mokyr did none of that. He was so surprised when he got that phone call early in the morning in Chicago. I think I read that he told someone he thought it was a prank because he liked his work. A lot of economic historians liked his work, but it just didn't seem to qualify. So it was just very exciting to me that he won it.
- 8.4 I want to lead with a quote that puts the whole couple of centuries of economic growth in perspective. It's from a woman named Rose Wilder Lane. Here's what she said: "Why did men die of hunger for 6,000 years? Why did they walk and carry goods and other men on their backs for 6,000 years? And suddenly, in one century, only on a sixth of this Earth's surface, they make steamships, railroads, motors, and are now flying around the Earth in its utmost heights of air? Why did families live thousands of years in floorless hovels without windows or chimneys? Then in 80 years, and only in these United States, they are taking floors, chimneys, glass windows for granted and regarding electric lights, porcelain toilets, and window screens as minimum necessities." This is from a book she wrote called "The Discovery of Freedom: Man's Struggle Against Authority." So clearly, she's attributing it to freedom. And that leads to what I call the hockey stick.
- 8.5 This gives the growth in world GDP per capita from 1000 to the present. If you look, you're moving along from the 11th through the 17th centuries without much happening. In the 18th century, and it's really in the late 18th century, you get a little blip. In the 19th century, you get a large increase. In the 20th century, you get a huge increase. That's why it's called the hockey stick. That increase, that line in the 20th century, is the blade of the stick.

- 8.6** There was an economist, an Austrian economist, even though he was not thought of as part of the Austrian School, who wrote a book called "Capitalism, Socialism, and Democracy," in the 1940s, a guy named Joseph Schumpeter. Here's what he said. "Queen Elizabeth I," the one in the 16th century, "owned silk stockings. The capitalist achievement does not typically consist in providing more silk stockings for queens, but in bringing them within the reach of factory girls in return for steadily decreasing amounts of efforts." In other words, normal people can now buy them, and buy them by using less and less of a percent of their income.
- 8.7** Let's look at life three and more centuries ago. Meat—that was a luxury. Tea— when economic historians want to track standards of living, they look at how commonly people use tea and how much of it they use. Tea was a luxury. There's an expression, "raining cats and dogs," and I found different sources on this. The one that seemed most likely is that, when it rained very hard, cats and dogs just got carried down because they didn't have shelter. And most seriously, life expectancy in those days was between 33 and 40 years. 33 and 40 years.
- 8.8** So think about wedding vows, "Until death do us part." Pretty different commitment back then. Now, it is true that a large part of the reason for that was child— deaths of babies or deaths of young children. Edward Gibbon, the famous historian who wrote "The [History of the] Decline and Fall of the Roman Empire," had five brothers named Edward. Why? Because the parents thought that, in the normal course of things, most of them would die, and they wanted the name Edward to carry on.
- 8.9** Here's the interesting thing: Schumpeter, in that quote about Queen Elizabeth, understated progress. Queen Elizabeth didn't have it so good either. What did she lack? Hot showers, high-quality food, fruits and vegetables in winter, good medical care, including antibiotics, the ability to talk to people 10 miles away, let alone 3,000 miles away. She lacked all of that.
- 8.10** There's a writer on travel named Bill Bryson. I love his work. I don't know if any of you have read it. He wrote a book. He wrote a book called, I think it's called "At Home." And I wrote a book review of it, and the title they gave it was "Home Economics." And he goes room to room in this old house, talking about how things have changed over a couple of centuries.
- 8.11** Here's what I said about his chapter on the bedroom: "It is impossible to read Bryson's chapter on the bedroom without emerging with an appreciation of economic growth and modern medicine. At inns, strangers often shared beds into the 19th century. And diaries frequently contain entries lamenting how the author was disappointed to find a late-arriving stranger clambering into bed with him." Can you imagine? "He tells of a squabble in 1776 between Benjamin Franklin and John Adams when they shared a bed in New Brunswick, New Jersey. The issue they disputed was not the role of the federal government. It was the far more important question of whether to have the window

open or not. And with the increasing wealth, of course, people no longer had to share beds."

- 8.12** I want to give you a feel for what Brad DeLong, in a piece called— he's a Berkeley economist—a piece called "Cornucopia," shows about the standards of living. It's just amazing. What he does is, he looks at the 1895 Montgomery Ward catalog and various items you could buy—well, I'll show you—various items you could buy. So, he looks at what the average earner could make per hour, and then how many hours of wages would it take to buy various things in 1895. And this is out of the Montgomery Ward catalog.
- 8.13** So, six volumes of Horatio Alger stories. It would take 21 hours to earn enough to buy them. And then what he does is look at what would it take for the average wage in the year 2000. And for that, it would take six-tenths of an hour, so not even an hour at the average wage to make enough to buy those books. So that helps him come up with something called a productivity multiple of 35.
- 8.14** A one-speed bicycle: 280 hours in 1895, 7.2 hours, less than a day, in 2000. A cushioned office chair: 24 hours in 1895 versus 2 hours, and so on. And you go down, and there's only one item that costs more in hours: a sterling silver teaspoon, 26 hours then, 34 hours now or in 2000. But here's the interesting thing, even with that one: why did you want a sterling silver teaspoon? You wanted it because you were worried about rust. Well, we have way cheaper ways f rust. Look in your cupboard, and you have things that aren't sterling silver for knives and forks, but they're not rust, they don't rust. So even that, there's been an improvement.
- 8.15** Why did we get that growth, starting in the last part of the 18th century and then hugely through the 19th century, and then even more hugely through the 20th century? The answer, and this is Joel Mokyr's answer, is the Industrial Revolution, which really got going in Britain more than anywhere else. And here's what these two authors, who I discovered after I wrote my piece in The Wall Street Journal, said very well: "The Industrial Revolution was a clustering of macro-inventions leading to an acceleration of micro-inventions. Two examples of macro-inventions are steam engines and smallpox vaccination. An example of a micro-invention is James Watt's separate condenser, which made steam engines cost-effective for industrial establishments. Why was Britain such a leader in industrialization? It had a comparative advantage in producing micro-inventions."
- 8.16** Now, some people talk about the age of this, the age of that. Mokyr quotes an economic historian, Deirdre McCloskey, who at the time was Donald McCloskey, and here's what McCloskey said: "The Industrial Revolution was not the age of cotton or the age of steam. It was the age of improvement."
- 8.17** So in his work, in his various books and articles, Mokyr delves into, we still want to know more. How did this happen? He comes up with the distinction between two types

of knowledge, what he calls propositional knowledge and prescriptive knowledge. Propositional knowledge is why things work. Prescriptive knowledge is how things work. In the Industrial Revolution era, there were positive feedback loops between the two. Mokyr says, "A virtuous circle was impossible in earlier days of engineering without mechanics." In other words, you had both of them working together. "Iron-making without metallurgy," so metallurgy is the science of it, and you need that as well. "Farming without organic chemistry, and medical practice without microbiology."

- 8.18** The two central drivers of the Industrial Revolution were the Industrial Enlightenment of the 18th century and the Scientific Revolution of the 17th century. And here's what those two authors who wrote on Mokyr, said about the Enlightenment: "The Scottish Enlightenment instilled the ideas that nature could be known and mastered, and that its laws could be elucidated through scientific research. At the same time, the proliferation of learned societies, periodicals, and informed intellectual networks fostered the spread of practical knowledge linking 'high' science and 'low' craft."
- 8.19** What Mokyr highlights is the importance of freedom. Britain had more freedom than the other countries on the continent. Governments were relatively tolerant of dissenting views and less censorious, less engaged in censorship, than regimes on the European continent. Mokyr writes about this in a book called "A Culture of Growth." The emergence of pan-European networks of knowledge creation and diffusion— that was an important part. Universities played a key role. The invention of the printing press reduced the costs of codifying and circulating knowledge.
- 8.20** There was something I hadn't known about before I did the research that morning for The Wall Street Journal article, the Republic of Letters. I'd never even heard of it, but it was this transnational community of scientists and craftsmen. So there was lots of pollinating of ideas, not just within Europe—within Britain, but also within Europe and across between the European continent and Britain. Very huge. It was very important that people be able to do these things across national boundaries.
- 8.21** At the same time, political fragmentation in Europe, that made it difficult for European governments to crack down. If you're in a city where the government is trying to crack down, that city government has a lot of power, but the next city over, they don't have as much power, or they're not cracking down, and you move over there.
- 8.22** Now, compare that to China, which cut itself off from the rest of the world. If you had been a Martian and you had come down to Earth in the 14th century and went around country to country and said, "Which country is going to make it big?", you probably would have said China. They were doing so many things right in terms of invention. And then their government or governments completely walled them off from the rest of the world, and they just lost out. So a huge part of what makes this work is cosmopolitanism, is sharing of ideas and goods and people across borders.

- 8.23** So there was an interview with Mokyr in 2009, and I want to quote some great quotes from that interview: "For the economy to function well, you don't just need good property rights; you also need what we would call, somewhat vaguely, economic freedoms. You need labor mobility; you need to get rid of guilds." Guilds were like occupational licensing. In order to practice a particular profession or a particular job, you needed the permission of people in that guild. "You need to get rid of monopolies." We talked about that, whether it be the Jones Act or other monopolies, both local and global. "You need to get rid of all kinds of regulations."
- 8.24** And here's what he says. "And above all, you need free trade." And then he said, "The more open the world is— the more free trade, the more ideas and people can move from one country to another—the less likely it is that technological progress will come to an end." He is concerned about whether it will not necessarily come to an end, but slow down, and part of it has to do with things like restrictions on immigration. We might be losing out on a lot of good scientists coming from China, for example.
- 8.25** The guy who invented Zoom came over on an H-1B visa from China to Silicon Valley. He applied for a green card nine times before he was accepted. I guess nine is the charm. And what would have happened if we hadn't had him? It doesn't take that many very productive immigrants to make a huge impact. So it is true that the immigrant who is running a restaurant or doing gardening or doing hairdressing or taking care of kids, those are all important, too. As far as technological change, these ones are really important.
- 8.26** So the moral of the story is that freedom mattered, and freedom matters. Now, we think of universities as being the place where ideas are generated. But I've got to ask this question: are universities—and the ones I know best are in America—are universities now places where people are free to pursue the truth without being censored or fired? And there's a lot of evidence that you can get in trouble for pursuing the truth.
- 8.27** I was at that Thomas Sowell event in October, where we were celebrating his work, and one of the speakers was a guy named Roland Fryer, an economist at Harvard University, a black economist at Harvard University. And he had won the John Bates Clark Medal. The John Bates Clark Medal is given once a year to the person who—I think it's in America, but half the economists are in America—the person who's the most promising economist under the age of 40. So it's on the level of the Nobel Prize. He's a curious person. That's the thing; if you're going to pursue research, you've got to be curious. You can't go in knowing what you want to find. You can go in thinking what you want to find. You can go in hoping what you want to find. But you've got to look at the data.
- 8.28** As I said, he's a black man. He did some research on whether police differentially killed, disproportionately killed, black people. I'm pretty sure he expected to find the answer was yes. He found the answer was no. He did find that police differentially treated

black people badly, but not killing them. And there was a big uproar at Harvard, where he was, because of that. And then they discovered that he might have been involved in some kind of sexual harassment, which hadn't been talked about before he came up with that result. They really cut a lot of his funding at Harvard, and so this kind of stuff happens.

- 8.29** So again, the moral of the story is that freedom mattered and freedom matters. And that, in a real nutshell, is Joel Mokyr, the Industrial Revolution, and the hockey stick. Thank you.
- 8.30** In the 14th century, why did China cut itself off from the rest of the world?
- 8.31** In the 14th century, why did China cut itself off from the rest of the world? Real fear, real paranoia about foreign influences, and would they lose their culture? Would they lose various things? So it's not that uncommon for governments to want to do that. They were the extreme in doing that.
- 8.32** We need free trade. We keep hearing that over and over. What are the potential setbacks with free trade? And we were talking about this yesterday, but with free trade, how would you balance that out? For example, if we were wanting to get something from China, and it's more than what they want from us, would that be balanced? How would we make that work?
- 8.33** So, free trade, I'm going to answer about free trade in two parts. One is, are there downsides of free trade? Yes. But the downsides are no different from the downsides of things happening domestically. So, for example, China produces this very cheap electric vehicle. If the US government allows it in without tariffs, that will hurt a lot of domestic companies producing electric vehicles, and some of those companies will lose jobs. But if some producer domestically comes up with a way of producing electric vehicles more cheaply, other companies here lose market share, lose jobs. So there's no difference in principle or effect between the effects of free trade across borders and the effects of free trade within a country.
- 8.34** And now I've forgotten your second question. It was [inaudible], oh, yes. So one of the things people worry about, that economists typically don't worry about, is whether we buy more from a given country than they buy from us. I have this huge trade deficit with Safeway. I spend about \$5,000 a year on Safeway. You know what those bums spend on my speeches and articles? They spend zero. I've got a \$5,000 trade deficit. Why do I not worry? Well, everyone understands why I don't worry. They buy other stuff, and I sell my stuff to other people.
- 8.35** Well, similarly, it doesn't matter whether you have a trade deficit with a particular country. We have trade deficits with China and Canada and Mexico, but we have trade surpluses with a bunch of countries like the Netherlands, but it doesn't matter. Now, it is true that, if you add up our trade in goods and our trade in services, we have an overall trade deficit of about just under a trillion dollars a year. But where does that

go? When they end up with that extra trillion, they invest it here, they buy US government bonds, they buy property, they buy companies, they make direct investments, setting up factories.

- 8.36** In 1988, Michael Dukakis was running for President. One of his big parts of his platform was to try to somehow crack down on foreign investment in the United States. He was really concerned about that. He went to a plant in a suburb of St. Louis, and he gave a talk about this. He said, "I know you don't want to be working for a foreign company, and we're going to try to make sure that doesn't happen very much." And some of the employees looked around quizzically. It turned out that what he didn't know was that they were working for a foreign company. Why is a foreign employer necessarily bad? I mean, they did all right with a foreign employer. And so, it's a lot of fear of the foreigner to some extent, which I think is unjustified.
- 8.37** Do you foresee technological progress coming to an end to some degree? And are you talking about more like AI and that kind of technological, or is it all of them, everything in general?
- 8.38** Do I see technological progress coming to an end? And what is the nature of modern technological progress? No, I don't see it coming to an end. I've always been kind of a glass-half-full person. I see ways that it will continue. Now, could the United States lose some of the edge if we regulate too much, if we don't allow more immigrants? Yes, but then it can happen elsewhere. But that's the great thing. A technological improvement, wherever it happens, we gain a little more if it happens here, but we gain almost as much if it happens somewhere else, because we get to still buy it or use it.
- 8.39** AI, of course, is the one people are talking about a lot now, and I do fear that it will be regulated in certain ways. There are these people who are thought of as being serious people—I just read one the other day, and I can't remember the name—who was saying that if we don't get our hands on AI and control it, it's going to take over within a decade or two and kill us all. Kill us. I don't mean metaphorically. I mean, it's going to be like Hal in 2001: A Space Odyssey, where it's going to just say, "OK, sucker, we don't like what you're doing. You're dead."
- 8.40** Now, do I know enough to say that can't be? No. But do I have enough native skepticism to say that's really, really unlikely? Yes. And by the way, if a government takes it over, what's to stop the government from using AI to kill us all? We do have a lot of experience with government killing people like the Holocaust, like Mao's China, and so on. And so, yeah, I don't fear AI the way a lot of people do.
- 8.41** I think that the median position of people who work with AI is more like mine. In other words, we can use it as a tool, and it's going to be very effective. So, yeah, I think technological progress will continue. I think Elon Musk isn't done yet, for example, and we'll get more from him. We'll get more from other people. So, yeah, I think it'll work out pretty well.

Lecture 9: Public Choice School

At a glance — The Public Choice School.

- 9.1 So far, we've seen a number of schools of thought, the Austrians, University of Chicago, the Chicago School, the UCLA School, and also the mainstream people who aren't really part of a school.
- 9.2 This is the final school I want to talk about: it's called the public choice school. Now, the public choice school is not free market per se. But if you look at their reasoning, their reasoning leads people to be more skeptical than otherwise about government intervention, and therefore, implicitly more favorable to economic freedom and free markets. The two most prominent members of the school of public choice were James Buchanan and Gordon Tullock, both of whom died.
- 9.3 Public choice is the application of the tools of economics to political behavior. James Buchanan, who won the Nobel Prize in economics in 1986 for his contributions to public choice, called public choice "politics without romance." In other words, I don't know if you've ever seen "The American President." I love that movie, but I love it as a romance. I don't love it in the sense that it's telling me much about the effectiveness of government policy in a good way. I even wrote a piece on that after the director was murdered very recently.
- 9.4 So let me say more about the public choice perspective versus the traditional perspective. Many economists, noting ways that free markets fail to deliver what they want, advocate giving more power to government. Most of them take what Harold Demsetz calls the nirvana approach, which we saw earlier. Public choice scholars say, "No, we assume that producers and consumers in the private sector are self-interested. Let's assume the same about politicians, bureaucrats, and voters."
- 9.5 Are we able, by assuming that and applying the standard tools of economic analysis to actors in the political sphere, to predict and understand behavior, and especially important, our pillar number two, incentives matter, and pillar number five, information is costly and valuable? These scholars who've looked at that, their basic answer is, "Yes, we can understand a lot of behavior that way." There are exceptions that I'll talk about, especially having to do with voters, but there's a lot of behavior that we can understand.
- 9.6 Here are two major insights from public choice. First, the individual becomes the fundamental unit of analysis. Public choice rejects the construction of organic decision-making units such as the people, community, or society. Bastiat, who I talked about, who is, in a way, an early public choice person without that term being used, once said, when someone told him society wanted something, he replied, "Bring society to lunch, and we'll talk about it."

- 9.7 Second, public and private choice processes differ not because the motivations of the actors are different, but because of stark differences in the incentives and constraints that channel the pursuit of self-interest in the two settings. For example, imagine a person buying a house. That person will look at the benefits of the house, what surrounds the house, what the neighbors are like, what the roads are like, what the schools are like, and takes all of that account in deciding whether to buy the house, versus that same person deciding whether to support a dam paid for by taxes, so they can get more water.
- 9.8 Well, now that person might come up with a decision. "Hey, I want the dam because my taxes for it will be less than the benefits." But other people might not want the dam because their taxes are greater than the benefits. So it's a very different choice. And that person will then probably look narrowly at whether, does the dam benefit me on net? Whereas with the house, it's, does the house benefit me on net? But the house is the whole decision. The dam is a bigger decision, and he just looks at how it affects him.
- 9.9 Now, one of the really interesting things that came out of public choice is what's called the median voter theorem. And the median voter theorem explains why the two main political parties are usually similar. It says this: if voters are fully informed, if their preferred outcomes can be arrayed along one dimension, for example, left to right, if each voter has a single most preferred outcome, and if decisions are made by simple majority rule, then the median voter will be decisive. The policy that wins is the one the median voter favors. This insight was from an early public choice person in 1958 named Duncan Black.
- 9.10 So, take a recent example. Kamala Harris and Donald Trump look pretty different on many issues. But look at two of the big issues: Social Security and Medicare, which together take a huge percent of the federal budget. Did they disagree on that? No. Neither of them wanted to do anything to trim Social Security and Medicare. As a test of the median voter theorem, let's see what happens when one of the major party candidates—the two-party system we have here—when one candidate positions himself far from the median voter. I'll take two examples.
- 9.11 One is Barry Goldwater versus Lyndon Johnson in the 1964 presidential election. Goldwater was more extreme on the right. LBJ was not more extreme on the left. So let's say we draw a bell curve of people's preferences, and it looks something like this. The median voter is here. And one of the big issues in the 1964 election, a huge issue, was Vietnam. It was early. Should we intervene in Vietnam? Goldwater was extreme on that. So Goldwater is, say, right here. In his view, I'll call it AuH₂O, Goldwater.
- 9.12 Now, if LBJ wants to do well, where should he position himself? A little this way, enough to look different enough, but then to capture a whole lot of the vote. So LBJ was somewhat hawkish. And the line at the time, the famous joke at the time, was this voter saying, "I was told if I voted for Goldwater, we'd bomb North Vietnam. So I voted

for Goldwater, and sure enough, we bombed North Vietnam." Goldwater lost. That was a joke. So Goldwater got— sorry, LBJ got 61% of the vote, and Goldwater got 38.5%.

9.13 We saw something similar with Nixon versus George McGovern in 1972. There was a huge difference. Well, McGovern was very left on economic policy. He wanted something called "demogrants," where he gave everyone \$1,000, which was a lot of money in those days. So again, the bell curve. Here's McGovern. Nixon's best strategy is to go different enough to look different and motivate people to vote, but not way different. And so he collects all those votes. So sure enough, Nixon got approximately 60% of the vote, 60.7%, and McGovern got 37.5%. So the median voter holds up. It's not a sure thing, but it holds up some of the time. It holds up when there's one main issue, let's put it that way, and two parties.

9.14 One of the other things that has come out of public choice is the issue of special interests, concentrated benefits and dispersed costs. We have sugar import quotas in this country that are very tight. In other words, our federal government limits the amount of sugar that can be imported from other countries. We have tighter import quotas than many other countries have. Of course, that makes the price of sugar domestically higher than otherwise. Our price of sugar is roughly two times the world price of sugar.

9.15 Now, 340 million consumers pay an average of \$8 more each for sugar in a year. We lose, in other words. A handful of producers gain tens of millions of dollars each because the quotas drive the price higher, and those producers make out. Economists have shown that the loss to consumers is substantially higher than the gain to producers because we're getting very high-cost sugar, instead of cheaper sugar from other countries. So it doesn't pass a cost-benefit analysis test.

9.16 Why does it come about? Because the loss to each of us consumers is so low, we don't care much, and we probably don't even know about it. As a result, our voice in this controversy is a whisper, at best. When I've taught this, and I've asked my students, "Who knows what the quota in sugar is from some country?", they don't know the answer. Of course, they don't know the answer. It doesn't matter to them. But if you ask some of this handful of producers what the quota is, they'd be able to tell you.

9.17 By the way, one of the ways the inefficiency shows up: Coca-Cola. They don't use sucrose. They use— what do they use? Corn syrup. Coca-Cola, they use corn syrup, which is cheaper. By the way, I can go to Costco and buy Mexican Coca-Cola because they use sugar, because it's cheaper. Now, it's more expensive for me, but it's much better. And by the way, just one other interesting fact—and again, this gets to the issue of free trade—Life Savers, which used to be produced in Michigan, moved to Quebec, because they could get a major input, namely sugar, much cheaper.

9.18 So, special interests, concentrated benefits and dispersed costs: the producers gain a lot per person, so they have a very loud voice in the discussion. The consumers lose a

lot per person, so they have a very quiet voice in the discussion, if at all. And this is just one example, but there's so many programs like this. This explains a lot of government.

- 9.19** I was talking earlier about occupational licensing and restricting who can do what and so on. If you look at the occupational licensing boards, state by state, they're made up primarily of people in that occupation who have a strong incentive to keep people out. If you look at subsidies to farmers, we have a couple of million farmers and 340 million food consumers and roughly 200 and some million taxpayers, paying for those subsidies. And so we, as taxpayers, as consumers, don't pay much attention. The farmers do, so they manage to dominate in the political sector. The participation is very lopsided.
- 9.20** So there's a book by a political scientist named James Payne called "The Culture of Spending." And what he did was, he looked at a whole lot of—he looked at 14 House and Senate committees in various years in the mid-1980s and looked at who testified in front of them, and what percent of them were testifying in favor of the government programs, and what percent were testifying against. 1,014 of them, or 95.7%, were in favor of the government programs. There were seven opponents, and they were 0.7% of the overall number of witnesses. So again, we see that lopsided thing in testimony.
- 9.21** Also, when public choice people started talking about these issues, they thought in terms of producer interests being the concentrated interest. But often, the concentrated interest is the government employees themselves. So here's the other thing he found: of the 1,060 Congressional witnesses mentioned, 673 were government officials, and 497 of those were federal administrators, and 65 were members of Congress. In other words, almost two-thirds of those testifying were government officials.
- 9.22** When I worked in the Labor Department, that was where I worked before I went to the Reagan Council of Economic Advisors, I would hear people who had been in that building a long time talk about the public. I came in with this idea of what the public were. And their idea of the public was whoever came and talked to them. So it typically was labor union officials and so on. It wasn't just what most of us would think of as the public. Again, that's the concentrated interest versus the dispersed interest.
- 9.23** Now, one way to see that this concentrated interest thing matters, I found two measures that I'd followed closely at the time, that I voted on in California. Here's the thing. Take any issue. If there's a vote, there's a chance that the median voter dominates. So if the issue is unpopular with the median voter, there are going to be more votes against it than for it.
- 9.24** There was a referendum in 1996 on Proposition 214 in California, the Health Care Patient Protection Act. This would have restricted health insurers' power to limit patient care. Now, why did health insurers want to do that? Well, think about the incentives within the health insurance system. The doctor doesn't bear the cost of

ordering tests. The patient bears only a tiny fraction of the cost. So the insurance company bears most of the cost. And the insurance company wants to somehow be a gatekeeper and say, "Well, is this test really necessary?" So they wanted to limit things like that.

- 9.25** Who were the major supporters of this initiative? Nurses and other medical professionals. In other words, not sick people. We had a referendum, as I said. The vote was 61.2% against and 38.8% in favor. So it went down. Insurance companies were now free to impose the limits they'd had. And by the way, we as subscribers to insurance gain somewhat because, of course, insurance rates are lower than otherwise to reflect those limits. So, the people who wanted those restrictions went to the California legislature, and they got the Department of Managed Health Care set up in 1999. It established a special department to provide stricter oversight of patient rights.
- 9.26** Concentrated interest versus dispersed interest—the concentrated interest wanted this. It's not a referendum. It's a decision made within the government by legislators. And the ones they're hearing from, the ones making the noise, are the ones who want this restriction lifted. They want it to be harder for health insurance companies to restrict what's paid for. So, they lose because the median voter doesn't like it. So they go around it as a concentrated interest and get the legislature to pass it.
- 9.27** Now, one of the most interesting things that's come out of this public choice literature is the incentive to vote. One person's vote among millions or even tens of thousands hardly counts. The odds that your vote makes a difference in a Congressional election or a presidential election, those odds are extremely low, especially in the presidential election. And therefore, lots of people don't vote, and that's completely rational. So if you measure the payoff to voting solely by the outcome, that would say, don't vote. Well, there's a bit of a paradox here, because people do vote. So that must mean there are other motives for voting.
- 9.28** One way is to express your opinion. "I voted." You wear that little thing that says, "I voted." Some people see voting as a duty. I did my duty. Some people want to be seen as a good citizen, and that might explain why they say, "I voted." By the way, when I vote, and I think I'm in the minority, I never wear that thing because I don't want to encourage other people to vote.
- 9.29** But even then, people are rational. They'll vote less if the cost rises. Before poll taxes were made illegal in Southern states, a six-dollar poll tax in 1960—that was a lot of money; that would be like \$50 today—reduced the probability of an individual voting by 42%. So they still do respond to incentives. That whole idea of the gains from voting being so small has led to two terms that public choice economists use— rational ignorance and rational irrationality. Let me look at rational ignorance.
- 9.30** Because your probability of affecting the outcome is so low, you have very little incentive to be informed. A member of Congress or a presidential candidate takes

positions on hundreds of issues, and you might know none of them, or you might know a few of them. You don't have an incentive to be informed because you know, at a gut level, your vote won't affect the outcome. But it's also led to this later concept that came along. Rational ignorance came along in the 1960s and '70s.

- 9.31** This later thing came along in this century: rational irrationality. The idea is that not only is there little incentive to be informed, but there's little incentive to think rationally. For example, this candidate looks good, this candidate's handsome, or this candidate is beautiful, or this candidate is ugly, and it's irrational to vote on that basis, but there's little incentive to think rationally. Compared to what? You're buying a house, for example. You have a strong incentive to think rationally, but the outcome isn't really affected by you, so you have very little incentive to think rationally.
- 9.32** Now, one of the things that has come along as a result of the public choice insights is the whole idea of "taming Leviathan." So Leviathan is this idea of this overarching, big, powerful government that controls a lot of our lives. And of course, the Founding Fathers thought a lot about that. That's why they had the Constitution with its division of powers. And they drew in a Frenchman named Montesquieu, who'd written a lot about this. So the idea is that you tame Leviathan with a division of powers, with checks and balances, and explicitly in the United States with the Constitution, which has a division of powers and has checks and balances.
- 9.33** You don't need to know a lot of history to know that the Constitution is often ignored. For example, Article 1 of the Constitution gives Congress the power to declare war. The Constitution is a set of enumerated powers, which means, translation, if the Constitution doesn't give you that power, you don't have it. And yet, we've had President after President after President make war without Congress declaring war, and often without Congress having any input at all.
- 9.34** Public choice, though, buttresses the view that we should have some division of powers. And public choice people get creative with some of this. So I remember one of them at a conference saying, "You should have two votes on Congress. You should be able to vote for a candidate in your district, and you should be able to vote against a candidate in any other district." So you might single out these high-spenders in some county or some congressional district, probably in another state, who you don't like, you don't like what he's doing or she's doing, and you cast your negative vote on that person, that would be an interesting one.
- 9.35** So there are a lot of clever things people have come up with that, of course, no one has done yet. So that, in a nutshell, is the public choice school. Thank you.

Lecture 10: The West German Economic Miracle

At a glance — The West German economic miracle.

- 10.1** We're ending with one of my favorite topics: the West German economic miracle. It's a case where ideas really matter. We've talked about schools of thought, and I've talked about how, in various ways, they were important in effecting various policies. Here's the clearest-cut case where one particular school of free market economic thinking mattered a lot. It's called the West German economic miracle, but in fact, it's not a miracle if you understand economics, if you understand the bad effect of price controls, the bad effect of high marginal tax rates, and so on.
- 10.2** Let me give you the setting. The setting is postwar West Germany's economy. Food production per capita in 1947 is 51%—half, in other words, of its level in 1938 before World War II started. The official food ration, food was rationed, was 1,040 to 1,550 calories per day. Industrial output in 1947 was one-third its 1938 level. That's 1947. Twenty years later, Germany's economy was the world's envy. What happened?
- 10.3** Well, there were two main causes: currency reform and the end of price controls. Let me say more about the price controls. Hitler's price controls were stringent. There was a death penalty for extreme violations. Hermann Göring imposed rationing in 1938. Germany, of course, loses the war, and the parts of Germany get divided up. The Soviet Union has one piece; the US, Britain, and France have other pieces; and the Allied Control Authority is made up of all four of those.
- 10.4** There's a funny kind of thing that happens with humans. They have status quo bias. They keep what exists. So the Allied Control Authority kept price controls, kept rationing, and kept forced labor, so prices in May 1948 were only 31% of their level in 1938. Now, if you don't understand economics, you might think that's good. No, that just shows how out of line the prices were, because the money supply was five times as much. So with five times as much money, and say, the same amount of output, you should have five times the price level, rather than one-third the price level.
- 10.5** Prices that far out of kilter with what the free market prices would have been are going to cause extreme shortages, extreme. Let me tell you how extreme. People were growing their own food, and they were bartering. The money wasn't worth that much, because people selling the goods for that money weren't making much, because the prices were so low. So what people did was resort to barter.
- 10.6** There was this American economist who wrote a book about it. He was over there observing it, an American economist named Henry Wallich. And here's what he said: "Each day, and particularly on weekends, vast hordes of people trekked out to the country to barter food from the farmers. In dilapidated railway carriages, from which everything pilferable had long disappeared, on the roofs and on the running boards, hungry people traveled sometimes hundreds of miles at snail's pace to where they

hoped to find something to eat. They took their wares—personal effects, old clothes, sticks of furniture, whatever bombed-out remnants they had—and came back with grain or potatoes for a week or two." Remember when we talked about Menger—why money exists? Because barter is so inefficient. But of course, if prices are kept way below free market prices, money isn't going to be that valuable.

- 10.7 A German economist named Walter Eucken said that "barter reduced the economy to a primitive condition." But now we have a school of thought that comes along that's very important. I'm not sure how good my German is going to be. I'll try. It's the "Soziale Marktwirtschaft," or social free market school, based at Germany's University of Freiburg. It's kind of like the Chicago School of the same era. The members included the aforementioned Walter Eucken, Wilhelm Röpke, and Ludwig Erhard.
- 10.8 Röpke said, "We need to end repressed inflation." In other words, we need to end these price controls that are keeping prices down, making it look as if inflation is low, but there's no connection between prices and market prices. We need a currency reform to end inflation and price decontrol to end repression.
- 10.9 Ludwig Erhard said the same thing as Wilhelm Röpke. He had written a memo during World War II, laying out his vision of a free market economy. He made clear that he wanted the Nazis defeated. That takes guts. You're at a German university, and you write something that's out there, saying implicitly or explicitly, "The Nazis need to be defeated." That took guts.
- 10.10 Now, there was another side on this, the Social Democratic Party, and their main economic ideologue was Dr. Lothar Kreyssig. He argued against decontrolling currency reform. He wanted more of a British Labour Party or Scandinavian policy. And by the way, to put it in perspective, if you've ever seen "All Creatures, Great and Small," or other things like that of the postwar era, you hear them talk about rationing. Britain kept rationing and price controls until 1954, nine years after the war. So he wanted something like that.
- 10.11 What was the resolution? Well, General Lucius D. Clay, a US general, was military governor of the US zone, and he wanted non-Nazis, and they weren't easy necessarily to find. But Erhard had refused to join the Nazi Association of University Teachers, so that gave him points with Clay.
- 10.12 Also, Erhard was recommended by Konrad Adenauer, who, of course, later became the German Chancellor. If you do crossword puzzles, as I do, you'll see sometimes, "Name given to Konrad Adenauer, der Alte, the old man." Konrad Adenauer has a very interesting story that gave him his anti-Nazi credentials, big-time. He had been mayor of Cologne in the early 1930s, when Hitler was starting to consolidate his power. And when Hitler came to visit Cologne, Adenauer refused to fly the swastika. He and his wife were put in prison. His wife died in prison. He got out, obviously. But that meant that Lucius Clay could pretty much trust him as a non-Nazi, and even an anti-Nazi.

- 10.13** So, Clay, with the recommendation of Adenauer, made Erhard the director of the bizonal Office of Economic Opportunity. And here's what he did in a very short time: he did a currency reform on Sunday, June 20th, 1948. He substituted Deutsche Marks for Reichsmarks, and the money supply fell by 93%. So now it's more in line with prices. Also that Sunday, the German bizonal economic council adopted a price decontrol ordinance that gave permission against opposition from its SDP members, the Social Democratic Party members, to end price controls.
- 10.14** Erhard had a fun summer. He got rid of, basically, all the price controls. And on his own initiative, he abolished food rationing and ended price controls. And there was a journalist at the time who reported the conversation between General Clay and Ludwig Erhard. Clay: "Herr Erhard, my advisors tell me what you have done is a terrible mistake. What do you say to that?" Erhard: "Herr General, pay no attention to them. My advisors tell me the same thing."
- 10.15** Also, on the issue of rationing, Erhard had a confrontation with a US Army colonel the same month. The colonel said, "How dare you relax our rationing system when there is a widespread food shortage?" Erhard says, "But Herr Oberst, I have not relaxed rationing. I have abolished it. Henceforth, the only rationing ticket the people will need will be the Deutsche Mark, and they will work hard to get these Deutsche Marks, just wait and see." As you'll see in a couple of minutes, we don't have to wait long.
- 10.16** Also, they cut marginal tax rates. The top individual income tax rate going in was 95%, and they kept it at 95%, which sounds bad. But here's the big difference. That rate applied only to income above 250,000 Deutsche Marks annually. Previously, the 95% rate had applied to all income above 6,000 Deutsche Marks, so almost no incentive to produce. In 1950, after these tax cuts, the median-income German was in an 18% tax bracket. At that same real income in 1948, he would have been in an 85% tax bracket.
- 10.17** All this, by the way, was reported by a US economist named Walter Heller. That in itself is an interesting story. Walter Heller later became the chairman of the Council of Economic Advisers under LBJ, first of all under JFK, and then LBJ. He was seen as one of the main opponents of Milton Friedman on the issue of monetary policy. He thought monetary policy didn't matter much. But here's this guy, Walter Heller, saying it's good to have these tax cuts. And by the way, Kennedy, before he was assassinated, proposed cutting the top marginal tax rate from 91%, which is what we had, to 70%, and Walter Heller supported that.
- 10.18** And here's what Heller says in an article in the National Tax Journal in 1948. He said, "The reforms quickly re-established money as the preferred medium of exchange and monetary incentives as the prime mover of economic activity." I want to point something out. I talked earlier about hyperinflation in Germany in 1921-23, when prices went up millions of percent; and yet, people still used money. They had to use it quickly, but they used it.

- 10.19** The money supply was five times what it was before the war, rather than millions times more as in the early '20s. So you might think, well, money would be more valuable after World War II than during the hyperinflation. But no, what that shows is the incredible damage done by price controls, because price controls were not imposed during the hyperinflation. So money could still be used, and you could get it. You could get what you paid for. Whereas with price controls after World War II and the money supply just being five times the pre-war level, money was way less useful.
- 10.20** What were some of the other effects? In May 1948, before these reforms, absenteeism averaged nine and a half hours per week. Workers were spending time going out in the countryside and bartering, so they were taking time away from work. As early as October 1948, absenteeism averaged 4.2 hours per week. From June to December 1948, industrial production rose by more than 50%. Longer term, between the first half of 1948 and 1958, industrial production per capita at an annual rate tripled.
- 10.21** By the way, a personal story about Walter Heller: I talked earlier about Milton Friedman's role in helping end the draft. Well, there was this push to bring the draft back in 1979-1980, and I wrote an economist's statement against the draft, just a punchy one paragraph, and sent it to economists around the country asking them to sign it. And Milton Friedman signed and Alan Greenspan, later chairman of the Fed, signed, all these people.
- 10.22** I sent it to Heller, and I didn't get it back. So I called him up about two weeks later, reached his office at the University of Minnesota. His secretary told me he's out on the island he lives on in Washington State. She gave me his number. Isn't that funny? In those days, you could do that. So she gave me his number. I called him. I said, "Professor Heller, are you willing to sign this statement?" He goes, "No." I go, "Oh, that's too bad."
- 10.23** I always believe you try to get something out of someone, even if you don't get what you wanted. So, I said, "Why?" And he gave, I thought, a so-so answer. And then I said, "How come you moved to Washington State?" And he said, "Well, my friends accused me of trying to move as far away as possible from Milton Friedman." Milton Friedman, meanwhile, had moved from Chicago to the Hoover Institution. So I said, "That didn't work well, did it?" Nice man, though. Nice man.
- 10.24** The German economy is humming, and Erhard gets a reward. He becomes Germany's first Minister of Economic Affairs until 1963. And then he becomes Chancellor of Germany until 1966.
- 10.25** And now I want to tell another personal story: I became friends in college with this man who was about four years older than me, named Clancy Smith. Clancy knew everything. He'd read everything. His name doesn't imply it, but he was Jewish. And he had made peace with Germany because the people in Germany in the '60s weren't the people who did what those people did earlier to his relatives. He knew the whole story about

Erhard, and Erhard, meanwhile, had become Chancellor. And this is in the late, in about 1966 or '67, he came to give a talk at Winnipeg, and I went to the University of Winnipeg, where Clancy went.

- 10.26** So Clancy went to see him speak. And on Clancy's wall is a picture of Erhard that's autographed. Erhard liked cigars; Clancy liked cigars; and Erhard is reaching for a cigar. "I don't have one." And Clancy goes, "I've got one. Do you want one?" "Oh, sure." And I just get such a kick out of that because nowadays, "Does it have an explosive in it?" That kind of thing. But anyway, he gave Erhard a cigar.
- 10.27** Now, in telling the story of the German economic miracle, I haven't said a thing about the Marshall Plan. Remember the Marshall Plan? It was to give money to these countries to help their economies again. Why no mention of the Marshall Plan? Well, at its peak in 1948 and '49, Marshall Plan aid was less than 5% of German income, so it wasn't that big a deal. Moreover, other countries had higher Marshall Plan aid and lower growth. Britain was one of them, by the way. Belgium recovered fastest and relied on free markets, and its recovery predated the Marshall Plan.
- 10.28** So here was a case where ideas mattered. This school of thought in Germany that was ready for what happened once Germany lost the war had a huge effect, and it caused the German economy to recover, and not just recover, but flourish. Imagine what would have happened with the road not taken. Imagine what would have happened if they hadn't got rid of price controls and rationing and hadn't got rid of conscripted labor. That would have been a very bad story, and they avoided that. So ideas really mattered. Thank you.
- 10.29** How many people did we help with the Marshall Plan? How many countries?
- 10.30** How many people did we help with the Marshall Plan? I don't have it in my fingertips. The Marshall Plan aid went to Germany, Britain, France, I think Italy. So if you look at the population of those places, it would be 200 million, 150 million. But you can't say that everyone was helped. Remember, it's government aid. One of the requirements with Marshall Plan money was sometimes you had to buy things from the United States. It was partly implicit subject to exporters. So the Marshall Plan, we hear good things about it, but it wasn't that big a deal.